

# Security Review — Checkmarx False-Positive Responses

Managed package **Pledgivo** (namespace `pledgivo`) · responses prepared 2026-10-07 against source commit `073ee3bd`

## 1. Scan

|                          |                                           |
|--------------------------|-------------------------------------------|
| Scanner                  | Force.com Source Scanner (Checkmarx)      |
| Scan Id                  | a0OKX000001JT8J2AW                        |
| Engine / preset          | CxEngine 9.7 / PortalAll                  |
| Job type                 | ZIP_UPLOAD                                |
| Scan start / end         | 2026-10-06 04:53:53 / 2026-10-06 10:38:14 |
| Security issues reported | 273 paths, 91 distinct code locations     |
| Quality issues reported  | 102 paths (section 5)                     |

## 2. How to read this document

Checkmarx reports one *path* per source-to-sink dataflow, so a single line of code is often reported many times (one path per entry point that reaches it). Section 4 answers each **code location** once, with a response ID (**R-*nnn***). The **Location** column lists the query and the path numbers exactly as they appear in the scan report, the file and line at scan time, and the file and line in the submitted source. The **Response** column has a fixed shape: **Scanner's claim** restates what the query asserts about the flagged statement (the query description from the scan report, and the code it points at); **Why the claim is false** states, in one paragraph, the specific fact that makes the assertion untrue for this statement; **Supporting detail** traces every path that reaches the statement and the control that governs it; **Evidence** lists what was checked, including measurements taken in a scratch org; and **Sources** cites the Salesforce or vendor documentation each reason rests on. Where one location is flagged by more than one query, its full response appears under the first query and later queries refer back to it. Appendix A lists all 273 paths individually with their source, sink and response ID. The scan also reports 102 code-quality paths under three rules; they are not security findings, and section 5 gives each group of them one disposition, with every path listed in Appendix B.

Recurring platform facts the responses rely on:

- **Explicit user mode.** Queries and DML the scanner flags as missing `USER_MODE` often pass `AccessLevel.USER_MODE` as a method argument (the FFLib selector pattern); the query does not read that argument.
- **Deliberate system mode.** The `SYSTEM_MODE` / `as system` operations reported here fall into these families: (a) the Experience Cloud guest licence cannot hold the grant; (b) the object is the detail side of a master-detail relationship, where field permissions are inert and a user-mode query throws, so access is checked on the master; (c) asynchronous or install context (batch, queueable, platform-event subscriber, install handler) with no interactive user; (d) a standard object a second-generation package permission set cannot grant; (e) administrator-only setup-console operations on the package's own configuration and on setup objects such as permission-set assignments.
- **Sharing.** Every top-level class declares a sharing keyword. `without sharing` is used for scheduled/batch/queueable jobs, guest controllers, narrow inner wrappers and logging infrastructure, and each top-level declaration states its reason in the code. The package is built at API 67.0, where a class with no keyword runs `with sharing`, or in its parent class's mode when it extends one.

- **Email sends.** Every `Messaging.sendEmail` the scan reports sets `setSaveAsActivity(false)`, so the send creates no activity record, and its recipient is read from the donation record (the donor, or the tribute honoree the donor named at checkout), never from a request parameter.

### 3. Summary by query

| Query                            | Paths | False positive | Fixed since scan | Hardening before submission | Confirmed — fix |
|----------------------------------|-------|----------------|------------------|-----------------------------|-----------------|
| Apex CRUD Violation              | 28    | 28             | 0                | 0                           | 0               |
| Apex CRUD ContentDistribution    | 7     | 7              | 0                | 0                           | 0               |
| Client DOM Stored XSS            | 1     | 1              | 0                | 0                           | 0               |
| CRUD Delete                      | 1     | 1              | 0                | 0                           | 0               |
| FLS Create                       | 109   | 109            | 0                | 0                           | 0               |
| FLS Update                       | 69    | 69             | 0                | 0                           | 0               |
| Apex SOQL SOSL USER MODE MISSING | 29    | 29             | 0                | 0                           | 0               |
| Sharing                          | 27    | 27             | 0                | 0                           | 0               |
| Apex VF Dynamic JS Resource      | 2     | 2              | 0                | 0                           | 0               |
| Total                            | 273   | 273            | 0                | 0                           | 0               |

#### Query descriptions (from the scan report)

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Apex CRUD Violation           | This query looks for CRUD operations that are performed without checking for <code>isAccessible()</code> , <code>isUpdateable()</code> or <code>isDeletable()</code> . When an object uses <code>Whold</code> , <code>WhatId</code> , <code>ParentId</code> , <code>LinkedEntityId</code> fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields. |
| Apex CRUD ContentDistribution | USER_MODE is recommended for DML operations involving "ContentDistribution" object                                                                                                                                                                                                                                                                                                                                                         |
| Client DOM Stored XSS         | This query detects external data flowing into a javascript execution context when that data comes from the database or other stored parameters influenced by external users. This may be a false positive if the datatype is safe to render unescaped (such as an integer or rich text field).                                                                                                                                             |
| CRUD Delete                   | This query looks for CRUD Delete operations that are performed without checking for <code>isDeletable()</code> permisison before performing the delete operation.                                                                                                                                                                                                                                                                          |
| FLS Create                    | This query looks for CRUD Delete operations that are performed without checking for <code>isCreatable()</code> permisison before performing the Create operation on the Object and its fields.                                                                                                                                                                                                                                             |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FLS Update</b>                       | This query looks for CRUD Update operations that are performed without checking for isUpdateable() permisison before performing the Update operation on the Object and its fields.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Apex SOQL SOSL USER MODE MISSING</b> | USER_MODE is recommended for SOQL or SOSL operations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Sharing</b>                          | All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app. |
| <b>Apex VF Dynamic JS Resource</b>      | Static resources must be used load third-party javascript code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## 4. Responses

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### Apex CRUD Violation — 28 paths, 13 locations

**What this query claims:** *This query looks for CRUD operations that are performed without checking for isAccessible(), isUpdateable() or isDeletable(). When an object uses Whold, WhatId, ParentId, LinkedEntityId fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields.*

| Reported vulnerability                                | Location                                                                                                                              | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>R-001</b><br>Apex CRUD Violation<br>False positive | <b>Paths 1</b><br>At scan:<br>classes/CommunityService.cls:513<br>Submitted source:<br>classes/setup/service/CommunityService.cls:513 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD operations that are performed without checking for isAccessible(), isUpdateable() or isDeletable(). When an object uses Whold, WhatId, ParentId, LinkedEntityId fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields.</i> Flagged statement: Integer assignedCount = [</p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the statement is a read-only COUNT() of PermissionSetAssignment, a setup object with no sharing model, returning a yes/no verdict and no record data; it is reachable only from the administrator-only setup console, and a user-mode read of setup objects requires View Setup and Configuration, which the package's admin permission set cannot assume every administrator holds.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This is a read-only count of the site guest user's permission-set assignments, run from the administrator-only setup console to report whether the payment credential is granted to that guest user; it returns a yes/no verdict and no record data. It runs in system mode because PermissionSetAssignment is setup data that the running administrator can read only with 'View Setup and Configuration', which the package's admin permission set does not grant, and without it the console would falsely report the grant as missing. Only users holding the administrator permission set can reach it. The site carries its own SYSTEM_MODE comment saying so, as does every other setup-object read in the class.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/service/CommunityService.cls:511-516 - // SYSTEM_MODE: the guest user's assignments ... a count only, for the Fundraising_Admin-only Settings console. above the query</li> <li>python3 scripts/lint/system-mode-inventory.py -&gt; 0 unjustified sites (header-only count now excludes this file's reads)</li> <li>SetupController granted only in Fundraising_Admin; Fundraising_Admin carries no ViewSetup userPermission</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a> (system mode ignores the running user's FLS and object permissions; record sharing is governed by the class sharing keyword. Verified via search excerpt only; page body not fetchable)</li> </ul> |

| Reported vulnerability                                | Location                                                                                                                                                         | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>R-002</b><br>Apex CRUD Violation<br>False positive | <b>Paths 2, 7, 15</b><br>At scan:<br>classes/PaymentSetupGuideService.cls:1918<br>Submitted source:<br>classes/payment/service/PaymentSetupGuideService.cls:1918 | <b>SCANNER'S CLAIM</b><br>This query looks for CRUD operations that are performed without checking for <code>isAccessible()</code> , <code>isUpdateable()</code> or <code>isDeletable()</code> . When an object uses <code>Whold</code> , <code>WhatId</code> , <code>ParentId</code> , <code>LinkedEntityId</code> fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields. Flagged statement: <code>if (m.find()) {</code><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the flagged line is a regular-expression match on a string already in memory, not a query or DML statement, so no CRUD operation exists to check; the string was read with <code>WITH USER_MODE</code> , so the running administrator's permissions were already enforced.<br><b>SUPPORTING DETAIL</b><br>The flagged line is a regular-expression match on a string already in memory; it is not a query or DML statement. The data it examines was read with <code>WITH USER_MODE</code> , so the running administrator's object and field permissions were already enforced, and only credential and parameter names are retained for a setup checklist. The path is reachable only by the administrator permission set; the fake-Id generator source is test-only and the other is the SOQL builder.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/payment/service/PaymentSetupGuideService.cls:1917 <code>Matcher m = MERGE_FORMULA.matcher(...); if (m.find())</code></li> <li><code>queryNamedCredentials (:1583)</code> and <code>queryCustomHeaders (:1607)</code> <code>WITH USER_MODE</code></li> <li>base/main/default/classes/setup/controller/SetupController.cls:479; <code>SetupController</code> granted only in <code>Fundraising_Admin</code></li> <li><code>fplib_ApexMocks.mock/verify</code> and <code>fplib_IDGenerator.generate</code> are vendored test-support framework code: they are only ever invoked from <code>*_Test</code> classes to build stubs and fake Ids, and no production entry point reaches this sink through them.</li> <li><code>fplib_QueryFactory.setCondition/selectField(s)/addOrdering</code> is the SOQL builder the selectors use; it is a query-construction helper, not an entry point, and appears as a source only because the engine starts the trace at the builder's string parameters.</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a> (system mode ignores the running user's FLS and object permissions; record sharing is governed by the class sharing keyword. Verified via search excerpt only; page body not fetchable)</li> </ul> |

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| <b>R-003</b><br>Apex CRUD Violation<br>False positive | <b>Paths 3, 10, 20</b><br>At scan:<br>classes/ReceiptService.cls:103<br>Submitted source:<br>classes/donation/service/ReceiptService.cls:103 | <b>SCANNER'S CLAIM</b><br>This query looks for CRUD operations that are performed without checking for isAccessible(), isUpdateable() or isDeletable(). When an object uses Whold, WhatId, ParentId, LinkedEntityId fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields. Flagged statement: Messaging.sendEmail(new List<Messaging.SingleEmailMessage>{ msg });<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim that a CRUD check is missing before a create/update/delete is false because Messaging.sendEmail is not a DML statement: with setSaveAsActivity(false) it writes no record at all, so there is nothing for isCreateable() to guard. The recipient and merge-field source are the just-finalized gift's own donor Contact, not a request parameter.<br><br><b>SUPPORTING DETAIL</b><br>This sink sends a template receipt to the donor Contact of the gift that was just finalized; setTargetObjectId and setWhatId select the recipient and the merge-field sources, and setSaveAsActivity(false) is set, so the send writes no Task/activity record and performs no DML the user could be denied. The recipient is derived from the Opportunity, never from caller input. The only interactive reach is a staff retry of a stuck donation (a permission-set-granted console action), which re-runs the same scheduled finalization and sends the receipt only on first creation of the gift. The QueryFactory 'source' is the SOQL builder, not an entry point. The send was reviewed for a code change on 2026-10-01 and none applies: Messaging.sendEmail takes no access-level argument, and with setSaveAsActivity(false) there is no record operation for a CRUD check to guard.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/service/ReceiptService.cls:94-103 — setTemplateId, setTargetObjectId(targetContactId), setWhatId(opportunityId), setSaveAsActivity(false), applySender, sendEmail</li> <li>base/main/default/classes/donation/service/DonationService.cls:399 — sendDonationReceipt called on first creation only</li> <li>base/main/default/classes/donation/controller/DonationMonitorController.cls:57 retryDonationStagingNow -&gt; StagingReconciliationBatch.reconcileOne (without sharing)</li> <li>base/main/default/classes/donation/selector/OpportunitiesSelector.cls:333 selectDonorContactById SYSTEM_MODE with reason comment</li> <li>fflib_QueryFactory.setCondition/selectField(s)/addOrdering is the SOQL builder the selectors use; it is a query-construction helper, not an entry point, and appears as a source only because the engine starts the trace at the builder's string parameters.</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm">https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm</a> (Email base class: setSaveAsActivity is true by default; saving as activity applies to the targetObjectId record. Verify</li> </ul> |

| Reported vulnerability | Location | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |          | <p>d via search excerpt; the page body returned 403 to direct fetch and via r.jina.ai only a cookie shell, so the exact wording is not quoted. Whether saveAsActivity=false also suppresses an Enhanced Email EmailMessage row: unverified)</p> <ul style="list-style-type: none"><li>• <a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a> (system mode ignores the running user's FLS and object permissions; record sharing is governed by the class sharing keyword. Verified via search excerpt only; page body not fetchable)</li></ul> |

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| <b>R-004</b><br>Apex CRUD Violation<br>False positive | <b>Paths 4, 13</b><br>At scan:<br>classes/ReceiptService.cls:149<br>Submitted source:<br>classes/donation/service/ReceiptService.cls:149 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD operations that are performed without checking for isAccessible(), isUpdateable() or isDeletable(). When an object uses Whold, WhatId, ParentId, LinkedEntityId fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields.</i> Flagged statement: Messaging.sendEmail(new List&lt;Messaging.SingleEmailMessage&gt;{ msg });</p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because Messaging.sendEmail with setSaveAsActivity(false) performs no create, read, update or delete on any sObject, so no CRUD check applies; the recipient Contact Id is read from the finalized ticket order, not from a caller.</p> <p><b>SUPPORTING DETAIL</b></p> <p>The event-ticket receipt is sent to the purchaser's own Contact after the purchase is finalized; setSaveAsActivity(false) means the send creates no activity record, so there is no create/read/update/delete for a CRUD check to guard. The recipient Id comes from the purchase record, not from any request parameter. The only reported source is the SOQL builder class, which is a helper rather than an entry point. The send was reviewed for a code change on 2026-10-01 and none applies: Messaging.sendEmail takes no access-level argument, and with setSaveAsActivity(false) there is no record operation for a CRUD check to guard.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/service/ReceiptService.cls:143-149 — setTargetObjectId, setWhatId, setSaveAsActivity(false), sendEmail</li> <li>base/main/default/classes/event/service/EventPurchaseService.cls:533 caller (sendTicketReceiptQuietly, :1281)</li> <li>fflib_QueryFactory.setCondition/selectField(s)/addOrdering is the SOQL builder the selectors use; it is a query-construction helper, not an entry point, and appears as a source only because the engine starts the trace at the builder's string parameters.</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm">https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm</a> (Email base class: setSaveAsActivity is true by default; saving as activity applies to the targetObjectId record. Verified via search excerpt; the page body returned 403 to direct fetch and via r.jina.ai only a cookie shell, so the exact wording is not quoted. Whether saveAsActivity=false also suppresses an Enhanced Email EmailMessage row: unverified)</li> </ul> |

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| <b>R-005</b><br>Apex CRUD Violation<br>False positive | <b>Paths 5</b><br>At scan:<br>classes/ReceiptService.cls:734<br>Submitted source:<br>classes/donation/service/ReceiptService.cls:734 | <b>SCANNER'S CLAIM</b><br>This query looks for CRUD operations that are performed without checking for isAccessible(), isUpdateable() or isDeletable(). When an object uses Whold, WhatId, ParentId, LinkedEntityId fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields. Flagged statement: Messaging.sendEmail(new List<Messaging.SingleEmailMessage>{ msg });<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the send writes no record (setSaveAsActivity(false)), is reached only from the finalize path and a scheduled renewal job, and addresses the recurring schedule's own donor Contact; no user- or guest-callable entry point selects the recipient.<br><br><b>SUPPORTING DETAIL</b><br>Recurring-gift notices (confirmation, failed payment, dunning, re-authorization) are sent by the finalize path and a scheduled renewal job to the schedule's own donor Contact, with setSaveAsActivity(false), so the send writes no record. No user- or guest-callable entry point reaches this method. The reported sources are the SOQL builder's parameter methods. The send was reviewed for a code change on 2026-10-01 and none applies: Messaging.sendEmail takes no access-level argument, and with setSaveAsActivity(false) there is no record operation for a CRUD check to guard.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/service/ReceiptService.cls:709-734</li> <li>base/main/default/classes/recurring/async/RecurringRenewalBatch.cls:197-202;<br/>base/main/default/classes/recurring/service/RecurringDonationService.cls:956</li> <li>fflib_QueryFactory.setCondition/selectField(s)/addOrdering is the SOQL builder the selectors use; it is a query-construction helper, not an entry point, and appears as a source only because the engine starts the trace at the builder's string parameters.</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm">https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm</a> (Email base class: setSaveAsActivity is true by default; saving as activity applies to the targetObjectId record. Verified via search excerpt; the page body returned 403 to direct fetch and via r.jina.ai only a cookie shell, so the exact wording is not quoted. Whether saveAsActivity=false also suppresses an Enhanced Email EmailMessage row: unverified)</li> </ul> |

| Reported vulnerability                                | Location                                                                                                                                | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>R-006</b><br>Apex CRUD Violation<br>False positive | <b>Paths 6, 8</b><br>At scan:<br>classes/ReceiptService.cls:603<br>Submitted source:<br>classes/donation/service/ReceiptService.cls:603 | <b>SCANNER'S CLAIM</b><br>This query looks for CRUD operations that are performed without checking for isAccessible(), isUpdateable() or isDeletable(). When an object uses Whold, WhatId, ParentId, LinkedEntityId fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields. Flagged statement: Messaging.sendEmail(new List<Messaging.SingleEmailMessage>{ msg });<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the send has no target record (setTargetObjectId(null)) and setSaveAsActivity(false), so it writes nothing; it fires once during server-side finalization, to the honoree address the donor supplied with their own gift, and only when the donor opted in.<br><br><b>SUPPORTING DETAIL</b><br>The tribute notice is a one-time email to the honoree address the donor entered with their gift, sent during server-side finalization on first creation and only when the donor opted in. It has no target record (setTargetObjectId(null)) and setSaveAsActivity(false), so it writes nothing. The reported sources are a test-only fake-Id generator and the SOQL builder. The send was reviewed for a code change on 2026-10-01 and none applies: Messaging.sendEmail takes no access-level argument, and with setSaveAsActivity(false) there is no record operation for a CRUD check to guard.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/service/ReceiptService.cls:702-704 setTargetObjectId(null), setToAddresses(honoree), setSaveAsActivity(false)</li> <li>base/main/default/classes/donation/service/DonationService.cls:404 caller</li> <li>fflib_ApexMocks.mock/verify and fflib_IDGenerator.generate are vendored test-support framework code: they are only ever invoked from *_Test classes to build stubs and fake Ids, and no production entry point reaches this sink through them.</li> <li>fflib_QueryFactory.setCondition/selectField(s)/addOrdering is the SOQL builder the selectors use; it is a query-construction helper, not an entry point, and appears as a source only because the engine starts the trace at the builder's string parameters.</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm">https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm</a> (Email base class: setSaveAsActivity is true by default; saving as activity applies to the targetObjectId record. Verified via search excerpt; the page body returned 403 to direct fetch and via r.jina.ai only a cookie shell, so the exact wording is not quoted. Whether saveAsActivity=false also suppresses an Enhanced Email EmailMessage row: unverified)</li> </ul> |

| Reported vulnerability                                | Location                                                                                                                                 | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>R-007</b><br>Apex CRUD Violation<br>False positive | <b>Paths 9, 22</b><br>At scan:<br>classes/ReceiptService.cls:518<br>Submitted source:<br>classes/donation/service/ReceiptService.cls:518 | <b>SCANNER'S CLAIM</b><br>This query looks for CRUD operations that are performed without checking for isAccessible(), isUpdateable() or isDeletable(). When an object uses Whold, WhatId, ParentId, LinkedEntityId fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields. Flagged statement: <code>if (!msgs.isEmpty()) { Messaging.sendEmail(msgs); }</code><br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because this is an internal alert from a scheduled job using explicit addresses, no target record and setSaveAsActivity(false); it performs no DML and no user- or guest-reachable path leads to it.<br><br><b>SUPPORTING DETAIL</b><br>This is an internal alert sent by a scheduled reconciliation job to the owner of a disputed gift and the configured reply-to address; it uses explicit addresses, no target record, and setSaveAsActivity(false), so it performs no DML. No user- or guest-reachable code path reaches it; the reported sources are a test-only fake-Id generator and the SOQL builder's parameter methods. The send was reviewed for a code change on 2026-10-01 and none applies: Messaging.sendEmail takes no access-level argument, and with setSaveAsActivity(false) there is no record operation for a CRUD check to guard.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/service/ReceiptService.cls:575-579 setToAddresses + setSaveAsActivity(false)</li> <li>base/main/default/classes/donation/async/RefundReconciliationBatch.cls:726 sole caller</li> <li>base/main/default/classes/donation/selector/OpportunitiesSelector.cls:940 selectOwnerEmailById SYSTEM_MODE</li> <li>fflib_ApexMocks.mock/verify and fflib_IDGenerator.generate are vendored test-support framework code: they are only ever invoked from *_Test classes to build stubs and fake Ids, and no production entry point reaches this sink through them.</li> <li>fflib_QueryFactory.setCondition/selectField(s)/addOrdering is the SOQL builder the selectors use; it is a query-construction helper, not an entry point, and appears as a source only because the engine starts the trace at the builder's string parameters.</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm">https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm</a> (Email base class: setSaveAsActivity is true by default; saving as activity applies to the targetObjectId record. Verified via search excerpt; the page body returned 403 to direct fetch and via r.jina.ai only a cookie shell, so the exact wording is not quoted. Whether saveAsActivity=false also suppresses an Enhanced Email EmailMessage row: unverified)</li> </ul> |

| Reported vulnerability                                | Location                                                                                                                                           | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>R-008</b><br>Apex CRUD Violation<br>False positive | <b>Paths 11, 23, 27</b><br>At scan:<br>classes/DonorPortalService.cls:483<br>Submitted source:<br>classes/donor/service/DonorPortalService.cls:483 | <b>SCANNER'S CLAIM</b><br>This query looks for CRUD operations that are performed without checking for isAccessible(), isUpdateable() or isDeletable(). When an object uses Whold, WhatId, ParentId, LinkedEntityId fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields. Flagged statement: Messaging.sendEmail(new List<Messaging.SingleEmailMessage>{<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because Messaging.sendEmail with explicit addresses and setSaveAsActivity(false) writes no record, so there is no DML to guard; the address must already belong to an existing donor, the request is rate-limited, and the response is identical whether or not it matched.<br><br><b>SUPPORTING DETAIL</b><br>This sends a one-time sign-in code to a donor's own address after a guest asks for it; the address must match an existing donor, the request is rate-limited, and the response is the same whether or not it matched. The message uses explicit addresses with no target record and setSaveAsActivity(false), so no record is created. The staff controller methods listed as sources reach a different send site (the staff twin) and share only the message-building helper; the fake-Id generator source is test-only. The send was reviewed for a code change on 2026-10-01 and none applies: Messaging.sendEmail takes no access-level argument, and with setSaveAsActivity(false) there is no record operation for a CRUD check to guard.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donor/service/DonorPortalService.cls:481-490 sendCodeEmail</li> <li>buildCodeEmail: setToAddresses, setSaveAsActivity(false)</li> <li>fflib_ApexMocks.mock/verify and fflib_IDGenerator.generate are vendored test-support framework code: they are only ever invoked from *_Test classes to build stubs and fake Ids, and no production entry point reaches this sink through them.</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm">https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm</a> (Email base class: setSaveAsActivity is true by default; saving as activity applies to the targetObjectId record. Verified via search excerpt; the page body returned 403 to direct fetch and via r.jina.ai only a cookie shell, so the exact wording is not quoted. Whether saveAsActivity=false also suppresses an Enhanced Email EmailMessage row: unverified)</li> </ul> |

| Reported vulnerability                                | Location                                                                                                                                           | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>R-009</b><br>Apex CRUD Violation<br>False positive | <b>Paths 12, 24, 28</b><br>At scan:<br>classes/DonorPortalService.cls:548<br>Submitted source:<br>classes/donor/service/DonorPortalService.cls:548 | <b>SCANNER'S CLAIM</b><br>This query looks for CRUD operations that are performed without checking for <code>isAccessible()</code> , <code>isUpdateable()</code> or <code>isDeletable()</code> . When an object uses <code>Whold</code> , <code>WhatId</code> , <code>ParentId</code> , <code>LinkedEntityId</code> fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields. Flagged statement: <code>Messaging.sendEmail(new List&lt;Messaging.SingleEmailMessage&gt;{ code.message });</code><br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the send itself performs no DML ( <code>setSaveAsActivity(false)</code> ); the staff actions that reach it re-read the donor Contact in user mode, so the running user's CRUD, FLS and sharing have already been enforced on the only record involved.<br><br><b>SUPPORTING DETAIL</b><br>The CRUD finding on the send itself is a false positive: <code>setSaveAsActivity(false)</code> is set and no record is written by <code>Messaging.sendEmail</code> . The staff 'send portal link' and 'show portal code' actions accept a Contact Id, so both staff twins re-read the donor in the staff user's own mode ( <code>resolveStaffDonorEmail</code> , <code>USER_MODE</code> selectors for Contact and Person Account) and refuse a donor that user cannot see, before any code is minted, any email is sent or any address is returned. The guest path, which is authorized by possession of the code it produces, is unchanged. The send was reviewed for a code change on 2026-10-01 and none applies: <code>Messaging.sendEmail</code> takes no access-level argument, and with <code>setSaveAsActivity(false)</code> there is no record operation for a CRUD check to guard.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donor/service/DonorPortalService.cls:30-47 - class header: the staff twins re-read the donor in the STAFF user's mode (<code>resolveStaffDonorEmail</code>) and refuse a donor that user cannot see</li> <li>base/main/default/classes/donor/controller/DonorPortalAdminController.cls:72 <code>SendPortalLinkAction</code>, :93 <code>CreatePortalLinkAction</code>, :156 <code>ContactsSelector.selectById (USER_MODE)</code>, :175 <code>AccountsSelector.selectPersonDonorById (USER_MODE)</code></li> <li>DonorPortalService_Test / DonorPortalAdminController_Test: a Fundraising_User who cannot see the Contact gets 'not found' and no <code>Portal_Access_Token__c</code> row is created</li> <li>docs/security/appexchange-security-review-readiness.md F-10 (donor-portal IDOR closed 2026-09-25)</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm">https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm</a> (Email base class: <code>setSaveAsActivity</code> is true by default; saving as activity applies to the <code>targetObjectId</code> record. Verified via search excerpt; the page body returned 403 to direct fetch and via <code>r.jina.ai</code> only a cookie shell, so the exact wording is not quoted. Whether <code>saveAsActivity=false</code> also suppresses an Enhanced Email <code>EmailMessage</code> row: unverified)</li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a> (system mode ignores the running user's FLS and object permissions; record sharing is governed by the class sharing keyword. Verified via search excerpt only; page body not fetchable)</li> </ul> |

| Reported vulnerability                                | Location                                                                                                                                      | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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|                                                       |                                                                                                                                               | <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a> (sharing mode of the class where a method is DEFINED applies, not the caller's; inherited sharing takes the caller's. Page body not fetchable; the 'defined, not called' rule confirmed only via secondary sources, e.g. <a href="https://salesforcegeek.in/with-sharing-and-without-sharing-in-apex/">https://salesforcegeek.in/with-sharing-and-without-sharing-in-apex/</a> — treat as partially verified)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>R-010</b><br>Apex CRUD Violation<br>False positive | <b>Paths 14, 17, 25</b><br>At scan:<br>classes/ReceiptService.cls:329<br>Submitted source:<br>classes/donation/service/ReceiptService.cls:329 | <b>SCANNER'S CLAIM</b><br>This query looks for CRUD operations that are performed without checking for isAccessible(), isUpdateable() or isDeletable(). When an object uses Whold, WhatId, ParentId, LinkedEntityId fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields. Flagged statement: <code>if (!msgs.isEmpty()) { Messaging.sendEmail(msgs); }</code><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because every reported dataflow source is test-support framework code reachable only from test classes, and the sink is an email send with setSaveAsActivity(false) that creates no record; in production the method runs from the platform-event subscriber after a gift is finalized, to that gift's own donor.<br><b>SUPPORTING DETAIL</b><br>Every source reported for this sink is test-support framework code (the mocking library's mock/verify methods and its fake-Id generator), which is only called from test classes. In production the method runs from a platform-event subscriber after a gift is finalized and emails matching-gift instructions to that gift's own donor Contact. setSaveAsActivity(false) is set, so no activity record is written and there is no DML to check. The send was reviewed for a code change on 2026-10-01 and none applies: <code>Messaging.sendEmail</code> takes no access-level argument, and with setSaveAsActivity(false) there is no record operation for a CRUD check to guard.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/service/ReceiptService.cls:313-327 — loop builds messages, setSaveAsActivity(false)</li> <li>base/main/default/classes/donation/service/CompanyMatchService.cls:65 sole production caller</li> <li>fplib_ApexMocks.mock/verify and fplib_IDGenerator.generate are vendored test-support framework code: they are only ever invoked from *_Test classes to build stubs and fake Ids, and no production entry point reaches this sink through them.</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm">https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm</a> (Email base class: setSaveAsActivity is true by default; saving as activity applies to the targetObjectId record. Verified via search excerpt; the page body returned 403 to direct fetch and via r.jina.ai only a cookie shell, so the exact wording is not quoted. Whether saveAsActivity=false also suppresses an Enhanced Email EmailMessage row: unverfied)</li> </ul> |

| Reported vulnerability                                | Location                                                                                                                                      | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>R-011</b><br>Apex CRUD Violation<br>False positive | <b>Paths 16, 18, 26</b><br>At scan:<br>classes/ReceiptService.cls:408<br>Submitted source:<br>classes/donation/service/ReceiptService.cls:408 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD operations that are performed without checking for isAccessible(), isUpdateable() or isDeletable(). When an object uses Whold, WhatId, ParentId, LinkedEntityId fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields.</i> Flagged statement: <code>if (!msgs.isEmpty()) { Messaging.sendEmail(msgs); }</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the send writes no record (setSaveAsActivity(false)), it runs only after a refund has already been processed under the Process_Refunds custom permission or by the reconciliation job, and the recipient is the refunded gift's own donor Contact.</p> <p><b>SUPPORTING DETAIL</b></p> <p>The refund notice is sent only after a refund has been processed, to the refunded gift's own donor Contact, with setSaveAsActivity(false), so the send itself writes no record. The interactive path requires a dedicated custom permission checked in the controller before any work, and the controller declares with sharing, which the service inherits. Two of the reported sources are test-only mocking framework methods; the third is that permission-gated controller. The send was reviewed for a code change on 2026-10-01 and none applies: Messaging.sendEmail takes no access-level argument, and with setSaveAsActivity(false) there is no record operation for a CRUD check to guard.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/controller/RefundController.cls:22 refundDonation; doValidate checks FeatureManagement.checkPermission('Process_Refunds')</li> <li>base/main/default/classes/donation/service/RefundService.cls:15 public inherited sharing class RefundService; :328-334 sendRefundNotification after commit</li> <li>base/main/default/classes/donation/service/ReceiptService.cls:404-408 setSaveAsActivity(false)</li> <li>base/main/default/classes/donation/selector/OpportunitiesSelector.cls:865 selectByIdForRefund SYSTEM_MODE FOR UPDATE</li> <li>fplib_ApexMocks.mock/verify and fplib_IDGenerator.generate are vendored test-support framework code: they are only ever invoked from *_Test classes to build stubs and fake Ids, and no production entry point reaches this sink through them.</li> <li>base/main/default/classes/donation/async/RefundReconciliationBatch.cls:734 batch caller (sendRefundNotifications, ReceiptService:380)</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm">https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm</a> (Email base class: setSaveAsActivity is true by default; saving as activity applies to the targetObjectId record. Verified via search excerpt; the page body returned 403 to direct fetch and via r.jina.ai only a cookie shell, so the exact wording is not quoted. Whether saveAsActivity=false also suppresses an Enhanced Email EmailMessage row: unverified)</li> </ul> |

| Reported vulnerability | Location | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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|                        |          | <ul style="list-style-type: none"> <li>• <a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a><br/>(system mode ignores the running user's FLS and object permissions; record sharing is governed by the class sharing keyword. Verified via search excerpt only; page body not fetchable)</li> <li>• <a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a><br/>(sharing mode of the class where a method is DEFINED applies, not the caller's; inherited sharing takes the caller's. Page body not fetchable; the 'defined, not called' rule confirmed only via secondary sources, e.g. <a href="https://salesforcegeek.in/with-sharing-and-without-sharing-in-apex/">https://salesforcegeek.in/with-sharing-and-without-sharing-in-apex/</a> — treat as partially verified)</li> </ul> |

| Reported vulnerability                                | Location                                                                                                                              | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>R-012</b><br>Apex CRUD Violation<br>False positive | <b>Paths 19</b><br>At scan:<br>classes/ReceiptService.cls:173<br>Submitted source:<br>classes/donation/service/ReceiptService.cls:173 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD operations that are performed without checking for isAccessible(), isUpdateable() or isDeletable(). When an object uses Whold, WhatId, ParentId, LinkedEntityId fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields.</i> Flagged statement: Messaging.sendEmail(new List&lt;Messaging.SingleEmailMessage&gt;{ msg });</p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the send writes no record (setSaveAsActivity(false)), and the only records involved, the gift and its donor Contact, are read in user mode before the message is built, so a staff user who cannot see them gets nothing to send.</p> <p><b>SUPPORTING DETAIL</b></p> <p>Resending a receipt is a staff action whose record access is enforced before the message is built: the gift and the donor Contact are read in user mode (object, field and sharing checks of the running user), so a user who cannot see either gets nothing to send. The recipient is always the gift's own donor Contact, not a caller-supplied address, and setSaveAsActivity(false) means the send writes no record. The other two sources reported are the SOQL builder's parameters, not entry points. The send was reviewed for a code change on 2026-10-01 and none applies: Messaging.sendEmail takes no access-level argument, and with setSaveAsActivity(false) there is no record operation for a CRUD check to guard.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/service/ReceiptService.cls:196 oppSel.selectById (USER_MODE), :211 conSel.selectById (USER_MODE)</li> <li>base/main/default/classes/donation/service/ReceiptService.cls:223-227 setTargetObjectId + setSaveAsActivity(false)</li> <li>base/main/default/classes/donation/controller/ReceiptAdminController.cls:38 resendReceipt, class declared with sharing</li> <li>Fundraising_User / Fundraising_Admin permission sets grant ReceiptAdminController</li> <li>fflib_QueryFactory.setCondition/selectField(s)/addOrdering is the SOQL builder the selectors use; it is a query-construction helper, not an entry point, and appears as a source only because the engine starts the trace at the builder's string parameters.</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm">https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm</a> (Email base class: setSaveAsActivity is true by default; saving as activity applies to the targetObjectId record. Verified via search excerpt; the page body returned 403 to direct fetch and via r.jina.ai only a cookie shell, so the exact wording is not quoted. Whether saveAsActivity=false also suppresses an Enhanced Email EmailMessage row: unverified)</li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a> (system mode ignores the running user's FLS and object permissions; record sharing is governed by the class sharing)</li> </ul> |

| Reported vulnerability                                | Location                                                                                                                                              | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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|                                                       |                                                                                                                                                       | <p>ng keyword. Verified via search excerpt only; page body not fetchable)</p> <ul style="list-style-type: none"> <li>• <a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a> (sharing mode of the class where a method is DEFINED applies, not the caller's; inherited sharing takes the caller's. Page body not fetchable; the 'defined, not called' rule confirmed only via secondary sources, e.g. <a href="https://salesforcegeek.in/with-sharing-and-without-sharing-in-apex/">https://salesforcegeek.in/with-sharing-and-without-sharing-in-apex/</a> — treat as partially verified)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>R-013</b><br>Apex CRUD Violation<br>False positive | <b>Paths 21</b><br>At scan:<br>classes/RefundRequestHandler.cls:138<br>Submitted source:<br>classes/guestrequest/handler/RefundRequestHandler.cls:138 | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD operations that are performed without checking for isAccessible(), isUpdateable() or isDeletable(). When an object uses Whold, WhatId, ParentId, LinkedEntityId fields an additional CRUD check should also be performed on the parent object along with the CRUD and FLS checks on the child object and fields.</i> Flagged statement: <code>Messaging.sendEmail(new List&lt;Messaging.SingleEmailMessage&gt;{ msg });</code><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the send performs no DML (setSaveAsActivity(false), explicit addresses); the gift Id it refers to was validated against the authenticated donor's own giving history when the request was recorded, and the message goes only to that gift's owner and the configured reply-to address.<br><b>SUPPORTING DETAIL</b><br>This notifies staff that a donor asked for a refund through the donor portal; the gift Id was checked against the authenticated donor's own giving history when the request was recorded, and the message goes to that gift's owner and the configured reply-to address, never to a caller-supplied address. It is plain text with explicit addresses and setSaveAsActivity(false), so it writes no record. The reported sources are the SOQL builder's parameter methods, not entry points. The send was reviewed for a code change on 2026-10-01 and none applies: <code>Messaging.sendEmail</code> takes no access-level argument, and with <code>setSaveAsActivity(false)</code> there is no record operation for a CRUD check to guard.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>• base/main/default/classes/guestrequest/handler/RefundRequestHandler.cls:194-199 setToAddresses + setSaveAsActivity(false)</li> <li>• DonorPortalService.submitRefundRequest -&gt; requireOwnedDonation</li> <li>• fflib_QueryFactory.setCondition/selectField(s)/addOrdering is the SOQL builder the selectors use; it is a query-construction helper, not an entry point, and appears as a source only because the engine starts the trace at the builder's string parameters.</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li>• <a href="https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm">https://developer.salesforce.com/docs/atlas.en-us.apexref.meta/apexref/apex_classes_email_outbound_base.htm</a> (Email base class: setSaveAsActivity is true by default; saving as activity applies to the targetObjectId record. Verified via search excerpt; the page body returned 403 to direct fetch and via r.jina.ai only a cookie shell, so the exact wording is not quoted. Whether saveAsActivity=false also suppresses an Enhanced Email EmailMessage row: unverified)</li> </ul> |

## Apex CRUD ContentDistribution — 7 paths, 3 locations

**What this query claims:** *USER\_MODE is recommended for DML operations involving "ContentDistribution" object*

| Reported vulnerability                                             | Location                                                                                                                                         | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>R-014</b><br>Apex CRUD<br>ContentDistribution<br>False positive | <b>Paths 1–2</b><br>At scan:<br>classes/CampaignPostService.cls:555<br>Submitted source:<br>classes/campaign/service/CampaignPostService.cls:555 | <p><b>SCANNER'S CLAIM</b></p> <p><i>USER_MODE is recommended for DML operations involving "ContentDistribution" object</i> Flagged statement: <code>delete as system revokeDistributions;</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim that this delete should run in user mode is false because the user was already authorized for the action that triggers it (deleting or replacing a campaign post passes the controller's deletable/editable check on the post in user mode), and ContentDistribution is a standard object a 2GP permission set cannot grant: a user-mode delete of a public link created by a different staff member fails with <code>INSUFFICIENT_ACCESS_OR_READONLY</code> (measured), which would leave a revoked file publicly downloadable.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This delete revokes the public ContentDistribution link a campaign post served, after staff have deleted the post or replaced its file. It runs only after an action the user was already authorised for: deleting a post first passes <code>requireDeletable</code> on <code>Campaign_Post__c</code> and commits the delete through a user-mode Unit of Work, so the platform refuses it without delete access; replacing a file first filters the posts through <code>RecordAccess.editableIds</code>. The revocation set is narrow: <code>planRevocations</code> keeps any link that another post still serves, so only the link the released post exposed is removed. ContentDistribution is a standard object a second-generation package permission set cannot grant, so a user-mode delete would depend on a manual admin step. The path is never reachable by a guest, and revoking a public link reduces exposure; it never widens it. Converting this delete to user mode was tested and rejected because it breaks the feature for the users it is built for. Measured 2026-10-01 in a scratch org, running as a Standard User who holds the <code>Fundraising_Admin</code> permission set: deleting a ContentDistribution created by another staff member with <code>Database.delete(rows, AccessLevel.USER_MODE)</code> fails with <code>INSUFFICIENT_ACCESS_OR_READONLY</code>. In user mode the public link of a post one colleague published could not be revoked when another colleague deletes that post, and the file would stay publicly reachable. System mode is what guarantees the link is withdrawn.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/campaign/controller/CampaignPostController.cls:156 - <code>requireDeletable(Campaign_Post__c.SObjectType, 'campaign posts')</code> before :166 <code>deletePost(postId)</code></li> <li>base/main/default/classes/campaign/service/CampaignPostService.cls:163-187 - <code>deletePost</code>: the <code>UserModeDML</code> delete commits at :179-183 before <code>retirement.execute(false)</code> at :186</li> <li>base/main/default/classes/campaign/service/CampaignPostService.cls:379-381 - <code>stampablePosts</code> filters through <code>RecordAccess.editableIds</code> before a file replace; :438 runs the retirement</li> <li>base/main/default/classes/campaign/service/CampaignPostService.cls:515-545 - <code>planRevocations</code> keeps any link another post still serves (<code>selectByFileUrlsSystemMode</code>, :530)</li> </ul> |

| Reported vulnerability | Location | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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|                        |          | <ul style="list-style-type: none"><li>base/main/default/classes/campaign/service/CampaignPostService.cls:547-555 - execute() rationale and the SYSTEM_MODE comment at the site</li><li>base/main/default/classes/campaign/service/CampaignPostService.cls:453 - <code>private with sharing class FileRetirement;</code> outer class <code>with sharing</code> at:28</li></ul> <b>SOURCES</b> <ul style="list-style-type: none"><li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a> (user-mode DML enforces the running user's object permissions and sharing)</li><li>AGENTS.md Change Management: 2GP strips standard-object grants (project finding 2026-08-16)</li></ul> |

| Reported vulnerability                                             | Location                                                                                                                                     | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>R-015</b><br>Apex CRUD<br>ContentDistribution<br>False positive | <b>Paths 3–4, 6</b><br>At scan: <code>classes/AppLogger.cls:125</code><br>Submitted source:<br><code>classes/shared/AppLogger.cls:125</code> | <p><b>SCANNER'S CLAIM</b></p> <p><i>USER_MODE is recommended for DML operations involving "ContentDistribution" object</i> Flagged statement: <code>insert as system new Log__c(</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the insert is the package's diagnostic log, which must succeed from catch blocks in every context including guest controllers; no permission set grants Create on Log__c by design, so a user-mode insert fails for every user (measured: <code>insert as user new Log__c</code> throws <code>System.TypeException "DML operation INSERT not allowed on Log__c"</code>), and the row holds only a class name, level and message with no caller-controlled routing.</p> <p><b>SUPPORTING DETAIL</b></p> <p>AppLogger persists diagnostic entries from catch blocks in every context, including guest controllers whose users cannot hold Create on any internal object; no permission set grants Create on Log__c by design, so logs are writable only through this class. Values are truncated to field length, Log__c is Private with read granted only to internal staff sets, and no client-callable method takes a log payload - the message text comes from server-side exceptions and developer-written strings. The class-level comment documents why it is <code>without sharing</code> and bypasses the Unit of Work. Converting this insert to user mode was tested and rejected. Measured 2026-10-01 in a scratch org, running as a Standard User who holds the Fundraising_Admin permission set: <code>insert as user new Log__c(...)</code> throws <code>System.TypeException: DML operation INSERT not allowed on Log__c</code>, because no permission set grants Create on the log object - by design, so that no user can author or forge diagnostic rows. A user-mode insert would therefore lose every log entry, including the ones that record security-relevant failures.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/shared/AppLogger.cls:1-9 - <code>without sharing</code> with rationale</li> <li>base/main/default/classes/shared/AppLogger.cls:117-135 - DML-limit guard; left() truncation; failures swallowed</li> <li>Log__c OWD Private; Fundraising_Admin/User/ReadOnly: Read only, no Create; GuestDonor: no grant</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a> (user-mode DML throws when the running user lacks object Create)</li> <li>Measured in a scratch org (2026-09-30): <code>`insert as user new Log__c(...)`</code> throws <code>System.TypeException "DML operation INSERT not allowed on Log__c"</code></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                               | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>R-016</b><br>Apex CRUD<br>ContentDistribution<br>False positive | <b>Paths 5, 7</b><br>At scan:<br>classes/SettingsService.cls:345<br>Submitted source:<br>classes/setup/service/SettingsService.cls:345 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for DML operations involving "ContentDistribution" object</i> Flagged statement: <code>upsert as system settings;</code><br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false on both counts: the upsert is shared by the InstallHandler (package system user, no permission set) and the admin console (gated by the Manage_Fundraising_Settings custom permission), and in both contexts a user-mode write on the package's hierarchy custom setting requires Customize Application, which neither the install user nor the package's admin permission set can hold (measured: <code>upsert as user</code> throws <code>SecurityException</code> ). The ContentDistribution query reads only a Content Delivery the same code created from a packaged static resource.<br><br><b>SUPPORTING DETAIL</b><br>This upsert is shared by the package install handler, which runs as the package's system user with no permission set, and by the settings console controller, which is granted only in the Fundraising_Admin permission set. System mode is kept because the install path has no FLS to enforce and because DML on a Hierarchy Custom Setting in user mode requires Customize Application, a system permission no packaged permission set can carry. The console path does not rely on Apex class access alone: every controller action that reaches this write (SaveGeneralSettings, DetectOrgFeatures, ResetToDefaults, SaveExperienceSite) first checks the packaged custom permission Manage_Fundraising_Settings, granted only in Fundraising_Admin, and refuses the save with a generic error when the running user does not hold it. The site carries a SYSTEM_MODE comment naming that gate. The user-mode alternative was tested and does not work for the people who use this screen. Measured 2026-10-01 in a scratch org, running as a Standard User who holds the Fundraising_Admin permission set: <code>Schema.sObjectType.Settings__c</code> reports <code>isCreateable</code> , <code>isUpdateable</code> and <code>isAccessible</code> all false, and <code>upsert as user</code> on the settings row throws <code>System.SecurityException: Access to entity 'Settings__c' denied</code> . The platform ties custom-setting DML to Customize Application, which is a system permission the fundraising administrator is not expected to hold, so a user-mode write would lock every non-system-administrator out of the settings console. The custom-permission check before the write is the access control; it is enforced in Apex on every action that reaches this line.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/service/SettingsService.cls:338-345 - <code>// SYSTEM_MODE: reason naming the Manage_Fundraising_Settings gate, then upsert as system settings</code></li> <li>base/main/default/classes/setup/controller/SetupController.cls:1737-1742 - <code>SettingsWriteAction.requireSettingsPermission(); FeatureManagement.checkPermission('Manage_Fundraising_Settings') else addError(SETTINGS_NOT_AUTHORIZED)</code></li> </ul> |

| Reported vulnerability | Location | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|                        |          | <ul style="list-style-type: none"> <li>base/main/default/classes/setup/controller/SetupController.cls:1745/1754 SaveGeneralSettings, :2666-2667 DetectOrgFeatures, :2702-2703 ResetToDefaults, :2762/2771 SaveExperienceSite - each doValidate() starts with requireSettingsPermission()</li> <li>base/main/default/customPermissions/Manage_Fundraising_Settings.customPermission-meta.xml; Fundraising_Admin.permissionset-meta.xml grants it; Fundraising_User / Fundraising_ReadOnly / Fundraising_GuestDonor do not</li> <li>SetupController_Test: five deny tests (bare Standard User), one deny test (Fundraising_User), one allow test (Fundraising_Admin); 312 tests pass, SetupController 96%, SettingsService 98%</li> <li>PostInstallHandler.onInstall -&gt; seedRecordTypeMappings / ensureDefaultLogoSeeded still reach this write with no permission check: install context, no running user to check</li> <li>docs/security/apexchange-security-review-readiness.md F-13</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://help.salesforce.com/s/articleView?id=sf.cs_schema_settings.htm">https://help.salesforce.com/s/articleView?id=sf.cs_schema_settings.htm</a></li> <li><a href="https://applicationperfection.com/2024/02/29/control-who-gets-read-access-to-custom-settings/">https://applicationperfection.com/2024/02/29/control-who-gets-read-access-to-custom-settings/</a></li> <li><a href="https://github.com/patronmanager/apex-dml-manager/issues/2">https://github.com/patronmanager/apex-dml-manager/issues/2</a></li> <li><a href="https://help.salesforce.com/s/articleView?language=en_US&amp;id=release-notes.rn_apex_default_user_mode.htm&amp;release=262&amp;type=5">https://help.salesforce.com/s/articleView?language=en_US&amp;id=release-notes.rn_apex_default_user_mode.htm&amp;release=262&amp;type=5</a></li> <li><a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5</a></li> </ul> |

## Client DOM Stored XSS — 1 paths, 1 locations

**What this query claims:** This query detects external data flowing into a javascript execution context when that data comes from the database or other stored parameters influced by external users. This may be a false positive if the datatype is safe to render unescaped (such as an integer or rich text field).

| Reported vulnerability                                               | Location                                                                                                                                              | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>R-017</b><br>Client DOM Stored XSS<br><span>False positive</span> | <b>Paths 1</b><br>At scan:<br>lwc/pfGivingStatement/pfGivingStatement.html:34<br>Submitted source:<br>lwc/pfGivingStatement/pfGivingStatement.html:34 | <b>SCANNER'S CLAIM</b><br><i>This query detects external data flowing into a javascript execution context when that data comes from the database or other stored parameters influenced by external users. This may be a false positive if the datatype is safe to render unescaped (such as an integer or rich text field). Flagged statement: &lt;p class="gs-err" role="alert"&gt;{errorMessage}&lt;/p&gt;</i><br><b>WHY THE CLAIM IS FALSE</b><br>The claim that stored, externally influenced data reaches a JavaScript execution context is false because the reported source (the donor's portal session token) is only passed as an Apex parameter and never written to the DOM; the value rendered at the flagged line is the response envelope's error text or a packaged constant, bound through an LWC {expression}, which the framework renders as escaped text and never as HTML or script.<br><b>SUPPORTING DETAIL</b><br>The flagged source is the donor's own portal session token read from sessionStorage; it is passed as an Apex parameter to DonorPortalController.getGivingStatement and is never written into the DOM. The value rendered at base/main/default/lwc/pfGivingStatement/pfGivingStatement.html:34 is either the error string of the controller's response envelope or a package-authored literal. The controller's action classes extend GuestAction, whose callerSafeMessage returns an AppException message (every one on this path is a constant: the invalid-session message, 'Choose a tax year.', 'That tax year is not available.', 'A donor is required to build a giving statement.') or one fixed generic sentence for any other exception - no record field, request parameter or token is echoed. Finally, the sink is an LWC {expression} text binding, which the framework renders as a text node, not HTML; neither pfGivingStatement nor publicDonorDashboard uses innerHTML, lwc:inner-html or lwc:dom="manual". The flagged source is also shape-checked: the value read from sessionStorage is accepted only when it matches the exact shape of a session token the server issues (64 lowercase hexadecimal characters); anything else is discarded and removed from storage before it reaches component state or an Apex call. The finding is a false positive - the value is never rendered - and storage content that the component did not write cannot enter it at all.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/lwc/publicDonorDashboard/publicDonorDashboard.js:314-331 - stored = window.sessionStorage.getItem(SESSION_KEY) (SESSION_KEY = 'pf-donor-portal-session', :41); assigned to this.token then loadDashboard(); not bound into markup</li> <li>base/main/default/lwc/publicDonorDashboard/publicDonorDashboard.js:45 and :320-330 - SESSION_TOKEN_SHAPE = /^[0-9a-f]{64}\$/ ; a stored value that fails the test is removed and the sign-in form is shown</li> <li>base/main/default/classes/donor/service/DonorPortalService.cls:809-811 - generateToken() returns EncodingUtil.convertToHex(Crypto.generateAesKey(256)), i.e. 64 lowercase hex characters</li> </ul> |

| Reported vulnerability | Location | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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|                        |          | <ul style="list-style-type: none"> <li>base/main/default/lwc/publicDonorDashboard/publicDonorDashboard.html:389 - token={token} passed as a component property only; the only other {token} use is if:false={token} (line 47), a boolean</li> <li>base/main/default/lwc/pfGivingStatement/pfGivingStatement.js:96-104 and :115-121 - errorMessage = res?.error    literal, or the printStatement() return value</li> <li>base/main/default/lwc/uiStatementPrint/uiStatementPrint.js:249 - printStatement returns only the literal 'Pop-up blocked. Allow pop-ups ...' or null; the statement document itself is built with createElement/textContent (el) helper :31-35), never innerHTML</li> <li>base/main/default/classes/donor/controller/DonorPortalController.cls:126-128, :383-399, :472-478 - getGivingStatement -&gt; TokenAction.doValidate requireNotBlank(token, 'token') -&gt; validateToken -&gt; service</li> <li>base/main/default/classes/shared/GuestAction.cls:36-38 - callerSafeMessage returns AppException message verbatim, otherwise GENERIC_FAILURE constant</li> <li>base/main/default/classes/shared/ControllerAction.cls:45-57 - validation errors joined from addError literals; exceptions routed through callerSafeMessage; full message to AppLogger only</li> <li>base/main/default/classes/donor/service/DonorPortalService.cls:54-55, :696-711 - validateToken throws AppException(INVALID_MSG) constant for every failure</li> <li>base/main/default/classes/donation/service/GivingStatementService.cls:298, :303, :310-312 - the only AppExceptions on the statement path are constant strings</li> <li>grep lwc:dom lwc:inner-html innerHTML in lwc/pfGivingStatement and lwc/publicDonorDashboard - zero hits</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://lwc.dev/guide/html_templates">https://lwc.dev/guide/html_templates</a></li> <li><a href="https://trailhead.salesforce.com/content/learn/modules/secure-clientside-development/mitigate-crosssite-scripting">https://trailhead.salesforce.com/content/learn/modules/secure-clientside-development/mitigate-crosssite-scripting</a></li> <li><a href="https://developer.salesforce.com/docs/platform/lwc/guide/security-lwsec-sanitize.html">https://developer.salesforce.com/docs/platform/lwc/guide/security-lwsec-sanitize.html</a></li> </ul> |

## CRUD Delete — 1 paths, 1 locations

**What this query claims:** *This query looks for CRUD Delete operations that are performed without checking for isDeletable() permisison before performing the delete operation.*

| Reported vulnerability                                     | Location                                                                                                                                 | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>R-018</b><br>CRUD Delete<br><span>False positive</span> | <b>Paths 1</b><br>At scan:<br>classes/RetentionPurgeBatch.cls:90<br>Submitted source:<br>classes/shared/async/RetentionPurgeBatch.cls:90 | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Delete operations that are performed without checking for isDeletable() permisison before performing the delete operation.</i> Flagged statement: delete as system scope;<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim that an isDeletable() check is missing is false because the delete runs in batch Apex, which the scan's own rule exempts, over a scope produced only by the retention selectors (diagnostic, staging and request rows past the administrator-configured window); no caller chooses the rows, and a job owner without Delete on an internal object would otherwise silently leave expired data in place.<br><br><b>SUPPORTING DETAIL</b><br>This is the data-retention purge: the scope is produced only by the per-target selector locators (Log__c and Transaction_Log__c older than the admin-configured retention window, terminal Donation_Staging__c and Guest_Request__c older than it, and terminal or stale portal tokens after 7 days). It runs as a scheduled batch or when an administrator runs it from the setup console; no caller supplies record Ids. No packaged permission set grants Delete on these append-only tables by design, so the purge must delete in system mode.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/shared/async/RetentionPurgeBatch.cls:1-16 - rationale; without sharing</li> <li>base/main/default/classes/shared/async/RetentionPurgeBatch.cls:41-78 - fixed selector locators per Target; else throws</li> <li>base/main/default/classes/setup/service/SettingsService.cls:927-932 - dataRetentionDays default 90 (no lower floor; admin-controlled)</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li>unverified against primary docs (developer.salesforce.com apex_scheduler.htm returned 403 / cookie wall); secondary: <a href="https://help.salesforce.com/s/articleView?id=sf.code_schedule_batch_apex.htm">https://help.salesforce.com/s/articleView?id=sf.code_schedule_batch_apex.htm</a> (search snippet: scheduled Apex executes under the user who scheduled the job)</li> </ul> |

## FLS Create — 109 paths, 22 locations

**What this query claims:** *This query looks for CRUD Delete operations that are performed without checking for isCreatable() permisison before performing the Create operation on the Object and its fields.*

| Reported vulnerability                                    | Location                                                                                                                                                               | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>R-019</b><br>FLS Create<br><span>False positive</span> | <b>Paths 1</b><br>At scan: <code>classes/ReconciliationHeartbeat.cls:59</code><br>Submitted source: <code>classes/donation/async/ReconciliationHeartbeat.cls:59</code> | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD Delete operations that are performed without checking for <code>isCreatable()</code> permission before performing the Create operation on the Object and its fields.</i> Flagged statement: <code>insert as system new Transaction_Log__c(Event_Type__c = 'HEARTBEAT_ALERT', Status__c = 'Rejected',</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the insert is an operations alarm written by a scheduled job with no user input, every field a constant or a count of stuck rows; the job runs as whoever scheduled it, including the install context that holds no permission set, and Salesforce's own guidance accepts system-mode DML in scheduled context.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This is an operations alarm written by a scheduled job with no user input: every field is a constant or a message built from a count of stuck staging rows. The job is scheduled at package install (post-install context, which holds no permission sets) or by an administrator from the setup console, so the insert must not depend on the scheduling user's <code>Transaction_Log__c</code> grants; <code>Transaction_Log__c</code> is an internal audit object not granted to guests. No record Id or value from a caller reaches the insert.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/async/ReconciliationHeartbeat.cls:8-11 - <code>without sharing</code> with rationale (scheduled job; runs as whoever scheduled it)</li> <li>base/main/default/classes/donation/async/ReconciliationHeartbeat.cls:26-62 - <code>execute()</code>: selector count -&gt; constant-valued log insert -&gt; admin email</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:495 - <code>scheduleIfAbsent</code> schedules it at install</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li>unverified against primary docs (developer.salesforce.com apex_scheduler.htm returned 403 / cookie wall); secondary: <a href="https://help.salesforce.com/s/articleView?id=sf.code_schedule_batch_apex.htm">https://help.salesforce.com/s/articleView?id=sf.code_schedule_batch_apex.htm</a> (search snippet: scheduled Apex executes under the user who scheduled the job)</li> </ul> |
| <b>R-015</b><br>FLS Create<br><span>False positive</span> | <b>Paths 2-7</b><br>At scan: <code>classes/AppLogger.cls:125</code><br>Submitted source: <code>classes/shared/AppLogger.cls:125</code>                                 | See <b>R-015</b> under “Apex CRUD ContentDistribution” — same code location, same response.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Reported vulnerability                                    | Location                                                                                                                                                                | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>R-020</b><br>FLS Create<br><span>False positive</span> | <b>Paths 8–12</b><br>At scan:<br><code>classes/StripePaymentSweep.cls:203</code><br>Submitted source:<br><code>classes/donation/async/StripePaymentSweep.cls:203</code> | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields.</i> Flagged statement:<br><code>insert as system new Transaction_Log__c(</code><br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the insert records the outcome of a scheduled repair sweep with constant values, the staging Id being repaired and a caught exception message; it runs in batch context, including when scheduled at install by a user with no permission set, and no caller-supplied value reaches it.<br><br><b>SUPPORTING DETAIL</b><br>logSweep records the outcome of the scheduled Stripe-to-Salesforce repair sweep; its arguments are constants chosen by the sweep, the staging Id it is repairing, and a caught exception message. The sweep is scheduled at install or started by an administrator from the setup console, and the install-context owner holds no permission sets, so the audit insert runs in system mode. No caller-supplied value reaches it.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/async/StripePaymentSweep.cls:23-26 - <code>without sharing</code> with rationale</li> <li>base/main/default/classes/donation/async/StripePaymentSweep.cls:200-210 - private logSweep, all values from sweep internals</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li>unverified against primary docs (developer.salesforce.com apex_scheduler.htm returned 403 / cookie wall); secondary: <a href="https://help.salesforce.com/s/articleView?id=sf.code_schedule_batch_apex.htm">https://help.salesforce.com/s/articleView?id=sf.code_schedule_batch_apex.htm</a> (search snippet: scheduled Apex executes under the user who scheduled the job)</li> </ul> |

| Reported vulnerability                       | Location                                                                                                                                                                                                            | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>R-021</b><br>FLS Create<br>False positive | <b>Paths 13–14, 24, 41, 48–49</b><br>At scan:<br>classes/StagingReconciliationBatch.cls:281<br>Submitted source:<br>classes/donation/async/StagingReconciliationBatch.cls:281                                       | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields.</i> Flagged statement:<br>insert as system log;<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the insert writes the job's own audit row with every field system-derived (lds created in the same transaction, a constant event type, an exception message, a timestamp), in a scheduled batch context the scan's own rule exempts and that may run as a user with no permission set.<br><br><b>SUPPORTING DETAIL</b><br>This inserts the reconciliation audit row (Transaction_Log__c) for each staging the job processes; every field is system-derived (staging Id, Opportunity Id created by the same transaction, a constant event type/status, an exception message, a timestamp). It runs in the scheduled reconciler and the internal retry action; the scheduled run can be owned by the install context user, which holds no permission sets, so the audit write must not depend on the running user's grants. Transaction_Log__c is internal and not granted to the guest profile.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/async/StagingReconciliationBatch.cls:186-250 - log fields populated from staging Id, oppld, constants, e.getMessage()</li> <li>base/main/default/classes/donation/async/StagingReconciliationBatch.cls:281 - single insert per staging</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li>unverified against primary docs (developer.salesforce.com apex_scheduler.htm returned 403 / cookie wall); secondary: <a href="https://help.salesforce.com/s/articleView?id=sf.code_schedule_batch_apex.htm">https://help.salesforce.com/s/articleView?id=sf.code_schedule_batch_apex.htm</a> (search snippet: scheduled Apex executes under the user who scheduled the job)</li> </ul> |
| <b>R-016</b><br>FLS Create<br>False positive | <b>Paths 15, 28, 50–54, 56–61, 63–65, 67–68, 74–75, 77–78, 81, 88, 90, 92, 94, 96, 98, 100</b><br>At scan:<br>classes/SettingsService.cls:345<br>Submitted source:<br>classes/setup/service/SettingsService.cls:345 | See <b>R-016</b> under “Apex CRUD ContentDistribution” — same code location, same response.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Reported vulnerability                                    | Location                                                                                                                                                         | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>R-022</b><br>FLS Create<br><span>False positive</span> | <b>Paths 16–17, 29</b><br>At scan:<br>classes/RefundReconciliationBatch.cls:344<br>Submitted source:<br>classes/donation/async/RefundReconciliationBatch.cls:344 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields. Flagged statement:</i></p> <pre>upsert as system s;</pre> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim that this write bypasses a required createable/updateable check is false: the row is the package's own job cursor (two timestamps computed from the payment gateway's responses), the statement runs only inside a scheduled batch with no interactive user, and Salesforce's own guidance accepts system-mode DML in batch context as a false positive. A user-mode write would make the job's correctness depend on which administrator happened to schedule it.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This write stores two Datetime high-water marks (refund and dispute cursors) computed by the batch from the payment gateway's list responses; no user-supplied value reaches it. It runs only in the scheduled reconciliation job or when an administrator starts that job from the settings console, and the cursor must advance regardless of the scheduling user's field permissions or the next run re-reads the whole window. Enforcing FLS here would not protect any user data; it would only let a missing grant silently stall reconciliation.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/async/RefundReconciliationBatch.cls:334-344 finish(): only when !hadError; sets Refund_Reconcile_Cursor__c / Dispute_Reconcile_Cursor__c from cursorTarget/disputeCursorTarget (system-computed), then <code>upsert as system s</code></li> <li>base/main/default/classes/donation/async/RefundReconciliationBatch.cls:170-171 reads the same cursors with a 2h overlap</li> <li>Scheduled Apex executes in the context of the user who scheduled it (CronTrigger owner), in system mode - not the Automated Process user</li> <li>Fundraising_Admin holds edit on both cursor fields; the only interactive trigger (runJobNow) is Fundraising_Admin-only</li> <li>base/main/default/classes/donation/async/RefundReconciliationBatch.cls:155 and :336 use Settings__c.getOrgDefaults() (the org-default row only)</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://help.salesforce.com/s/articleView?id=platform.code_schedule_batch_apex.htm&amp;language=en_US&amp;type=5">https://help.salesforce.com/s/articleView?id=platform.code_schedule_batch_apex.htm&amp;language=en_US&amp;type=5</a></li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                      | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>R-023</b><br>FLS Create<br><span>False positive</span> | <b>Paths 18–19, 43</b><br>At scan:<br>classes/SettingsService.cls:1026<br>Submitted source:<br>classes/setup/service/SettingsService.cls:1026 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields. Flagged statement:</i></p> <pre>upsert as system s;</pre> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the write is to the org-default row of the package's own custom setting, reached only through an administrator console gated by the Manage_Fundraising_Settings custom permission, with both the field name and the value validated against fixed allow-lists; and a user-mode upsert on a hierarchy custom setting requires Customize Application, which the package's own admin permission set cannot grant (measured: <code>upsert as user</code> throws <code>SecurityException</code> for a <code>Fundraising_Admin</code> user).</p> <p><b>SUPPORTING DETAIL</b></p> <p>This writes one of two scheduling-interval fields on the org-default <code>Settings__c</code> row; the family is checked against two fixed names and the interval against a fixed allow-list before the write. It is reached only from the settings console controller's <code>ScheduleJobFamily</code> action, which is granted only in the <code>Fundraising_Admin</code> permission set. System mode is retained because DML on a Hierarchy Custom Setting in user mode requires Customize Application, which a packaged permission set cannot grant. <code>ScheduleJobFamily.doValidate()</code> first checks the packaged custom permission <code>Manage_Fundraising_Settings</code> (granted only in <code>Fundraising_Admin</code>) and refuses the save when the running user does not hold it; the site carries a <code>SYSTEM_MODE</code> comment naming that gate. The user-mode alternative was tested and does not work for the people who use this screen. Measured 2026-10-01 in a scratch org, running as a Standard User who holds the <code>Fundraising_Admin</code> permission set:</p> <p><code>Schema.sObjectType.Settings__c</code> reports <code>isCreateable</code>, <code>isUpdateable</code> and <code>isAccessible</code> all false, and <code>upsert as user</code> on the settings row throws <code>System.SecurityException: Access to entity 'Settings__c' denied</code>. The platform ties custom-setting DML to Customize Application, which is a system permission the fundraising administrator is not expected to hold, so a user-mode write would lock every non-system-administrator out of the settings console. The custom-permission check before the write is the access control; it is enforced in Apex on every action that reaches this line.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/service/SettingsService.cls:1022-1026 - <code>// SYSTEM_MODE:</code> reason naming the <code>ScheduleJobFamily</code> gate, then <code>upsert as system s</code></li> <li>base/main/default/classes/setup/controller/SetupController.cls:2192-2201 - <code>ScheduleJobFamily</code> extends <code>SettingsWriteAction</code>; <code>doValidate()</code> calls <code>requireSettingsPermission()</code> before the family / interval checks</li> <li><code>SetupController_Test.test_scheduleJobFamily_userWithoutSettingsPermission_isRefused</code></li> <li><code>Fundraising_Admin</code> is the only permission set with <code>SetupController</code> classAccess and the only one carrying <code>Manage_Fundraising_Settings</code></li> </ul> |

| Reported vulnerability | Location | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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|                        |          | <ul style="list-style-type: none"><li>docs/security/appexchange-security-review-readiness.md F-13</li></ul> <b>SOURCES</b> <ul style="list-style-type: none"><li><a href="https://help.salesforce.com/s/articleView?id=sf.cs_schema_settings.htm">https://help.salesforce.com/s/articleView?id=sf.cs_schema_settings.htm</a></li><li><a href="https://github.com/patronmanager/apex-dml-manager/issues/2">https://github.com/patronmanager/apex-dml-manager/issues/2</a></li><li><a href="https://help.salesforce.com/s/articleView?language=en_US&amp;id=release-notes.rn_apex_default_user_mode.htm&amp;release=262&amp;type=5">https://help.salesforce.com/s/articleView?language=en_US&amp;id=release-notes.rn_apex_default_user_mode.htm&amp;release=262&amp;type=5</a></li></ul> |

| Reported vulnerability                                    | Location                                                                                                                                            | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>R-024</b><br>FLS Create<br><span>False positive</span> | <b>Paths 20, 44</b><br>At scan:<br>classes/PostInstallHandler.cls:1515<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:1515 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields.</i> Flagged statement: <code>upsert as system s;</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the write targets the org-default row of the package's own hierarchy custom setting, runs only inside the InstallHandler as the package's system user (no permission set), and writes an Id the same install step just produced; custom-setting DML in user mode would additionally require Customize Application, which no install user holds.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This stores the Id of the event pricebook that the same install step just found or created on the org-default custom setting row, only when that field is blank. It runs only from the InstallHandler as the package's system user, which has no permission set to enforce, and never overwrites an administrator's value. No user input reaches the write.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:18 <code>public without sharing class PostInstallHandler implements InstallHandler;</code> lines 1-4 header: 'runs in package install/upgrade context'</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:169-173 doc: 'the script runs as a per-package system user that does not exist before runtime'</li> <li>Salesforce: a post-install script 'runs as a special system user that represents your package' (search-result excerpt of the help page; page body itself not fetchable, so the quote is from the indexed snippet)</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:183-186 onInstall calls each seed* step in its own try/catch; only logStepFailure on error</li> <li>grep of non-test classes: seedDefaultSettings, seedDefaultImages, seedDefaultPageDesign, seedEventPricebook, seedEventDonationProduct, seedEventFeeProduct, seedContentTemplates have NO callers outside PostInstallHandler (mentions in PricebookService/SettingsService are comments/strings only)</li> <li>Class is <code>public</code>, not <code>global</code> (changed 2026-09-05), so subscriber anonymous Apex / other namespaces cannot invoke these methods</li> <li>Only backfills when the field is blank ('only backfills the Settings lookup when blank, so an admin's own configuration is never overwritten on upgrade' - method Javadoc)</li> <li>Value is the Id of a record this same method just found or inserted; no user input</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5</a></li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                            | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>R-025</b><br>FLS Create<br><span>False positive</span> | <b>Paths 21, 45</b><br>At scan:<br>classes/PostInstallHandler.cls:1557<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:1557 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields. Flagged statement:</i></p> <pre>upsert as system s;</pre> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the write is to the org-default row of the package's own custom setting, from the InstallHandler's system user with no permission set, storing an Id the install step itself created; there is no running user whose field permissions could meaningfully be checked.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This records the Id of the event donation product, found or created by the same install step, on the org-default custom setting row when that field is blank. It runs only from the InstallHandler as the package's system user and does not overwrite an administrator's value. No user input reaches the write.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:18 public without sharing class PostInstallHandler implements InstallHandler; lines 1-4 header: 'runs in package install/upgrade context'</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:169-173 doc: 'the script runs as a per-package system user that does not exist before runtime'</li> <li>Salesforce: a post-install script 'runs as a special system user that represents your package' (search-result excerpt of the help page; page body itself not fetchable, so the quote is from the indexed snippet)</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:183-186 onInstall calls each seed* step in its own try/catch; only logStepFailure on error</li> <li>grep of non-test classes: seedDefaultSettings, seedDefaultImages, seedDefaultPageDesign, seedEventPricebook, seedEventDonationProduct, seedEventFeeProduct, seedContentTemplates have NO callers outside PostInstallHandler (mentions in PricebookService/SettingsService are comments/strings only)</li> <li>Class is public, not global (changed 2026-09-05), so subscriber anonymous Apex / other namespaces cannot invoke these methods</li> <li>Only backfills when the field is blank ('only backfills the Settings lookup when blank, so an admin's own configuration is never overwritten on upgrade' - method Javadoc)</li> <li>Value is the Id of a record this same method just found or inserted; no user input</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5</a></li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                            | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>R-026</b><br>FLS Create<br><span>False positive</span> | <b>Paths 22, 46</b><br>At scan:<br>classes/PostInstallHandler.cls:1601<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:1601 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields.</i> Flagged statement:<br/>           upsert as system s;</p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the write is to the org-default row of the package's own custom setting, from the InstallHandler's system user with no permission set, storing an Id the install step itself created; the administrator's existing value is never overwritten.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This records the Id of the event processing-fee product, found or created by the same install step, on the org-default custom setting row when that field is blank. It runs only from the InstallHandler as the package's system user and never overwrites an administrator's value. No user-supplied data reaches it.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:18 public without sharing class PostInstallHandler implements InstallHandler; lines 1-4 header: 'runs in package install/upgrade context'</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:169-173 doc: 'the script runs as a per-package system user that does not exist before runtime'</li> <li>Salesforce: a post-install script 'runs as a special system user that represents your package' (search-result excerpt of the help page; page body itself not fetchable, so the quote is from the indexed snippet)</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:183-186 onInstall calls each seed* step in its own try/catch; only logStepFailure on error</li> <li>grep of non-test classes: seedDefaultSettings, seedDefaultImages, seedDefaultPageDesign, seedEventPricebook, seedEventDonationProduct, seedEventFeeProduct, seedContentTemplates have NO callers outside PostInstallHandler (mentions in PricebookService/SettingsService are comments/strings only)</li> <li>Class is public, not global (changed 2026-09-05), so subscriber anonymous Apex / other namespaces cannot invoke these methods</li> <li>Only backfills when the field is blank ('only backfills the Settings lookup when blank, so an admin's own configuration is never overwritten on upgrade' - method Javadoc)</li> <li>Value is the Id of a record this same method just found or inserted; no user input</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5</a></li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                             | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>R-027</b><br>FLS Create<br><span>False positive</span> | <b>Paths 23, 25–27</b><br>At scan:<br>classes/PostInstallHandler.cls:260<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:260 | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields.</i> Flagged statement:<br>upsert as system s;<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the write seeds the org-default row of the package's own hierarchy custom setting from compiled constants, runs only in the InstallHandler as the package's system user (no permission set), and fills only blank fields; custom-setting DML in user mode requires Customize Application, which the install user does not hold, so a user-mode write would fail the install.<br><br><b>SUPPORTING DETAIL</b><br>This seeds the org-default row of the package's hierarchy custom setting with constant defaults during package install or upgrade, filling only blank fields. It runs only from the InstallHandler, which executes as the package's special system user; that user holds no permission set, so there is no user FLS to enforce. The method is not reachable from any controller, guest page or subscriber Apex.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:18 public without sharing class PostInstallHandler implements InstallHandler; lines 1-4 header: 'runs in package install/upgrade context'</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:169-173 doc: 'the script runs as a per-package system user that does not exist before runtime'</li> <li>Salesforce: a post-install script 'runs as a special system user that represents your package' (search-result excerpt of the help page; page body itself not fetchable, so the quote is from the indexed snippet)</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:183-186 onInstall calls each seed* step in its own try/catch; only logStepFailure on error</li> <li>grep of non-test classes: seedDefaultSettings, seedDefaultImages, seedDefaultPageDesign, seedEventPricebook, seedEventDonationProduct, seedEventFeeProduct, seedContentTemplates have NO callers outside PostInstallHandler (mentions in PricebookService/SettingsService are comments/strings only)</li> <li>Class is public, not global (changed 2026-09-05), so subscriber anonymous Apex / other namespaces cannot invoke these methods</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:206-260 seedDefaultSettings: creates the org-default Settings__c only if absent (SetupOwnerId = org Id) and backfills blank colour/text defaults from constants; adoptTributeMasterSwitch reads Campaign WITH_SYSTEM_MODE</li> <li>All values are package constants or derived from existing org data; no user input</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5</a></li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>R-028</b><br>FLS Create<br><span>False positive</span> | <b>Paths 30–32</b><br>At scan:<br>classes/SettingsService.cls:563<br>Submitted source:<br>classes/setup/service/SettingsService.cls:563 | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields.</i> Flagged statement:<br>Database.insert(cv, AccessLevel.SYSTEM_MODE);<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the insert runs during install as the package's system user, which holds no permission set, and copies the package's own static-resource image into ContentVersion with constant Title and PathOnClient; ContentVersion is a standard object a 2GP permission set cannot grant, and no subscriber data is involved.<br><br><b>SUPPORTING DETAIL</b><br>PackagedAssetSeeder copies the package's own static-resource images (default logo, hero image, social-share image) into ContentVersion so the public pages have a default; Title/PathOnClient are constants and VersionData is the packaged resource body. It runs in the post-install context, whose user holds no permission sets, and ContentVersion is a standard object a second-generation package permission set cannot grant. The interactive admin upload of a custom logo uses user mode instead.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/service/SettingsService.cls:543-594 - PackagedAssetSeeder, inner <code>without sharing</code> with rationale; content from DefaultOrgLogo/DefaultHeroImage/DefaultSocialShareImage</li> <li>base/main/default/classes/setup/service/SettingsService.cls createLogoDistribution - USER_MODE for the admin upload path</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm</a> (post-install script runs as a package-specific system user)</li> <li>AGENTS.md Change Management: 2GP strips standard-object grants (project finding 2026-08-16)</li> </ul> |

| Reported vulnerability                              | Location                                                                                                                                | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>R-029</b><br>FLS Create<br><b>False positive</b> | <b>Paths 33–37</b><br>At scan:<br>classes/SettingsService.cls:581<br>Submitted source:<br>classes/setup/service/SettingsService.cls:581 | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields.</i> Flagged statement:<br>Database.insert(cd, AccessLevel.SYSTEM_MODE);<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the insert runs during install as the package's system user, which holds no permission set, and publishes a public link for the file the previous statement created from a packaged static resource; ContentDistribution is a standard object a 2GP permission set cannot grant, and the public link is the feature (default images on unauthenticated donation pages and receipts), not a leak.<br><br><b>SUPPORTING DETAIL</b><br>This publishes a public link for the package's own default image, which must be publicly reachable because it is rendered on unauthenticated donation pages and in receipt emails. The file is created one line earlier from a packaged static resource, so no subscriber or user file can be published through this path. It runs in the post-install context (no permission sets) or from an administrator's reset-to-default action.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"><li>base/main/default/classes/setup/service/SettingsService.cls:570-581 - ContentDistribution on the ContentVersion just inserted from a static resource</li></ul><br><b>SOURCES</b> <ul style="list-style-type: none"><li><a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm</a> (post-install script runs as a package-specific system user)</li><li>AGENTS.md Change Management: 2GP strips standard-object grants (project finding 2026-08-16)</li></ul> |

| Reported vulnerability                                    | Location                                                                                                                                        | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>R-030</b><br>FLS Create<br><span>False positive</span> | <b>Paths 38</b><br>At scan:<br>classes/PostInstallHandler.cls:1503<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:1503 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields.</i> Flagged statement:<br/> <code>insert as system pb;</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the insert runs only inside the package InstallHandler as the package's system user, which holds no permission set, and Pricebook2 is a standard object a 2GP permission set cannot grant; every value is a constant, so no user input reaches the write.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This creates the package's event pricebook once, at install, with a constant name, only if it does not already exist. Pricebook2 is a standard object that a packaged permission set cannot grant, and the method runs only from the InstallHandler as the package's system user. No controller, guest page or subscriber code can call it.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:18 <code>public without sharing class PostInstallHandler implements InstallHandler</code>; lines 1-4 header: 'runs in package install/upgrade context'</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:169-173 doc: 'the script runs as a per-package system user that does not exist before runtime'</li> <li>Salesforce: a post-install script 'runs as a special system user that represents your package' (search-result excerpt of the help page; page body itself not fetchable, so the quote is from the indexed snippet)</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:183-186 onInstall calls each seed* step in its own try/catch; only logStepFailure on error</li> <li>grep of non-test classes: seedDefaultSettings, seedDefaultImages, seedDefaultPageDesign, seedEventPricebook, seedEventDonationProduct, seedEventFeeProduct, seedContentTemplates have NO callers outside PostInstallHandler (mentions in PricebookService/SettingsService are comments/strings only)</li> <li>Class is <code>public</code>, not <code>global</code> (changed 2026-09-05), so subscriber anonymous Apex / other namespaces cannot invoke these methods</li> <li>Pricebook2 and Product2 are standard objects; AGENTS.md / readiness FP-1 item 2 record that a 2GP permission set cannot ship a standard-object grant (build strips it silently), and the install user holds no permission set anyway</li> <li>base/main/default/classes/event/service/PricebookService.cls:11-24 doc: install-testing (2026-08-09) showed a <code>with sharing</code> path could not even resolve Pricebook2 as the per-package system user</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:1501 <code>new Pricebook2(Name = EVENT_PRICEBOOK_NAME, IsActive = true)</code> - constant values; inserted only when neither the current nor the legacy pricebook exists</li> </ul> <p><b>SOURCES</b></p> |

| Reported vulnerability | Location | Response                                                                                                                                                                                                                                                                              |
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|                        |          | <ul style="list-style-type: none"><li>• <a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5</a></li></ul> |

| Reported vulnerability                                    | Location                                                                                                                                        | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>R-031</b><br>FLS Create<br><span>False positive</span> | <b>Paths 39</b><br>At scan:<br>classes/PostInstallHandler.cls:1537<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:1537 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields.</i> Flagged statement:<br/> <code>insert as system p;</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the insert runs only inside the package InstallHandler as the package's system user, which holds no permission set, and Product2 is a standard object a 2GP permission set cannot grant; every field value is a compiled constant.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This creates the package's generic event donation product once, at install, with constant field values and only if it does not already exist. Product2 is a standard object that a packaged permission set cannot grant, and the code runs only from the InstallHandler as the package's system user. It is not reachable from any user-facing entry point.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:18 <code>public without sharing class PostInstallHandler implements InstallHandler</code>; lines 1-4 header: 'runs in package install/upgrade context'</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:169-173 doc: 'the script runs as a per-package system user that does not exist before runtime'</li> <li>Salesforce: a post-install script 'runs as a special system user that represents your package' (search-result excerpt of the help page; page body itself not fetchable, so the quote is from the indexed snippet)</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:183-186 onInstall calls each seed* step in its own try/catch; only logStepFailure on error</li> <li>grep of non-test classes: seedDefaultSettings, seedDefaultImages, seedDefaultPageDesign, seedEventPricebook, seedEventDonationProduct, seedEventFeeProduct, seedContentTemplates have NO callers outside PostInstallHandler (mentions in PricebookService/SettingsService are comments/strings only)</li> <li>Class is <code>public</code>, not <code>global</code> (changed 2026-09-05), so subscriber anonymous Apex / other namespaces cannot invoke these methods</li> <li>Pricebook2 and Product2 are standard objects; AGENTS.md / readiness FP-1 item 2 record that a 2GP permission set cannot ship a standard-object grant (build strips it silently), and the install user holds no permission set anyway</li> <li>base/main/default/classes/event/service/PricebookService.cls:11-24 doc: install-testing (2026-08-09) showed a <code>with sharing</code> path could not even resolve Pricebook2 as the per-package system user</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:1534 <code>new Product2(Name = EVENT_DONATION_PRODUCT_NAME, IsActive = true)</code> - constants; only when no Product2 with that name exists (query WITH SYSTEM_MODE)</li> </ul> <p><b>SOURCES</b></p> |

| Reported vulnerability | Location | Response                                                                                                                                                                                                                                                                              |
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|                        |          | <ul style="list-style-type: none"><li>• <a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5</a></li></ul> |

| Reported vulnerability                                    | Location                                                                                                                                        | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>R-032</b><br>FLS Create<br><span>False positive</span> | <b>Paths 40</b><br>At scan:<br>classes/PostInstallHandler.cls:1581<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:1581 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields.</i> Flagged statement:<br/> <code>insert as system p;</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the insert runs only inside the package InstallHandler as the package's system user, which holds no permission set, and Product2 is a standard object a 2GP permission set cannot grant; the values are constants.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This creates the package's event processing-free product once, at install, with constant values and only if it does not already exist. Product2 is a standard object a packaged permission set cannot grant, and the method runs only from the InstallHandler as the package's system user. It has no interactive or guest caller.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:18 <code>public without sharing class PostInstallHandler implements InstallHandler</code>; lines 1-4 header: 'runs in package install/upgrade context'</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:169-173 doc: 'the script runs as a per-package system user that does not exist before runtime'</li> <li>Salesforce: a post-install script 'runs as a special system user that represents your package' (search-result excerpt of the help page; page body itself not fetchable, so the quote is from the indexed snippet)</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:183-186 onInstall calls each seed* step in its own try/catch; only logStepFailure on error</li> <li>grep of non-test classes: seedDefaultSettings, seedDefaultImages, seedDefaultPageDesign, seedEventPricebook, seedEventDonationProduct, seedEventFeeProduct, seedContentTemplates have NO callers outside PostInstallHandler (mentions in PricebookService/SettingsService are comments/strings only)</li> <li>Class is <code>public</code>, not <code>global</code> (changed 2026-09-05), so subscriber anonymous Apex / other namespaces cannot invoke these methods</li> <li>Pricebook2 and Product2 are standard objects; AGENTS.md / readiness FP-1 item 2 record that a 2GP permission set cannot ship a standard-object grant (build strips it silently), and the install user holds no permission set anyway</li> <li>base/main/default/classes/event/service/PricebookService.cls:11-24 doc: install-testing (2026-08-09) showed a <code>with sharing</code> path could not even resolve Pricebook2 as the per-package system user</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:1578 <code>new Product2(Name = EVENT_FEE_PRODUCT_NAME, IsActive = true)</code> - constants; only when absent</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5</a></li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                       | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>R-033</b><br>FLS Create<br><span>False positive</span> | <b>Paths 42</b><br>At scan:<br>classes/EventPurchaseService.cls:590<br>Submitted source:<br>classes/event/service/EventPurchaseService.cls:590 | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permisison before performing the Create operation on the Object and its fields.</i> Flagged statement:<br>insert as system toAdd; // bulk – never per-attendee DML<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the insert runs in the reconciliation job that finalizes a paid ticket order, a context the scan's own rule exempts, and CampaignMember is a standard object a 2GP permission set cannot grant; the Contact and Campaign Ids come from the finalized order, not from a caller.<br><br><b>SUPPORTING DETAIL</b><br>After a paid ticket order is finalized by the reconciliation job, attendees with a Contact are added to the event Campaign as CampaignMembers. CampaignMember is a standard object that a second-generation package permission set cannot grant, and the finalizer runs in scheduled context, so the insert runs in system mode. The Campaign comes from the server-side staging record and the Contacts from the attendees created in the same finalization; existing members are skipped, and no browser input chooses either.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/event/service/EventPurchaseService.cls:570-590 - campaign from staging, contacts from attendees, existing members excluded</li> <li>Comment cites 'CampaignMemberSyncService.syncForOpportunities', which does not exist (method is syncForDonations); 'after a guest order is committed' - it actually runs in the batch/retry finalizer</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_batch_interface.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_batch_interface.htm</a> (batch Apex runs asynchronously; Salesforce false-positive guidance exempts batch and async context)</li> <li>AGENTS.md Change Management: 2GP strips standard-object grants (project finding 2026-08-16)</li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                                  | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>R-034</b><br>FLS Create<br><span>False positive</span> | <b>Paths 47</b><br>At scan:<br>classes/CampaignMemberSyncService.cls:80<br>Submitted source:<br>classes/donation/service/CampaignMemberSyncService.cls:80 | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields.</i> Flagged statement:<br><pre>List&lt;Database.SaveResult&gt; results = Database.insert(toAdd, false, AccessLevel.SYSTEM_MODE);</pre> <b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the insert runs inside the platform-event subscriber trigger as the Automated Process entity, which can hold no permission set, and CampaignMember is a standard object a 2GP permission set cannot grant; the Contact and Campaign Ids come from the finalized donation itself, not from a caller. <b>SUPPORTING DETAIL</b><br>This adds the donor to the donation's campaign as a CampaignMember after a gift is finalized; it runs in the platform-event subscriber trigger, which executes as the Automated Process entity. CampaignMember is a standard object that a second-generation package permission set cannot grant, and the Automated Process context has no user permissions to enforce. CampaignId and ContactId are read from the finalized Opportunity itself, and existing memberships are skipped, so the insert cannot target arbitrary campaigns or contacts. <b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/service/CampaignMemberSyncService.cls:68 - CampaignMember built from Opportunity CampaignId + resolved contact</li> <li>base/main/default/classes/donation/service/CampaignMemberSyncService.cls:80 - comment present and accurate</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.platform_events.meta/platform_events/platform_events_subscribe_apex.htm">https://developer.salesforce.com/docs/atlas.en-us.platform_events.meta/platform_events/platform_events_subscribe_apex.htm</a></li> <li>AGENTS.md Change Management: 2GP strips standard-object grants (project finding 2026-08-16)</li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                                   | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>R-035</b><br>FLS Create<br><span>False positive</span> | <b>Paths 55, 80, 83–87</b><br>At scan:<br>classes/PostInstallHandler.cls:1645<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:1645 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields.</i> Flagged statement: insert as system toInsert;</p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the insert runs only inside the package InstallHandler as the package's system user, which holds no permission set, and the rows are starter templates whose text is literal in the package source; no user supplies any value and no user's permissions exist to check.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This inserts a starter set of question and FAQ templates, whose text is literal in the package source, the first time the package is installed and only when the template table is empty. It runs only from the InstallHandler as the package's system user, which holds no permission set. Administrators edit or delete these templates afterwards through the normal, permission-checked console.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:18 public without sharing class PostInstallHandler implements InstallHandler; lines 1-4 header: 'runs in package install/upgrade context'</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:169-173 doc: 'the script runs as a per-package system user that does not exist before runtime'</li> <li>Salesforce: a post-install script 'runs as a special system user that represents your package' (search-result excerpt of the help page; page body itself not fetchable, so the quote is from the indexed snippet)</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:183-186 onInstall calls each seed* step in its own try/catch; only logStepFailure on error</li> <li>grep of non-test classes: seedDefaultSettings, seedDefaultImages, seedDefaultPageDesign, seedEventPricebook, seedEventDonationProduct, seedEventFeeProduct, seedContentTemplates have NO callers outside PostInstallHandler (mentions in PricebookService/SettingsService are comments/strings only)</li> <li>Class is public, not global (changed 2026-09-05), so subscriber anonymous Apex / other namespaces cannot invoke these methods</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:1615-1672 seedContentTemplates inserts starter question/FAQ Content_Template__c rows (RecordTypeId, Name, Prompt__c, Question_Type__c, Is_Active__c, Picklist_Options__c, Answer__c) only when the table is empty; all text is literal in source</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5</a></li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                                                                           | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>R-036</b><br>FLS Create<br><span>False positive</span> | <b>Paths 62, 66, 73, 76, 79, 82, 89, 91, 93, 95, 97, 99, 101–109</b><br>At scan:<br>classes/PostInstallHandler.cls:758<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:758 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permission before performing the Create operation on the Object and its fields.</i> Flagged statement: <code>upsert as system toUpsert;</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the upsert creates or refreshes the six packaged theme records from constants compiled into the package, runs only in the InstallHandler as the package's system user with no permission set, and cannot be invoked by any user at runtime.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This creates or refreshes the six packaged page-design theme records from constants compiled into the package, during install or upgrade. It runs only from the InstallHandler as the package's system user, which holds no permission set, and every assigned value is a literal from the package source. Packaged themes are read-only to administrators, who clone them to customise, so no administrator data is overwritten.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:18 <code>public without sharing class PostInstallHandler implements InstallHandler</code>; lines 1-4 header: 'runs in package install/upgrade context'</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:169-173 doc: 'the script runs as a per-package system user that does not exist before runtime'</li> <li>Salesforce: a post-install script 'runs as a special system user that represents your package' (search-result excerpt of the help page; page body itself not fetchable, so the quote is from the indexed snippet)</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:183-186 <code>onInstall</code> calls each <code>seed*</code> step in its own <code>try/catch</code>; only <code>logStepFailure</code> on error</li> <li>grep of non-test classes: <code>seedDefaultSettings</code>, <code>seedDefaultImages</code>, <code>seedDefaultPageDesign</code>, <code>seedEventPricebook</code>, <code>seedEventDonationProduct</code>, <code>seedEventFeeProduct</code>, <code>seedContentTemplates</code> have NO callers outside <code>PostInstallHandler</code> (mentions in <code>PricebookService/SettingsService</code> are comments/strings only)</li> <li>Class is <code>public</code>, not <code>global</code> (changed 2026-09-05), so subscriber anonymous Apex / other namespaces cannot invoke these methods</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:687-758 <code>seedDefaultPageDesign</code> upserts the six packaged <code>Campaign_Design__c</code> theme rows; <code>buildTheme</code> (:1196+) assigns colours, fonts, radius, widths from compile-time theme constants; reads <code>WITH SYSTEM_MODE</code>; followed by <code>retireDeletedDesigns</code></li> <li>Packaged themes are read-only in the console (clone to edit), so overwriting their style fields on upgrade is the intended contract</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5</a></li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                   | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>R-037</b><br>FLS Create<br><span>False positive</span> | <b>Paths 69, 71</b><br>At scan:<br>classes/PricebookService.cls:125<br>Submitted source:<br>classes/event/service/PricebookService.cls:125 | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permisison before performing the Create operation on the Object and its fields.</i> Flagged statement: <code>if (!newStd.isEmpty()) { insert as system newStd; } // MUST precede the custom-pricebook insert below</code><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because PricebookEntry is a standard object a 2GP permission set cannot grant, and the insert runs from install (package system user, no permission set), admin ticket-tier saves and guest purchases, where the Guest licence cannot hold Create; every value is fixed by Apex from a Product2 the package itself created.<br><b>SUPPORTING DETAIL</b><br>This creates the standard pricebook PricebookEntry a ticket Product2 needs before an OpportunityLineItem can reference it. Values are fixed by Apex (the Product2 the package created for the ticket tier, the pricebook Id, a placeholder unit price, IsActive=true, the org currency) - the real price lives on the line item. PricebookEntry/Pricebook2 are standard objects the package permission sets cannot grant, and the same method runs in the post-install context and the scheduled finalizer; the staff entry point first verifies the user can edit the event Campaign and that the tier belongs to it.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/event/service/PricebookService.cls - class without sharing with doc rationale (Pricebook2 unresolvable under with sharing in post-install context)</li> <li>base/main/default/classes/event/service/PricebookService.cls:142-143 - PricebookEntry(Product2Id, Pricebook2Id, UnitPrice=PLACEHOLDER_PRICE, IsActive=true)</li> <li>base/main/default/classes/event/controller/EventTicketAdminController.cls:156-176, 197-263 - requireCampaignWritable + requireTiersBelongToCampaign before ensureEntriesFor; wrapped non-fatal</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li>AGENTS.md Change Management: 2GP strips standard-object grants (project finding 2026-08-16)</li> </ul> |

| Reported vulnerability                       | Location                                                                                                                                   | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>R-038</b><br>FLS Create<br>False positive | <b>Paths 70, 72</b><br>At scan:<br>classes/PricebookService.cls:135<br>Submitted source:<br>classes/event/service/PricebookService.cls:135 | <b>SCANNER'S CLAIM</b><br>This query looks for CRUD Delete operations that are performed without checking for isCreatable() permisison before performing the Create operation on the Object and its fields. Flagged statement:<br>insert as system newEvt;<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because PricebookEntry is a standard object a 2GP permission set cannot grant, and the insert runs from install (package system user, no permission set), admin ticket-tier saves and guest purchases, where the Guest licence cannot hold Create; every value is fixed by Apex from a Product2 the package itself created.<br><br><b>SUPPORTING DETAIL</b><br>This creates the package event pricebook PricebookEntry a ticket Product2 needs before an OpportunityLineItem can reference it. Values are fixed by Apex (the Product2 the package created for the ticket tier, the pricebook Id, a placeholder unit price, IsActive=true, the org currency) - the real price lives on the line item. PricebookEntry/Pricebook2 are standard objects the package permission sets cannot grant, and the same method runs in the post-install context and the scheduled finalizer; the staff entry point first verifies the user can edit the event Campaign and that the tier belongs to it.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/event/service/PricebookService.cls - class without sharing with doc rationale (Pricebook2 unresolvable under with sharing in post-install context)</li> <li>base/main/default/classes/event/service/PricebookService.cls:142-143 - PricebookEntry(Product2Id, Pricebook2Id, UnitPrice=PLACEHOLDER_PRICE, IsActive=true)</li> <li>base/main/default/classes/event/controller/EventTicketAdminController.cls:156-176, 197-263 - requireCampaignWritable + requireTiersBelongToCampaign before ensureEntriesFor; wrapped non-fatal</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li>AGENTS.md Change Management: 2GP strips standard-object grants (project finding 2026-08-16)</li> </ul> |

## FLS Update — 69 paths, 14 locations

**What this query claims:** This query looks for CRUD Update operations that are performed without checking for isUpdateable() permisison before performing the Update operation on the Object and its fields.

| Reported vulnerability                                    | Location                                                                                                                                                                                                       | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>R-039</b><br>FLS Update<br><span>False positive</span> | <b>Paths 1</b><br>At scan:<br>classes/StagingReconciliationBatch.cls:275<br>Submitted source:<br>classes/donation/async/StagingReconciliationBatch.cls:275                                                     | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Update operations that are performed without checking for isUpdateable() permission before performing the Update operation on the Object and its fields.</i> Flagged statement:<br>update as system upd;<br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the update runs in a scheduled batch (and the equivalent staff 'Retry now' path that re-reads the gateway), writing only the caught exception message and the gateway's own payment status onto a guest-created staging row that no running user is granted Edit on; the values never come from the caller.<br><b>SUPPORTING DETAIL</b><br>The reconciler writes the finalization error and, for an unpaid staging past its TTL, sets Status__c='Expired'; the values come from the caught exception and Stripe's PaymentIntent status, never from the caller. It runs in a scheduled batch and in an internal 'retry now' action; Fundraising_User can view and retry stagings but deliberately has no Edit on Donation_Staging__c (the row is system-owned state), so the status transition must run in system mode. The only caller input is the staging Id on the retry path, and the action it triggers is the same idempotent reconciliation the scheduler would perform anyway.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/async/StagingReconciliationBatch.cls:260-276 - upd built from e.getMessage() and pi.status; Expired only when past TTL and payment not succeeded</li> <li>base/main/default/permissionsets/Fundraising_User - Donation_Staging__c Create/Read, no Edit</li> <li>base/main/default/classes/donation/staging/DonationStagingSelector.cls selectById - SYSTEM_MODE read, so retry has no record-visibility gate (minor; controller granted only to staff sets)</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li>unverified against primary docs (developer.salesforce.com apex_scheduler.htm returned 403 / cookie wall); secondary: <a href="https://help.salesforce.com/s/articleView?id=sf.code_schedule_batch_apex.htm">https://help.salesforce.com/s/articleView?id=sf.code_schedule_batch_apex.htm</a> (search snippet: scheduled Apex executes under the user who scheduled the job)</li> </ul> |
| <b>R-016</b><br>FLS Update<br><span>False positive</span> | <b>Paths 2, 12–21, 26–27, 30–31, 35, 37–38, 41–42, 44, 46, 48, 50, 52, 54, 56, 58, 60</b><br>At scan:<br>classes/SettingsService.cls:345<br>Submitted source:<br>classes/setup/service/SettingsService.cls:345 | See <b>R-016</b> under “Apex CRUD ContentDistribution” — same code location, same response.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Reported vulnerability                                    | Location                                                                                                                                                                             | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>R-022</b><br>FLS Update<br><span>False positive</span> | <b>Paths 3–4</b><br>At scan:<br><code>classes/RefundReconciliationBatch.cls:344</code><br>Submitted source:<br><code>classes/donation/async/RefundReconciliationBatch.cls:344</code> | See <b>R-022</b> under “FLS Create” — same code location, same response.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>R-040</b><br>FLS Update<br><span>False positive</span> | <b>Paths 5</b><br>At scan:<br><code>classes/DonorPortalService.cls:383</code><br>Submitted source:<br><code>classes/donor/service/DonorPortalService.cls:383</code>                  | <p><b>SCANNER’S CLAIM</b></p> <p><i>This query looks for CRUD Update operations that are performed without checking for <code>isUpdateable()</code> permisison before performing the Update operation on the Object and its fields.</i> Flagged statement:<br/> <code>Database.update(bump, AccessLevel.SYSTEM_MODE);</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the write increments the failed-attempt counter on a portal access code and locks it, a brute-force defence that must run as the Experience Cloud guest user; the Guest User licence cannot hold Edit on any object, and the credential table carries no guest grant at all, so a user-mode update would disable the lockout.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This increments the failed-attempt counter on a donor-portal access code and locks it after the maximum attempts - a brute-force defence. The guest user cannot hold Edit on any object, and the credential table carries no guest grant at all, so the write must be system mode. The row is resolved server-side from the submitted email under FOR UPDATE, never from a client-supplied Id, and the only fields written are <code>Attempt_Count__c</code> and <code>Status__c='Locked'</code>, both computed by Apex. The worst a caller can do is lock a code, which the legitimate donor replaces by requesting a new one.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donor/service/DonorPortalService.cls:367-383 - bump built from selector row; attempts computed server-side</li> <li>base/main/default/classes/donor/service/DonorPortalService.cls class header - <code>without sharing</code> with rationale</li> <li>Portal_Access_Token__c has no Fundraising_GuestDonor grant (permset doc comment)</li> <li>ControllerAction.run has no savepoint rollback on caught error, so the increment persists after a failed verify (intended)</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li>project memory/platform limit: Guest User licence cannot be granted object Edit/Delete (AGENTS.md guest-user notes) - unverified against a primary URL in this pass</li> </ul> |

| Reported vulnerability                              | Location                                                                                                                                | Response                                                                 |
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| <b>R-023</b><br>FLS Update<br><b>False positive</b> | <b>Paths 6–7</b><br>At scan:<br>classes/SettingsService.cls:1026<br>Submitted source:<br>classes/setup/service/SettingsService.cls:1026 | See <b>R-023</b> under “FLS Create” — same code location, same response. |

| Reported vulnerability                                    | Location                                                                                                                                       | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>R-041</b><br>FLS Update<br><span>False positive</span> | <b>Paths 8</b><br>At scan:<br>classes/PostInstallHandler.cls:1498<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:1498 | <p><b>SCANNER'S CLAIM</b></p> <p><i>This query looks for CRUD Update operations that are performed without checking for isUpdateable() permission before performing the Update operation on the Object and its fields.</i> Flagged statement: update as system legacy;</p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the statement runs only inside the package InstallHandler, which executes as a package-specific system user that holds no permission set, so there is no user whose isUpdateable() could be consulted; and Pricebook2 is a standard object that a second-generation package permission set cannot grant, so no permission check could ever be satisfied by package metadata.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This renames the package's own event pricebook from its earlier name to the current one during upgrade, so existing price book entries are kept. Pricebook2 is a standard object that a packaged permission set cannot grant, and the code runs only from the InstallHandler as the package's system user. The new name is a constant; nothing user-supplied is written.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:18 public without sharing class PostInstallHandler implements InstallHandler; lines 1-4 header: 'runs in package install/upgrade context'</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:169-173 doc: 'the script runs as a per-package system user that does not exist before runtime'</li> <li>Salesforce: a post-install script 'runs as a special system user that represents your package' (search-result excerpt of the help page; page body itself not fetchable, so the quote is from the indexed snippet)</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:183-186 onInstall calls each seed* step in its own try/catch; only logStepFailure on error</li> <li>grep of non-test classes: seedDefaultSettings, seedDefaultImages, seedDefaultPageDesign, seedEventPricebook, seedEventDonationProduct, seedEventFeeProduct, seedContentTemplates have NO callers outside PostInstallHandler (mentions in PricebookService/SettingsService are comments/strings only)</li> <li>Class is public, not global (changed 2026-09-05), so subscriber anonymous Apex / other namespaces cannot invoke these methods</li> <li>Pricebook2 and Product2 are standard objects; AGENTS.md / readiness FP-1 item 2 record that a 2GP permission set cannot ship a standard-object grant (build strips it silently), and the install user holds no permission set anyway</li> <li>base/main/default/classes/event/service/PricebookService.cls:11-24 doc: install-testing (2026-08-09) showed a with sharing path could not even resolve Pricebook2 as the per-package system user</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:1477-1499 seedEventPricebook: if only the pre-rename pricebook exists, renames it in place to EVENT_PRICEBOOK_NAME ('Fundraising Events') so its PricebookEntry rows survive; the new name is a constant</li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                        | Response                                                                                                                                                                                                                                                                                             |
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|                                                           |                                                                                                                                                 | <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5</a></li> </ul> |
| <b>R-024</b><br>FLS Update<br><span>False positive</span> | <b>Paths 9</b><br>At scan:<br>classes/PostInstallHandler.cls:1515<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:1515  | See <b>R-024</b> under “FLS Create” — same code location, same response.                                                                                                                                                                                                                             |
| <b>R-025</b><br>FLS Update<br><span>False positive</span> | <b>Paths 10</b><br>At scan:<br>classes/PostInstallHandler.cls:1557<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:1557 | See <b>R-025</b> under “FLS Create” — same code location, same response.                                                                                                                                                                                                                             |
| <b>R-026</b><br>FLS Update<br><span>False positive</span> | <b>Paths 11</b><br>At scan:<br>classes/PostInstallHandler.cls:1601<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:1601 | See <b>R-026</b> under “FLS Create” — same code location, same response.                                                                                                                                                                                                                             |

| Reported vulnerability                                    | Location                                                                                                                                                       | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>R-042</b><br>FLS Update<br><span>False positive</span> | <b>Paths 22–23, 32</b><br>At scan:<br>classes/AttendeeNamingHandler.cls:129<br>Submitted source:<br>classes/guestrequest/handler/AttendeeNamingHandler.cls:129 | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Update operations that are performed without checking for isUpdateable() permission before performing the Update operation on the Object and its fields.</i> Flagged statement:<br>Database.SaveResult[] results = Database.update(toUpdate, false, AccessLevel.SYSTEM_MODE);<br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the write runs asynchronously from the request queue, after ownership of the ticket order was verified synchronously against the donor's portal session, and only the attendee-name fields of that order's own seats are written; the running user is the guest user or the sweep's scheduler, neither of whom can hold Edit on the object.<br><b>SUPPORTING DETAIL</b><br>A donor who is signed in to the donor portal with a verified access code may name the blank guest seats on their own ticket order. Ownership is checked synchronously before anything is queued (requireOwnedOrder against the portal session), the verified order Id is stored server-side in Subject_Id__c, and this handler re-reads the seats of that order only - seat Ids in the payload that are not on the order, cancelled seats, and seats already named are ignored. The only fields written are First_Name__c, Last_Name__c and (when blank) Email__c, length-clipped and email-validated, which are exactly the data the donor is entitled to supply; the Guest licence cannot hold Edit, hence system mode.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donor/service/DonorPortalService.cls:1231-1283 - requireOwnedOrder(ctx, orderId); payload rebuilt from owned seats, never echoed from the browser</li> <li>base/main/default/classes/guestrequest/handler/AttendeeNamingHandler.cls:83-117 - seats re-read from Subject_Id__c order; unmatched/cancelled/already-named seats skipped; NAME_MAX/EMAIL_MAX</li> <li>base/main/default/classes/guestrequest/handler/AttendeeNamingHandler.cls:10-14 - without sharing with rationale</li> <li>base/main/default/classes/guestrequest/handler/AttendeeNamingHandler.cls:19 - MAX_SEATS payload cap</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li>unverified in this pass: Guest User licence cannot hold object Edit (project-documented platform limit)</li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                                | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>R-043</b><br>FLS Update<br><span>False positive</span> | <b>Paths 24, 33</b><br>At scan:<br>classes/GuestRequestService.cls:138<br>Submitted source:<br>classes/guestrequest/service/GuestRequestService.cls:138 | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Update operations that are performed without checking for isUpdateable() permission before performing the Update operation on the Object and its fields.</i> Flagged statement:<br>Database.update(claimed, false, AccessLevel.SYSTEM_MODE);<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the write claims queued requests (status and attempt counter only) so the queueable and the sweep never process a row twice; it runs in the guest user's own queueable, where the Guest User licence cannot hold Edit on any object, or in the scheduled sweep, so a user-mode update could never succeed.<br><br><b>SUPPORTING DETAIL</b><br>This claims queued public requests (Status__c Pending -> Processing, Attempt_Count__c + 1) so the queueable and the sweep never process the same row twice. It runs in the guest user's own queueable - the Guest licence cannot hold Edit on any object - or in the scheduled sweep; the request Ids come from the row the server just inserted or from the sweep's query, and only rows still Pending are claimed. Both written values are computed by Apex.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/guestrequest/service/GuestRequestService.cls:120-138 - only Pending rows; status/attempts computed</li> <li>base/main/default/classes/guestrequest/async/GuestRequestQueueable.cls:31, base/main/default/classes/guestrequest/async/GuestRequestSweepBatch.cls:48 - sole callers of process()</li> <li>Fundraising_GuestDonor has no Edit on Guest_Request__c</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li>unverified against primary docs (developer.salesforce.com apex_scheduler.htm returned 403 / cookie wall); secondary: <a href="https://help.salesforce.com/s/articleView?id=sf.code_schedule_batch_apex.htm">https://help.salesforce.com/s/articleView?id=sf.code_schedule_batch_apex.htm</a> (search snippet: scheduled Apex executes under the user who scheduled the job)</li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                                    | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>R-044</b><br>FLS Update<br><span>False positive</span> | <b>Paths 25, 34, 39</b><br>At scan:<br>classes/GuestRequestService.cls:237<br>Submitted source:<br>classes/guestrequest/service/GuestRequestService.cls:237 | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Update operations that are performed without checking for isUpdateable() permission before performing the Update operation on the Object and its fields.</i> Flagged statement:<br>Database.update(resolving, false, AccessLevel.SYSTEM_MODE);<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the write closes the donor's own refund request only after the refund itself succeeded under the refund controller's Process_Refunds authorization; it selects rows by the refunded gift's Id and writes only a status, and the staff user is not granted Edit on the guest-request object by design, so a user-mode update would leave the portal showing a completed refund as pending.<br><br><b>SUPPORTING DETAIL</b><br>After staff refund a gift, this closes the donor's matching refund request that was parked for review so the donor portal stops showing it as pending. It runs only after the refund itself succeeded under the refund controller's own authorization, it selects requests by the refunded Opportunity Id and only those in Awaiting_Review of the refund type, and it writes a fixed status, a fixed donor-facing sentence and a timestamp. The request rows are created by the guest user and owned by the site's default owner, so an internal user generally has no sharing on them; system mode is what lets the refund close its own request.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/service/RefundService.cls:352-357 - sole caller, constant status/message</li> <li>base/main/default/classes/guestrequest/service/GuestRequestService.cls:214-238 - filters to Awaiting_Review + type</li> <li>Fundraising_User has Edit but not View All on Guest_Request__c; Admin has Modify All - so the real blocker for a user-mode update is record sharing, not the object grant</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a> (user-mode DML requires the running user's Edit permission on the object)</li> <li><a href="https://help.salesforce.com/s/articleView?id=sf.perm_sets_custom.htm">https://help.salesforce.com/s/articleView?id=sf.perm_sets_custom.htm</a> (custom permission Process_Refunds is the authorization the refund controller enforces; unverified exact URL)</li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                      | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>R-045</b><br>FLS Update<br><span>False positive</span> | <b>Paths 28</b><br>At scan:<br>classes/PostInstallHandler.cls:409<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:409 | <b>SCANNER'S CLAIM</b><br><i>This query looks for CRUD Update operations that are performed without checking for isUpdateable() permission before performing the Update operation on the Object and its fields.</i> Flagged statement: <code>if (!toUpdate.isEmpty()) { update as system toUpdate; }</code><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the update runs only inside the InstallHandler as the package's system user, which holds no permission set, touches only the package's own theme records (selected by their packaged names) and writes a URL computed from the org's own site settings; no user input and no user permissions are involved.<br><b>SUPPORTING DETAIL</b><br>This backfills the default hero image URL on the package's own theme records during install or upgrade, limited to the packaged theme names. It runs only from the InstallHandler as the package's system user, which has no permission set, and the value is a URL the install step itself generated from a packaged static resource. No interactive or guest caller reaches it.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:18 <code>public without sharing class PostInstallHandler implements InstallHandler</code>; lines 1-4 header: 'runs in package install/upgrade context'</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:169-173 doc: 'the script runs as a per-package system user that does not exist before runtime'</li> <li>Salesforce: a post-install script 'runs as a special system user that represents your package' (search-result excerpt of the help page; page body itself not fetchable, so the quote is from the indexed snippet)</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:183-186 <code>onInstall</code> calls each <code>seed*</code> step in its own <code>try/catch</code>; only <code>logStepFailure</code> on error</li> <li>grep of non-test classes: <code>seedDefaultSettings</code>, <code>seedDefaultImages</code>, <code>seedDefaultPageDesign</code>, <code>seedEventPricebook</code>, <code>seedEventDonationProduct</code>, <code>seedEventFeeProduct</code>, <code>seedContentTemplates</code> have NO callers outside <code>PostInstallHandler</code> (mentions in <code>PricebookService/SettingsService</code> are comments/strings only)</li> <li>Class is <code>public</code>, not <code>global</code> (changed 2026-09-05), so subscriber anonymous Apex / other namespaces cannot invoke these methods</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:383-409 <code>seedDefaultImages</code>: copies the seeded default hero image URL from <code>Settings__c</code> onto <code>Campaign_Design__c.Default_Hero_Image_URL__c</code> for the packaged theme names only, and only when changed; the query is <code>WITH SYSTEM_MODE</code></li> <li>Value originates from the packaged static-resource distribution URL seeded by <code>SettingsService.ensureDefaultImagesSeeded</code>, not user input</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5">https://help.salesforce.com/s/articleView?id=sf.apex_post_install_script.htm&amp;language=en_US&amp;type=5</a></li> </ul> |

| Reported vulnerability                                    | Location                                                                                                                                                                                         | Response                                                                 |
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| <b>R-036</b><br>FLS Update<br><span>False positive</span> | <b>Paths 29, 36, 40, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61–69</b><br>At scan:<br>classes/PostInstallHandler.cls:758<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:758 | See <b>R-036</b> under “FLS Create” — same code location, same response. |

## Apex SOQL SOSL USER MODE MISSING — 29 paths, 20 locations

**What this query claims:** *USER\_MODE* is recommended for SOQL or SOSL operations

| Reported vulnerability                                             | Location                                                                                                                       | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>R-046</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 1</b><br>At scan:<br>classes/CurrencyResolver.cls:265<br>Submitted source:<br>classes/shared/CurrencyResolver.cls:265 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement:<br>Database.queryWithBinds(soql, new Map<String, Object>{ 'recordIds' => recordIds }, AccessLevel.SYSTEM_MODE),<br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the query reads only Id and CurrencyIsoCode for parent records whose Ids a trigger-driven rollup already holds, in order to convert amounts before writing the rollup total; the object name comes from the Id's schema token, not from input, and the read runs in trigger context as the DML writer, where a user without the parent's currency field would otherwise see the rollup silently fail.<br><b>SUPPORTING DETAIL</b><br>The query reads only Id and CurrencyIsoCode for parent records whose Ids the trigger-driven rollup already holds, so it can convert gift amounts into the parent's currency before writing the total. The object name comes from the Id's own schema token, not from input, so the dynamic FROM cannot be injected. The result is a map of Id to ISO code that is used in the arithmetic and never returned to a user. Running in user mode would make a rollup fail or mis-convert for a staff user who can save a gift but lacks read FLS on a standard CurrencyIsoCode field. The code catches any exception and falls back to the corporate currency.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/shared/CurrencyResolver.cls:258-269 - dynamic FROM built from Id.getSObjectType().getDescribe().getName(); bind is Set&lt;Id&gt;; SYSTEM_MODE; catch returns empty map</li> <li>base/main/default/classes/shared/CurrencyResolver.cls - isoCodesByRecordId doc states the reason (currency code is not customer data)</li> <li>base/main/default/classes/donation/selector/OpportunitiesSelector.cls:189 - upstream SYSTEM_MODE rollup read, itself commented</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                                  | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>R-047</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 2, 4</b><br>At scan:<br>classes/AccountsSelector.cls:311<br>Submitted source:<br>classes/donor/selector/AccountsSelector.cls:311 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: <code>public List&lt;Account&gt; selectPersonDonorDetailById(Id accountId) {</code><br><b>WHY THE CLAIM IS FALSE</b><br>The claim that USER_MODE is missing is false: the query is executed with Database.queryWithBinds(..., AccessLevel.USER_MODE), which enforces the running user's object permissions, field-level security and sharing. The scanner recognises only the literal WITH USER_MODE clause, and adding that clause to this query is impossible: the platform rejects a dynamic query that carries both (measured: "Cannot use the WITH AccessLevel clause in dynamic queries that also specify an access level").<br><b>SUPPORTING DETAIL</b><br>This query already runs in user mode: it is executed with Database.queryWithBinds(..., AccessLevel.USER_MODE), which enforces object permissions, field-level security and sharing for the running user. The scanner only recognises the literal WITH USER_MODE clause and does not read the AccessLevel argument. The caller, DonorDetailController.requireDonor, uses this user-mode, sharing-enforced read as its IDOR gate: if the record Id supplied by the component is not visible to the user, zero rows come back and the controller throws "Donor not found" before any related data is read. A literal WITH USER_MODE clause cannot be added to satisfy the pattern: the query is built by the FFLib query factory as a dynamic string, and the platform rejects a dynamic query that carries both the clause and an AccessLevel argument. Measured 2026-10-01 in a scratch org, running as a Standard User who holds the Fundraising_Admin permission set: Database.queryWithBinds('SELECT Id FROM Account WITH USER_MODE', binds, AccessLevel.USER_MODE) throws System.SecurityException: Cannot use the WITH AccessLevel clause in dynamic queries that also specify an access level. The AccessLevel argument is the documented way to run a dynamic query in user mode.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donor/selector/AccountsSelector.cls:311~~322 - Database.queryWithBinds(..., AccessLevel.USER_MODE); class is inherited sharing</li> <li>base/main/default/classes/donor/controller/DonorDetailController.cls:352 - requireDonor calls the Account/Contact detail selector and throws QueryException("Donor not found") on an empty result</li> <li>base/main/default/classes/donor/controller/DonorDetailController.cls:102,142,176,205,238,272 - every @AuraEnabled action calls requireDonor before any other read</li> <li>base/main/default/classes/donor/controller/DonorDetailController.cls header - documents requireDonor as the gate in front of the SYSTEM_MODE reads of Donation_Designation__c and Guest_Request__c</li> <li>base/main/default/classes/campaign/controller/FundraiserDetailController.cls:39 - getRecurring never calls AccountsSelector; Checkmarx path 4/6 source is spurious</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a></li> </ul> |

| Reported vulnerability | Location | Response                                                                                                                                                                                                                                                                              |
|------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |          | <ul style="list-style-type: none"><li>• <a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a></li></ul> |

| Reported vulnerability                                             | Location                                                                                                                                  | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>R-048</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 3, 5</b><br>At scan:<br>classes/ContactsSelector.cls:228<br>Submitted source:<br>classes/donor/selector/ContactsSelector.cls:228 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: <code>public List&lt;Contact&gt; selectDonorDetailById(Id contactId) {</code><br><b>WHY THE CLAIM IS FALSE</b><br>The claim that USER_MODE is missing is false: the query factory is built with CRUD and FLS enforcement on and the SOQL is executed with Database.queryWithBinds(..., AccessLevel.USER_MODE), which enforces object permissions, FLS and sharing for the running user. The scanner recognises only the literal WITH USER_MODE clause, which the platform forbids in a dynamic query that already specifies an access level (measured).<br><b>SUPPORTING DETAIL</b><br>The SOQL string is built by the query factory and executed with Database.queryWithBinds(..., AccessLevel.USER_MODE), so object permissions, FLS and sharing are enforced for the running user. The factory is also constructed with CRUD enforcement on. The scanner does not recognise the AccessLevel argument. The only caller that passes a client-supplied Id is DonorDetailController.requireDonor, which throws "Donor not found" when this user-mode read returns nothing, so a user cannot reach another donor's gifts, recurring plans or requests by guessing an Id. A literal WITH USER_MODE clause cannot be added to satisfy the pattern: the query is built by the FFLib query factory as a dynamic string, and the platform rejects a dynamic query that carries both the clause and an AccessLevel argument. Measured 2026-10-01 in a scratch org, running as a Standard User who holds the Fundraising_Admin permission set: Database.queryWithBinds('SELECT Id FROM Account WITH USER_MODE', binds, AccessLevel.USER_MODE) throws System.SecurityException: Cannot use the WITH AccessLevel clause in dynamic queries that also specify an access level. The AccessLevel argument is the documented way to run a dynamic query in user mode.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donor/selector/ContactsSelector.cls:228-~242 - newQueryFactory() (fflib defaults enforceCRUD=true) then queryWithBinds with AccessLevel.USER_MODE</li> <li>base/main/default/classes/donor/controller/DonorDetailController.cls:352 - requireDonor gate; empty result -&gt; QueryException</li> <li>base/main/default/classes/donor/controller/DonorDetailController.cls:102,142,176,205,238,272 - every action gated by requireDonor first</li> <li>base/main/default/classes/campaign/controller/FundraiserDetailController.cls:39 - getRecurring uses RecurringDonationsSelector.selectByCampaignPaged, not ContactsSelector</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a></li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                                 | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>R-049</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 6, 8</b><br>At scan:<br>classes/CommunityService.cls:269<br>Submitted source:<br>classes/setup/service/CommunityService.cls:269 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: <code>protected virtual List&lt;NetworkRow&gt; queryNetworkById(String networkId) {</code><br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the read returns only the Experience site's Id, name, URL prefix and status, org configuration rather than record data, and is reachable only from the administrator-only setup console; Network is a setup object, and a user-mode read of setup objects requires View Setup and Configuration, which the package's admin permission set cannot assume every administrator holds.<br><br><b>SUPPORTING DETAIL</b><br>This read resolves the Experience Cloud site that the administrator is configuring. It returns only the Network's Id, name, URL prefix and status, which are org configuration and not record data. It is reachable only from the package's setup console, whose controller is granted solely by the admin permission set. Network visibility in user mode is limited to sites the user is a member of, unless the user holds Create and Set Up Experiences, View All Data or Modify All Data. The admin must be able to resolve a site they are about to repoint the package at, including one still in Preview. The Id binds as a variable, and the query sits in a try/catch that reports "site not found" instead of exposing an error.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/service/CommunityService.cls:269-277 - Database.query with AccessLevel.SYSTEM_MODE; bind variable, not concatenation</li> <li>base/main/default/classes/setup/service/CommunityService.cls:251-266 - resolveByNetworkId wraps it in try/catch and logs via AppLogger.info</li> <li>base/main/default/classes/setup/controller/SetupController.cls:632,642 - saveExperienceSite / checkGuestPermissionSet are @AuraEnabled on a with-sharing controller</li> <li>base/main/default/permissionsets - only Fundraising_Admin grants classAccess to SetupController; Fundraising_User/ReadOnly deliberately omit it</li> <li>base/main/default/classes/setup/service/CommunityService.cls:6-13 - public virtual without sharing; header rationale: User is the only record-shared read; Network and the other setup objects carry per-site SYSTEM_MODE reasons</li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.object_reference.meta/object_reference/sforce_api_objects_network.htm">https://developer.salesforce.com/docs/atlas.en-us.object_reference.meta/object_reference/sforce_api_objects_network.htm</a> - special access rules: users with Create and Set Up Experiences / View All Data / Modify All Data see all Network rows; others see only sites they are members of</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.object_reference.meta/object_reference/sforce_api_objects_network.htm">https://developer.salesforce.com/docs/atlas.en-us.object_reference.meta/object_reference/sforce_api_objects_network.htm</a></li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a></li> </ul> |

| Reported vulnerability                                                | Location                                                                                                                                       | Response                                                                                    |
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| <b>R-016</b><br>Apex SOQL SOSL USER<br>MODE MISSING<br>False positive | <b>Paths 7, 15–16, 27</b><br>At scan:<br>classes/SettingsService.cls:345<br>Submitted source:<br>classes/setup/service/SettingsService.cls:345 | See <b>R-016</b> under “Apex CRUD ContentDistribution” — same code location, same response. |
| <b>R-001</b><br>Apex SOQL SOSL USER<br>MODE MISSING<br>False positive | <b>Paths 9</b><br>At scan:<br>classes/CommunityService.cls:513<br>Submitted source:<br>classes/setup/service/CommunityService.cls:513          | See <b>R-001</b> under “Apex CRUD Violation” — same code location, same response.           |

| Reported vulnerability                                             | Location                                                                                                                               | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>R-050</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 10</b><br>At scan:<br>classes/CommunityService.cls:328<br>Submitted source:<br>classes/setup/service/CommunityService.cls:328 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement:<br>List<PermissionSetAssignment> assignments = [<br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the read is a diagnostic over PermissionSetAssignment, a setup object with no sharing model, returning only a boolean and a remediation message to the administrator-only setup console; the guest user Id is resolved on the server, and a user-mode read would require View Setup and Configuration, which the package cannot grant.<br><b>SUPPORTING DETAIL</b><br>This is a read-only diagnostic in the admin setup console. It checks whether the site's guest user has been assigned the package's guest permission set and returns only a boolean and a remediation message. The guest user Id is resolved on the server from the selected site, not taken from the client. No record data is returned, and the read is wrapped in a try/catch that degrades to 'could not verify'.<br>PermissionSetAssignment is a setup object whose visibility is governed by View Setup and Configuration rather than record sharing, which the package's admin permission set does not grant; the read stays in system mode so an administrator without it gets a true answer. The site carries its own SYSTEM_MODE comment stating this.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/service/CommunityService.cls:325-331 - per-site // SYSTEM_MODE: comment directly above the query (setup data about the site GUEST user; only a Boolean / Set&lt;Id&gt; leaves the method)</li> <li>python3 scripts/lint/system-mode-inventory.py -&gt; 0 unjustified sites</li> <li>SetupController.cls saveExperienceSite / checkGuestPermissionSet are @AuraEnabled on a with-sharing controller granted only in Fundraising_Admin</li> <li>docs/security/appexchange-security-review-readiness.md F-1 (third pass, 2026-09-25)</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.object_reference.meta/object_reference/sforce_api_objects_permissionsetassignment.htm">https://developer.salesforce.com/docs/atlas.en-us.object_reference.meta/object_reference/sforce_api_objects_permissionsetassignment.htm</a></li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a></li> <li><a href="https://developer.salesforce.com/blogs/2021/05/boost-security-by-auditing-your-privileged-users-with-soql">https://developer.salesforce.com/blogs/2021/05/boost-security-by-auditing-your-privileged-users-with-soql</a></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                               | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>R-051</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 11</b><br>At scan:<br>classes/CommunityService.cls:569<br>Submitted source:<br>classes/setup/service/CommunityService.cls:569 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: <code>for (PermissionSetAssignment psa : [</code><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the read checks the guest user's permission-set assignments for a standard-object grant the administrator must add by hand and returns only a boolean to the administrator-only setup console; the objects are setup objects with no sharing model, and a user-mode read would require View Setup and Configuration, which the package cannot grant.<br><b>SUPPORTING DETAIL</b><br>This read-only check verifies that the site guest user has read/create on User External Credential, a standard-object grant the admin must add by hand because a 2GP permission set cannot carry it. The method returns only a boolean for the setup console checklist. The guest user Id is derived on the server, the read never runs as or for a guest request, and failures degrade to a 'check manually' message. The reads are setup data governed by View Setup and Configuration, so they stay in system mode; each carries its own SYSTEM_MODE comment.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/service/CommunityService.cls:566-583 - per-site <code>// SYSTEM_MODE:</code> comment directly above the query (setup data about the site GUEST user; only a Boolean / Set&lt;Id&gt; leaves the method)</li> <li>python3 scripts/lint/system-mode-inventory.py -&gt; 0 unjustified sites</li> <li>SetupController.cls saveExperienceSite / checkGuestPermissionSet are @AuraEnabled on a with-sharing controller granted only in Fundraising_Admin</li> <li>docs/security/appexchange-security-review-readiness.md F-1 (third pass, 2026-09-25)</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.object_reference.meta/object_reference/sforce_api_objects_permissionsetassignment.htm">https://developer.salesforce.com/docs/atlas.en-us.object_reference.meta/object_reference/sforce_api_objects_permissionsetassignment.htm</a></li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                               | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>R-052</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 12</b><br>At scan:<br>classes/CommunityService.cls:608<br>Submitted source:<br>classes/setup/service/CommunityService.cls:608 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: for (SetupEntityAccess a : [<br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the read is over SetupEntityAccess, a setup object with no sharing model, returning only permission-set Ids used for an in-memory comparison in the administrator-only setup console; a user-mode read would require View Setup and Configuration, which the package's admin permission set cannot grant.<br><b>SUPPORTING DETAIL</b><br>This read-only check finds which permission sets grant access to the Stripe external credential's principal, so the setup console can tell the admin whether the guest user can make the payment callout. It returns only permission-set Ids used for an in-memory set intersection, never record data. SetupEntityAccess is a setup object whose visibility is governed by View Setup and Configuration, which the admin permission set does not grant, so the read stays in system mode. The inputs are resolved on the server, the call is reachable only from the admin-only setup controller, and the site carries its own SYSTEM_MODE comment.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/service/CommunityService.cls:605-611 - per-site // SYSTEM_MODE: comment directly above the query (setup data about the site GUEST user; only a Boolean / Set&lt;Id&gt; leaves the method)</li> <li>python3 scripts/lint/system-mode-inventory.py -&gt; 0 unjustified sites</li> <li>SetupController.cls saveExperienceSite / checkGuestPermissionSet are @AuraEnabled on a with-sharing controller granted only in Fundraising_Admin</li> <li>docs/security/appexchange-security-review-readiness.md F-1 (third pass, 2026-09-25)</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.object_reference.meta/object_reference/sforce_api_objects_setupentityaccess.htm">https://developer.salesforce.com/docs/atlas.en-us.object_reference.meta/object_reference/sforce_api_objects_setupentityaccess.htm</a></li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                               | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>R-053</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 13</b><br>At scan:<br>classes/CommunityService.cls:407<br>Submitted source:<br>classes/setup/service/CommunityService.cls:407 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: for (PermissionSetAssignment psa : [<br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the read counts ObjectPermissions rows on the guest user's permission sets and returns only a boolean to the administrator-only setup console; both are setup objects with no sharing model, and a user-mode read would require View Setup and Configuration, which the package's admin permission set cannot grant.<br><b>SUPPORTING DETAIL</b><br>This read-only check powers the setup console's 'guest user can read Campaign' step. It reads the guest user's permission-set assignments and counts ObjectPermissions rows granting Campaign read, returning only a boolean so the admin knows whether the documented manual grant is done (a managed package cannot ship that grant on a standard object). No record data leaves the method and the user Id is derived on the server. Both reads are setup data governed by View Setup and Configuration, which the admin permission set does not grant, so they stay in system mode; each carries its own SYSTEM_MODE comment.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/service/CommunityService.cls:404-418 - per-site // SYSTEM_MODE: comment directly above the query (setup data about the site GUEST user; only a Boolean / Set&lt;Id&gt; leaves the method)</li> <li>python3 scripts/lint/system-mode-inventory.py -&gt; 0 unjustified sites</li> <li>SetupController.cls saveExperienceSite / checkGuestPermissionSet are @AuraEnabled on a with-sharing controller granted only in Fundraising_Admin</li> <li>docs/security/appexchange-security-review-readiness.md F-1 (third pass, 2026-09-25)</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.object_reference.meta/object_reference/sforce_api_objects_permissionsetassignment.htm">https://developer.salesforce.com/docs/atlas.en-us.object_reference.meta/object_reference/sforce_api_objects_permissionsetassignment.htm</a></li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a></li> </ul> |
| <b>R-023</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 14</b><br>At scan:<br>classes/SettingsService.cls:1026<br>Submitted source:<br>classes/setup/service/SettingsService.cls:1026 | See <b>R-023</b> under “FLS Create” — same code location, same response.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| <b>R-054</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 17</b><br>At scan:<br>classes/EventPurchaseService_Test.cls:906<br>Submitted source:<br>classes/event/service/EventPurchaseService_Test.cls:906 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: delete as system tier;<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the statement is a fixture delete inside an @IsTest class, which no user can invoke at runtime, and must succeed under whatever user the subscriber's org-wide test run executes as; it is also DML, not a query, so the SOQL/SOSL rule does not apply.<br><br><b>SUPPORTING DETAIL</b><br>This is a fixture write inside an @IsTest class. It adjusts a test ticket-tier product (for example, sets it sold out or outside its sale window) to arrange the scenario for the service method under test. The class runs only in Apex test execution and is not reachable by any runtime user. The dataflow source is a schema record-type describe, not external input. The statement stays in system mode on purpose: the package's tests are also run in subscriber orgs (an org-wide test run there includes managed-package tests), where the running user may lack a permission the fixture would need in user mode (edit access to Campaign, the master of the ticket-type object, which the platform ties to the Marketing User flag). A user-mode fixture would fail the subscriber's test run for a reason unrelated to the behaviour under test, while protecting nothing, because a test class has no runtime caller.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/event/service/EventPurchaseService_Test.cls:1 - class annotated @IsTest(SeeAllData=false)</li> <li>base/main/default/classes/shared/TestDataFactory.cls:743 - the dataflow source is the Product2 record-type describe in TestDataFactory.createTicketProduct (:741), a test fixture helper</li> <li>base/main/default/classes/event/service/EventPurchaseService_Test.cls:906 - fixture mutation of a test ticket tier (capacity, sold count, or window) before the method under test runs; DML, not SOQL</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm</a></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                                                 | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>R-055</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 18</b><br>At scan:<br>classes/EventPurchaseService_Test.cls:261<br>Submitted source:<br>classes/event/service/EventPurchaseService_Test.cls:261 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: update as system tier;<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the statement is a fixture write inside an @IsTest class, which no user can invoke at runtime, and must succeed under whatever user the subscriber's org-wide test run executes as; it is also DML, not a query, so the SOQL/SOSL rule does not apply.<br><br><b>SUPPORTING DETAIL</b><br>This is a fixture write inside an @IsTest class. It adjusts a test ticket-tier product (for example, sets it sold out or outside its sale window) to arrange the scenario for the service method under test. The class runs only in Apex test execution and is not reachable by any runtime user. The dataflow source is a schema record-type describe, not external input. The statement stays in system mode on purpose: the package's tests are also run in subscriber orgs (an org-wide test run there includes managed-package tests), where the running user may lack a permission the fixture would need in user mode (edit access to Campaign, the master of the ticket-type object, which the platform ties to the Marketing User flag). A user-mode fixture would fail the subscriber's test run for a reason unrelated to the behaviour under test, while protecting nothing, because a test class has no runtime caller.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/event/service/EventPurchaseService_Test.cls:1 - class annotated @IsTest(SeeAllData=false)</li> <li>base/main/default/classes/shared/TestDataFactory.cls:743 - the dataflow source is the Product2 record-type describe in TestDataFactory.createTicketProduct (:741), a test fixture helper</li> <li>base/main/default/classes/event/service/EventPurchaseService_Test.cls:261 - fixture mutation of a test ticket tier (capacity, sold count, or window) before the method under test runs; DML, not SOQL</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm</a></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                                                 | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>R-056</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 19</b><br>At scan:<br>classes/EventPurchaseService_Test.cls:578<br>Submitted source:<br>classes/event/service/EventPurchaseService_Test.cls:578 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: update as system tier;<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the statement is a fixture write inside an @IsTest class, which no user can invoke at runtime, and must succeed under whatever user the subscriber's org-wide test run executes as; it is also DML, not a query, so the SOQL/SOSL rule does not apply.<br><br><b>SUPPORTING DETAIL</b><br>This is a fixture write inside an @IsTest class. It adjusts a test ticket-tier product (for example, sets it sold out or outside its sale window) to arrange the scenario for the service method under test. The class runs only in Apex test execution and is not reachable by any runtime user. The dataflow source is a schema record-type describe, not external input. The statement stays in system mode on purpose: the package's tests are also run in subscriber orgs (an org-wide test run there includes managed-package tests), where the running user may lack a permission the fixture would need in user mode (edit access to Campaign, the master of the ticket-type object, which the platform ties to the Marketing User flag). A user-mode fixture would fail the subscriber's test run for a reason unrelated to the behaviour under test, while protecting nothing, because a test class has no runtime caller.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/event/service/EventPurchaseService_Test.cls:1 - class annotated @IsTest(SeeAllData=false)</li> <li>base/main/default/classes/shared/TestDataFactory.cls:743 - the dataflow source is the Product2 record-type describe in TestDataFactory.createTicketProduct (:741), a test fixture helper</li> <li>base/main/default/classes/event/service/EventPurchaseService_Test.cls:578 - fixture mutation of a test ticket tier (capacity, sold count, or window) before the method under test runs; DML, not SOQL</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm</a></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                                                 | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>R-057</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 20</b><br>At scan:<br>classes/EventPurchaseService_Test.cls:610<br>Submitted source:<br>classes/event/service/EventPurchaseService_Test.cls:610 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: update as system tier;<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the statement is a fixture write inside an @IsTest class, which no user can invoke at runtime, and must succeed under whatever user the subscriber's org-wide test run executes as; it is also DML, not a query, so the SOQL/SOSL rule does not apply.<br><br><b>SUPPORTING DETAIL</b><br>This is a fixture write inside an @IsTest class. It adjusts a test ticket-tier product (for example, sets it sold out or outside its sale window) to arrange the scenario for the service method under test. The class runs only in Apex test execution and is not reachable by any runtime user. The dataflow source is a schema record-type describe, not external input. The statement stays in system mode on purpose: the package's tests are also run in subscriber orgs (an org-wide test run there includes managed-package tests), where the running user may lack a permission the fixture would need in user mode (edit access to Campaign, the master of the ticket-type object, which the platform ties to the Marketing User flag). A user-mode fixture would fail the subscriber's test run for a reason unrelated to the behaviour under test, while protecting nothing, because a test class has no runtime caller.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/event/service/EventPurchaseService_Test.cls:1 - class annotated @IsTest(SeeAllData=false)</li> <li>base/main/default/classes/shared/TestDataFactory.cls:743 - the dataflow source is the Product2 record-type describe in TestDataFactory.createTicketProduct (:741), a test fixture helper</li> <li>base/main/default/classes/event/service/EventPurchaseService_Test.cls:610 - fixture mutation of a test ticket tier (capacity, sold count, or window) before the method under test runs; DML, not SOQL</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm</a></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                                                 | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>R-058</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 21</b><br>At scan:<br>classes/EventPurchaseService_Test.cls:641<br>Submitted source:<br>classes/event/service/EventPurchaseService_Test.cls:641 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: update as system tier;<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the statement is a fixture write inside an @IsTest class, which no user can invoke at runtime, and must succeed under whatever user the subscriber's org-wide test run executes as; it is also DML, not a query, so the SOQL/SOSL rule does not apply.<br><br><b>SUPPORTING DETAIL</b><br>This is a fixture write inside an @IsTest class. It adjusts a test ticket-tier product (for example, sets it sold out or outside its sale window) to arrange the scenario for the service method under test. The class runs only in Apex test execution and is not reachable by any runtime user. The dataflow source is a schema record-type describe, not external input. The statement stays in system mode on purpose: the package's tests are also run in subscriber orgs (an org-wide test run there includes managed-package tests), where the running user may lack a permission the fixture would need in user mode (edit access to Campaign, the master of the ticket-type object, which the platform ties to the Marketing User flag). A user-mode fixture would fail the subscriber's test run for a reason unrelated to the behaviour under test, while protecting nothing, because a test class has no runtime caller.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/event/service/EventPurchaseService_Test.cls:1 - class annotated @IsTest(SeeAllData=false)</li> <li>base/main/default/classes/shared/TestDataFactory.cls:743 - the dataflow source is the Product2 record-type describe in TestDataFactory.createTicketProduct (:741), a test fixture helper</li> <li>base/main/default/classes/event/service/EventPurchaseService_Test.cls:641 - fixture mutation of a test ticket tier (capacity, sold count, or window) before the method under test runs; DML, not SOQL</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm</a></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                                                 | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>R-059</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 22</b><br>At scan:<br>classes/EventPurchaseService_Test.cls:672<br>Submitted source:<br>classes/event/service/EventPurchaseService_Test.cls:672 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: update as system tier;<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the statement is a fixture write inside an @IsTest class, which no user can invoke at runtime, and must succeed under whatever user the subscriber's org-wide test run executes as; it is also DML, not a query, so the SOQL/SOSL rule does not apply.<br><br><b>SUPPORTING DETAIL</b><br>This is a fixture write inside an @IsTest class. It adjusts a test ticket-tier product (for example, sets it sold out or outside its sale window) to arrange the scenario for the service method under test. The class runs only in Apex test execution and is not reachable by any runtime user. The dataflow source is a schema record-type describe, not external input. The statement stays in system mode on purpose: the package's tests are also run in subscriber orgs (an org-wide test run there includes managed-package tests), where the running user may lack a permission the fixture would need in user mode (edit access to Campaign, the master of the ticket-type object, which the platform ties to the Marketing User flag). A user-mode fixture would fail the subscriber's test run for a reason unrelated to the behaviour under test, while protecting nothing, because a test class has no runtime caller.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/event/service/EventPurchaseService_Test.cls:1 - class annotated @IsTest(SeeAllData=false)</li> <li>base/main/default/classes/shared/TestDataFactory.cls:743 - the dataflow source is the Product2 record-type describe in TestDataFactory.createTicketProduct (:741), a test fixture helper</li> <li>base/main/default/classes/event/service/EventPurchaseService_Test.cls:672 - fixture mutation of a test ticket tier (capacity, sold count, or window) before the method under test runs; DML, not SOQL</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm</a></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                                                                        | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>R-060</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 23</b><br>At scan:<br>classes/EventRegistrationController_Test.cls:171<br>Submitted source:<br>classes/fundraising/controller/EventRegistrationController_Test.cls:171 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: <code>update as system t;</code><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the statement is a fixture update inside an @IsTest class, which cannot run outside Apex test execution and must succeed under whatever user the subscriber's org-wide test run executes as; it is DML, not a query, so the SOQL/SOSL rule does not apply, and the reported dataflow source is a schema describe, not user input.<br><b>SUPPORTING DETAIL</b><br>This line is a test-fixture update inside an @IsTest class. It prepares a ticket product so the controller method under test sees a particular state. It cannot run outside Apex test execution. The dataflow source is a schema describe. The statement stays in system mode on purpose: the package's tests are also run in subscriber orgs (an org-wide test run there includes managed-package tests), where the running user may lack a permission the fixture would need in user mode (edit access to Campaign, the master of the ticket-type object, which the platform ties to the Marketing User flag). A user-mode fixture would fail the subscriber's test run for a reason unrelated to the behaviour under test, while protecting nothing, because a test class has no runtime caller.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/fundraising/controller/EventRegistrationController_Test.cls:1 - class annotated @IsTest(SeeAllData=false)</li> <li>base/main/default/classes/shared/TestDataFactory.cls:743 - the dataflow source is the Product2 record-type describe in TestDataFactory.createTicketProduct (:741), a test fixture helper</li> <li>base/main/default/classes/fundraising/controller/EventRegistrationController_Test.cls:171 - fixture update of a test ticket product; DML, not SOQL</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_annotation_isTest.htm</a></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                                                      | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>R-061</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 24–26, 29</b><br>At scan:<br>classes/ReceiptSequenceSelector.cls:63<br>Submitted source:<br>classes/donation/selector/ReceiptSequenceSelector.cls:63 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: <code>return (List&lt;Receipt_Sequence__c&gt;) Database.queryWithBinds(</code><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the row is an internal per-tax-year receipt counter holding only a year and the last number issued, not donor data; the FOR UPDATE lock keeps receipt numbers gapless, and the read must succeed from the guest finalize path and the renewal job, neither of whose running users can be granted the object.<br><b>SUPPORTING DETAIL</b><br>Receipt_Sequence__c is an internal per-tax-year counter row, not donor data. It holds only a year and the last receipt number issued. The query locks the counter (FOR UPDATE) so receipt numbers stay gapless and unique. Access must not depend on the running user, because a user without read access on the counter would otherwise break receipt numbering for their own gift. The only bind is a set of tax years derived from the Opportunity close date; no caller-supplied Id or filter reaches the query, and nothing from the counter row is returned to a UI.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/selector/ReceiptSequenceSelector.cls:60-65 - SYSTEM_MODE queryWithBinds, FOR UPDATE, bind is Set&lt;Decimal&gt; of tax years</li> <li>base/main/default/classes/donation/service/ReceiptNumberService.cls - lockCounters/assignWithoutUnitOfWork are the only callers; class is inherited sharing</li> <li>base/main/default/classes/donation/domain/Opportunities.cls:284-296 - stampOfflineReceiptNumber (before trigger, with sharing domain) calls assignWithoutUnitOfWork for manually entered Closed Won gifts; this runs as the interactive user, not in batch</li> <li>base/main/default/objects/Receipt_Sequence__c - sharingModel ReadWrite; Fundraising_Admin CRU+ViewAll, Fundraising_User and ReadOnly Read only (User cannot create a counter, so USER_MODE would break the offline-gift path for staff)</li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a> - USER_MODE with no object access throws a QueryException rather than returning zero rows</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a></li> </ul> |

| Reported vulnerability                                             | Location                                                                                                                        | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>R-062</b><br>Apex SOQL SOSL USER MODE MISSING<br>False positive | <b>Paths 28</b><br>At scan:<br>classes/CurrencyResolver.cls:264<br>Submitted source:<br>classes/shared/CurrencyResolver.cls:264 | <b>SCANNER'S CLAIM</b><br><i>USER_MODE is recommended for SOQL or SOSL operations</i> Flagged statement: <code>return toIsoMap(</code><br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the query reads only Id and CurrencyIsoCode for parent records whose Ids a trigger-driven rollup already holds, in order to convert amounts before writing the rollup total; the object name comes from the Id's schema token, not from input, and the read runs in trigger context as the DML writer, where a user without the parent's currency field would otherwise see the rollup silently fail.<br><br><b>SUPPORTING DETAIL</b><br>The query reads only Id and CurrencyIsoCode for parent records whose Ids the trigger-driven rollup already holds, so it can convert gift amounts into the parent's currency before writing the total. The object name comes from the Id's own schema token, not from input, so the dynamic FROM cannot be injected. The result is a map of Id to ISO code that is used in the arithmetic and never returned to a user. Running in user mode would make a rollup fail or mis-convert for a staff user who can save a gift but lacks read FLS on a standard CurrencyIsoCode field. The code catches any exception and falls back to the corporate currency.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/shared/CurrencyResolver.cls:258-269 - dynamic FROM built from <code>Id.getSObjectType().getDescribe().getName();</code> bind is <code>Set&lt;Id&gt;; SYSTEM_MODE;</code> catch returns empty map</li> <li>base/main/default/classes/shared/CurrencyResolver.cls - <code>isoCodesByRecordId</code> doc states the reason (currency code is not customer data)</li> <li>base/main/default/classes/donation/selector/OpportunitiesSelector.cls:189 - upstream <code>SYSTEM_MODE</code> rollup read, itself commented</li> </ul><br><b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_enforce_usermode.htm</a></li> </ul> |

## Sharing — 27 paths, 27 locations

**What this query claims:** All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.

| Reported vulnerability                                 | Location                                                                                                                                                     | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>R-063</b><br>Sharing<br><span>False positive</span> | <b>Paths 1</b><br>At scan:<br>classes/RecurringDonationsSelector.cls:11<br>Submitted source:<br>classes/recurring/selector/RecurringDonationsSelector.cls:11 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservice or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public inherited sharing class RecurringDonationsSelector extends fflib_SObjectSelector</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because <code>inherited sharing</code> is the documented mode for a class that is never an entry point: the selector takes the sharing of its caller, so a staff request is sharing-enforced and only the guest and job callers, which declare <code>without sharing</code> with a stated reason, bypass it.</p> <p><b>SUPPORTING DETAIL</b></p> <p><code>inherited sharing</code> is intentional on this selector. It is never the entry point of a transaction, so its record visibility is that of the caller: staff controllers are <code>with sharing</code>, and the guest controllers and reconciliation jobs are <code>without sharing</code> for their own documented reasons. The Apex Developer Guide recommends <code>inherited sharing</code> for 'service classes that must be flexible and support use cases with different sharing modes'. Each query also sets an explicit <code>AccessLevel</code>, and each <code>SYSTEM_MODE</code> query carries its own reason at the call site.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/recurring/selector/RecurringDonationsSelector.cls:11 - declaration</li> <li>base/main/default/classes/shared/Application.cls - resolved via the Selector factory only</li> <li>grep: no entry-point annotation or <code>async</code> interface in the class</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                                                         | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>R-064</b><br>Sharing<br><span>False positive</span> | <b>Paths 2</b><br>At scan:<br>classes/RecurringDesignationsSelector.cls:4<br>Submitted source:<br>classes/recurring/selector/RecurringDesignationsSelector.cls:4 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservice or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public inherited sharing class RecurringDesignationsSelector extends fflib_SObjectSelector</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because <code>inherited sharing</code> is the documented mode for a class that is never an entry point: the selector takes the sharing of its caller, so a staff request is sharing-enforced and only the guest and job callers, which declare <code>without sharing</code> with a stated reason, bypass it.</p> <p><b>SUPPORTING DETAIL</b></p> <p><code>inherited sharing</code> is intentional on this selector. It is never the entry point of a transaction, so its record visibility is that of the caller: staff controllers are <code>with sharing</code>, and the guest controllers and reconciliation jobs are <code>without sharing</code> for their own documented reasons. The Apex Developer Guide recommends <code>inherited sharing</code> for 'service classes that must be flexible and support use cases with different sharing modes'. Each query also sets an explicit <code>AccessLevel</code>, and each <code>SYSTEM_MODE</code> query carries its own reason at the call site.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/recurring/selector/RecurringDesignationsSelector.cls:4 - declaration</li> <li>base/main/default/classes/shared/Application.cls - resolved via the Selector factory only</li> <li>grep: no entry-point annotation or <code>async</code> interface in the class</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                                          | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>R-065</b><br>Sharing<br><span>False positive</span> | <b>Paths 3</b><br>At scan:<br>classes/OpportunitiesSelector.cls:19<br>Submitted source:<br>classes/donation/selector/OpportunitiesSelector.cls:19 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public inherited sharing class OpportunitiesSelector extends fflib_SObjectSelector</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because <code>inherited sharing</code> is the documented mode for a class that is never an entry point: the selector takes the sharing of its caller, so a staff controller's request is sharing-enforced and only the guest and job paths, which declare <code>without sharing</code> with a stated reason, bypass it.</p> <p><b>SUPPORTING DETAIL</b></p> <p><code>inherited sharing</code> is intentional on this selector. It is never the entry point of a transaction, so its record visibility is that of the caller: staff controllers are <code>with sharing</code>, and the guest controllers and reconciliation jobs are <code>without sharing</code> for their own documented reasons. The Apex Developer Guide recommends <code>inherited sharing</code> for 'service classes that must be flexible and support use cases with different sharing modes'. Each query also sets an explicit <code>AccessLevel</code>, and each <code>SYSTEM_MODE</code> query carries its own reason at the call site.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/selector/OpportunitiesSelector.cls:19 - declaration</li> <li>base/main/default/classes/shared/Application.cls - resolved via the Selector factory only</li> <li>grep: no entry-point annotation or <code>async</code> interface in the class</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                                    | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>R-066</b><br>Sharing<br><span>False positive</span> | <b>Paths 4</b><br>At scan:<br>classes/fflib_SObjectUnitOfWork.cls:191<br>Submitted source:<br>classes/fflib/fflib_SObjectUnitOfWork.cls:191 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: public inherited sharing virtual class SimpleDML implements IDML, IDMLUpsertable</p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because SimpleDML is declared inherited sharing, the documented mode for a non-entry-point class: a commit runs in the sharing mode of the class that called commitWork(), so a staff controller's commit is sharing-enforced by the platform and only callers that explicitly opt into the SystemDML subclass, each with a stated reason, bypass it.</p> <p><b>SUPPORTING DETAIL</b></p> <p>SimpleDML is the FFLib Unit of Work's default DML strategy. It is declared inherited sharing, so a commit follows the sharing mode of the class that called commitWork(). Every staff controller and service in the package is with sharing, so a staff commit runs with sharing and the platform enforces the running user's record access on the write. The writes that must bypass sharing (guest entry points, batch jobs, the platform-event subscriber and trigger-fired rollups) do not rely on this class: they construct their Unit of Work with SystemDML, a package-authored without sharing subclass, and each of the 24 construction sites states which of those families it belongs to. Object and field permissions are enforced one layer up, in the FFLib Selectors and by Security.stripInaccessible. SystemDML_Test proves, with a Minimum Access user, that a SimpleDML commit on a record the user is not shared is refused and a SystemDML commit succeeds.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/fflib/fflib_SObjectUnitOfWork.cls:64 - public virtual inherited sharing class fflib_SObjectUnitOfWork, rationale at :54-61</li> <li>base/main/default/classes/fflib/fflib_SObjectUnitOfWork.cls:165-191 - rationale and public inherited sharing virtual class SimpleDML implements IDML, IDMLUpsertable</li> <li>base/main/default/classes/shared/SystemDML.cls:30 - public without sharing class SystemDML extends fflib_SObjectUnitOfWork.SimpleDML, family list at :9-27</li> <li>grep: new SystemDML() at 24 production sites, each preceded by a // SystemDML: guest job rollup comment; none in a staff controller</li> </ul> |

| Reported vulnerability | Location | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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|                        |          | <ul style="list-style-type: none"><li>base/main/default/classes/shared/SystemDML_Test.cls:33 - test_simpleDml_withSharingCaller_cannotUpdateUnsharedRecord</li><li>base/main/default/classes/shared/SystemDML_Test.cls:60 - test_systemDml_withSharingCaller_updatesUnsharedRecord</li><li>DEPENDENCIES.md fork log row for fflib_SObjectUnitOfWork.cls</li><li><a href="https://github.com/apex-enterprise-patterns/fflib-apex-common/issues/530">https://github.com/apex-enterprise-patterns/fflib-apex-common/issues/530</a> - upstream proposal for the same inherited sharing declarations</li></ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"><li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li><li><a href="https://github.com/apex-enterprise-patterns/fflib-apex-common/issues/530">https://github.com/apex-enterprise-patterns/fflib-apex-common/issues/530</a></li></ul> |

| Reported vulnerability                                 | Location                                                                                                                                           | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>R-067</b><br>Sharing<br><span>False positive</span> | <b>Paths 5</b><br>At scan:<br>classes/EventTicketTypesSelector.cls:6<br>Submitted source:<br>classes/event/selector/EventTicketTypesSelector.cls:6 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservice or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public inherited sharing class EventTicketTypesSelector extends fflib_SObjectSelector</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because <code>inherited sharing</code> is the documented mode for a class that is never an entry point: the selector takes the sharing of its caller. Its reads are additionally gated on the master Campaign's visibility by the calling controller, because the object is a master-detail detail whose field permissions are inert.</p> <p><b>SUPPORTING DETAIL</b></p> <p><code>inherited sharing</code> is intentional on this selector. It is never the entry point of a transaction, so its record visibility is that of the caller: staff controllers are <code>with sharing</code>, and the guest controllers and reconciliation jobs are <code>without sharing</code> for their own documented reasons. The Apex Developer Guide recommends <code>inherited sharing</code> for 'service classes that must be flexible and support use cases with different sharing modes'. Each query also sets an explicit <code>AccessLevel</code>, and each <code>SYSTEM_MODE</code> query carries its own reason at the call site.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/event/selector/EventTicketTypesSelector.cls:6 - declaration</li> <li>base/main/default/classes/shared/Application.cls - resolved via the Selector factory only</li> <li>grep: no entry-point annotation or async interface in the class</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                                                             | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>R-068</b><br>Sharing<br><span>False positive</span> | <b>Paths 6</b><br>At scan:<br>classes/EventRegistrationController.cls:196<br>Submitted source:<br>classes/fundraising/controller/EventRegistrationController.cls:196 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>private inherited sharing class CreateTicketIntent extends GuestAction {</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the inner action class is <code>inherited sharing</code> (matching the <code>GuestAction</code> base it extends) and is invoked only from the public event-registration controller, where the Experience Cloud guest user holds no record sharing; the controller's <code>without sharing</code> carries its reason, and the action's reads are gated on an active, published campaign.</p> <p><b>SUPPORTING DETAIL</b></p> <p>The inner class extends <code>GuestAction</code>, which is <code>inherited sharing</code>, and declares the same keyword so its mode is readable at the declaration. Invoked from the <code>without sharing</code> guest controller, it runs without sharing. That is required: an Experience Cloud guest creates a ticket order against a Campaign it is not shared under sharing rules, and the Guest User licence holds no record sharing. Record visibility is decided by explicit predicates instead: <code>requireOpenCampaign</code> refuses the request unless the campaign is active, published and open for sales, and checkout is not blocked. <code>GuestAction</code> also collapses any unexpected error to one generic message, so nothing internal reaches the guest.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/fundraising/controller/EventRegistrationController.cls:194-196 - rationale and <code>private inherited sharing class CreateTicketIntent extends GuestAction</code></li> <li>base/main/default/classes/fundraising/controller/EventRegistrationController.cls:9 - <code>public without sharing class EventRegistrationController</code></li> <li>base/main/default/classes/fundraising/controller/EventRegistrationController.cls:283 - <code>requireOpenCampaign(campaignId)</code> gates the request</li> <li>base/main/default/classes/fundraising/controller/EventRegistrationController.cls:871-879 - <code>requireOpenCampaign: requireActiveCampaign, CampaignService.assertAcceptingDonations, SettingsService.assertCheckoutNotBlocked</code></li> <li>base/main/default/classes/shared/GuestAction.cls:30 - <code>public abstract inherited sharing class GuestAction extends ControllerAction</code></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                                | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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|                                                        |                                                                                                                                         | <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>R-069</b><br>Sharing<br><span>False positive</span> | <b>Paths 7</b><br>At scan:<br>classes/EmailSenderService.cls:10<br>Submitted source:<br>classes/setup/service/EmailSenderService.cls:10 | <b>SCANNER'S CLAIM</b><br><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public inherited sharing class EmailSenderService {</code><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the class is a stateless helper that is never a transaction entry point and reads only org email configuration; <code>inherited sharing</code> makes it run in the mode of the service that builds the message, which is the documented purpose of that keyword.<br><b>SUPPORTING DETAIL</b><br>This is a stateless helper that stamps the org's configured sender identity onto outbound donor email. It is never a transaction entry point, and it takes the sharing mode of the service that builds the message. It reads only org email configuration. The keyword was reviewed for a change to <code>with sharing</code> on 2026-10-01 and kept: the class is called from the guest donation path and from the platform-event subscriber as well as from staff actions, and <code>inherited sharing</code> is the declaration that lets each of those run in its caller's mode instead of forcing one mode on all of them.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/service/EmailSenderService.cls:1-10 - header + declaration</li> <li>grep: no entry-point annotation or async interface</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                           | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>R-070</b><br>Sharing<br><span>False positive</span> | <b>Paths 8</b><br>At scan:<br>classes/DonationService.cls:7<br>Submitted source:<br>classes/donation/service/DonationService.cls:7 | <b>SCANNER'S CLAIM</b><br><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app. Flagged statement: public inherited sharing class DonationService implements IDonationService {</i><br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because DonationService is never a transaction entry point; inherited sharing makes it take the mode of its callers, the guest controllers (where visibility is decided by token or campaign predicates) and the reconciliation jobs, each of which declares its own keyword with its reason.<br><br><b>SUPPORTING DETAIL</b><br>DonationService is never an entry point. Its public methods are reached from the guest donation controllers, where visibility is decided by token or campaign predicates, and from the reconciliation jobs. Both of those callers establish without sharing and document why. When a with sharing staff controller calls it, it runs with sharing. That is the documented behavior of inherited sharing and the reason it is used here.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/service/DonationService.cls:7 - declaration</li> <li>base/main/default/classes/donation/async/StagingReconciliationBatch.cls:216 - finalizeSucceededStaging caller</li> <li>base/main/default/classes/donation/async/StripePaymentSweep.cls:166 - caller</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                                                      | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>R-071</b><br>Sharing<br><span>False positive</span> | <b>Paths 9</b><br>At scan:<br>classes/DonationDesignationsSelector.cls:8<br>Submitted source:<br>classes/donation/selector/DonationDesignationsSelector.cls:8 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public inherited sharing class DonationDesignationsSelector extends fflib_SObjectSelector</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because <code>inherited sharing</code> is the mode Salesforce documents for non-entry-point classes and is accepted by its own false-positive guidance: the selector is never a transaction entry point, so it runs with the sharing of whichever controller or job called it, which is exactly the behaviour the rule asks for.</p> <p><b>SUPPORTING DETAIL</b></p> <p><code>inherited sharing</code> is intentional on this selector. It is never the entry point of a transaction, so its record visibility is that of the caller: staff controllers are <code>with sharing</code>, and the guest controllers and reconciliation jobs are <code>without sharing</code> for their own documented reasons. The Apex Developer Guide recommends <code>inherited sharing</code> for 'service classes that must be flexible and support use cases with different sharing modes'. Each query also sets an explicit <code>AccessLevel</code>, and each <code>SYSTEM_MODE</code> query carries its own reason at the call site.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/selector/DonationDesignationsSelector.cls:8 - declaration</li> <li>base/main/default/classes/shared/Application.cls - resolved via the Selector factory only</li> <li>grep: no entry-point annotation or async interface in the class</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                        | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>R-072</b><br>Sharing<br><span>False positive</span> | <b>Paths 10</b><br>At scan:<br>classes/CurrencyResolver.cls:8<br>Submitted source:<br>classes/shared/CurrencyResolver.cls:<br>8 | <b>SCANNER'S CLAIM</b><br><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: public inherited sharing class CurrencyResolver {<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the helper is never a transaction entry point; inherited sharing makes it run in the mode of the controller or trigger that called it, which is the documented purpose of that keyword.<br><br><b>SUPPORTING DETAIL</b><br>This helper resolves the currency for a campaign or payment account. It is never a transaction entry point and runs in whatever sharing mode its calling controller established, which is the purpose of inherited sharing.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/shared/CurrencyResolver.cls:8 - declaration</li> <li>grep: no entry-point annotation or async interface</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                                                | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>R-073</b><br>Sharing<br><span>False positive</span> | <b>Paths 11</b><br>At scan:<br>classes/CampaignMemberSyncService.cls:9<br>Submitted source:<br>classes/donation/service/CampaignMemberSyncService.cls:9 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public inherited sharing class CampaignMemberSyncService implements ICampaignMemberSyncService {</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the class has no @AuraEnabled, REST or asynchronous surface of its own, so it is not an entry point; <code>inherited sharing</code> makes it run in its caller's mode, which is the documented purpose of that keyword.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This is a service, never a transaction entry point. It takes the sharing mode of the controller or job that calls it, which is what <code>inherited sharing</code> is documented for. It has no @AuraEnabled, REST, or async surface of its own.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/service/CampaignMemberSyncService.cls:9 - declaration</li> <li>grep: no entry-point annotation or async interface</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                               | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>R-074</b><br>Sharing<br><span>False positive</span> | <b>Paths 12</b><br>At scan: <code>classes/SystemDML.cls:30</code><br>Submitted source:<br><code>classes/shared/SystemDML.cls:30</code> | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public without sharing class SystemDML extends fflib_SObjectUnitOfWork.SimpleDML {</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because SystemDML is a deliberately narrow wrapper, not an entry point: it is the package's explicit, per-site opt-in for commits where the running user is by design not the authority over the records (guest entry points, batch and queueable jobs, trigger-fired rollups), each opt-in carrying its reason in code; staff commits use the user-mode strategy instead. The reason is stated at the declaration.</p> <p><b>SUPPORTING DETAIL</b></p> <p>SystemDML is the package's explicit opt-in for writes where the running user is, by design, not the authority over the records: Experience Cloud guest entry points (the Guest User licence holds no record sharing), batch and queueable jobs that must write every donor's records whoever scheduled them, the platform-event subscriber that runs as Automated Process, and trigger-fired rollups written for whoever saved the gift. The default Unit of Work strategy follows its caller's sharing; this class is used only where a site asks for it, and each of the 24 construction sites carries a <code>// SystemDML: guest job rollup</code> comment naming its family. The class contains nothing but four DML statements. Object and field permissions are enforced one layer up, in the FFLib Selectors and by Security.stripInaccessible. SystemDML_Test proves the difference with a Minimum Access user.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/shared/SystemDML.cls:1-30 - class description, family list and <code>public without sharing class SystemDML extends fflib_SObjectUnitOfWork.SimpleDML</code></li> <li>grep: <code>new SystemDML()</code> at 24 production sites, each preceded by a <code>// SystemDML: guest job rollup</code> comment; none in a staff controller</li> <li>base/main/default/classes/shared/SystemDML_Test.cls:33 - SimpleDML commit on an unshared record is refused</li> <li>base/main/default/classes/shared/SystemDML_Test.cls:60 - SystemDML commit on the same record succeeds</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

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| <b>R-075</b><br>Sharing<br><span>False positive</span> | <b>Paths 13</b><br>At scan:<br>classes/StripePaymentSweep.cls:26<br>Submitted source:<br>classes/donation/async/StripePaymentSweep.cls:26 | <b>SCANNER'S CLAIM</b><br><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public without sharing class StripePaymentSweep</code><br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because this is batch Apex, which the scan's own rule exempts; the sweep compares the gateway ledger to every staging row and Opportunity in the org and takes no user input (the admin 'run now' button starts the same batch with no parameters).<br><br><b>SUPPORTING DETAIL</b><br>The sweep compares Stripe's payment ledger to the org's staging rows and Opportunities, and repairs gaps. That only works if it sees every row. It takes no input from users: the admin 'run now' button starts the same batch with no parameters.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/async/StripePaymentSweep.cls:26 - declaration</li> <li>base/main/default/classes/setup/controller/SetupController.cls:2142 - admin-only executeBatch, no parameters</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_scheduler.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_scheduler.htm</a></li> </ul> |

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| <b>R-076</b><br>Sharing<br><span>False positive</span> | <b>Paths 14</b><br>At scan:<br>classes/StagingReconciliationBatch.cls:31<br>Submitted source:<br>classes/donation/async/StagingReconciliationBatch.cls:31 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: public without sharing class StagingReconciliationBatch</p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because this is batch Apex, which the scan's own rule exempts; it must finalize staging rows created by the site guest user that no job owner is shared on. The single staff 'Retry now' entry point takes only a row Id and re-reads the charge from the gateway, so no caller-controlled data is written.</p> <p><b>SUPPORTING DETAIL</b></p> <p>The batch has to finalize donation staging rows created by the site guest user, and those rows are not owned by whoever runs the job. The one non-batch entry point is a single-row 'Retry now' action for staff. It re-reads the charge from Stripe and applies the same idempotent outcome logic as the batch. Donation_Staging__c ships with a Public Read/Write internal sharing model, so without sharing gives internal staff no visibility they do not already have.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/async/StagingReconciliationBatch.cls:134-148 - reconcileOne: selectById (SYSTEM_MODE) then Stripe retrieve then applyOutcome</li> <li>base/main/default/classes/donation/controller/DonationMonitorController.cls:196-212 - RetryDonationStagingNow calls reconcileOne(stagingId); only requireNotNull</li> <li>base/main/default/objects/Donation_Staging__c/Donation_Staging__c.object-meta.xml:11 - &lt;sharingModel&gt;ReadWrite&lt;/sharingModel&gt;</li> <li>base/main/default/permissionsets/Fundraising_User.permissionset-meta.xml - Donation_Staging__c allowRead=true, allowEdit=false</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_scheduler.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_scheduler.htm</a></li> </ul> |

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| <b>R-077</b><br>Sharing<br><span>False positive</span> | <b>Paths 15</b><br>At scan:<br>classes/SettingsService.cls:543<br>Submitted source:<br>classes/setup/service/SettingsService.cls:543 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app. Flagged statement: private without sharing class PackagedAssetSeeder {</i></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because this is a private inner helper, not an entry point, that runs only during install as the package's system user to copy the package's own static-resource images into public content distributions; that user cannot read StaticResource under sharing, and the helper never touches subscriber or donor data. The reason is stated at the declaration.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This private helper copies the package's own static-resource images into public content distributions during install, so the default logo and hero images exist. The install user cannot read StaticResource under sharing. The helper never touches admin- or donor-supplied files, which go through the user-mode upload path. The elevation covers two short methods and nothing else.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/service/SettingsService.cls:529-543 - rationale</li> <li>base/main/default/classes/setup/service/SettingsService.cls:547-552 - StaticResource read by packaged name only</li> <li>base/main/default/classes/setup/service/SettingsService.cls:378,141 - only instantiation sites</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

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| <b>R-078</b><br>Sharing<br><span>False positive</span> | <b>Paths 16</b><br>At scan:<br>classes/RetentionPurgeBatch.cls:16<br>Submitted source:<br>classes/shared/async/RetentionPurgeBatch.cls:16 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public without sharing class RetentionPurgeBatch implements Database.Batchable&lt;SObject&gt;, Database.Stateful {</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because this is batch Apex, which the scan's own rule exempts; it enforces the configured retention policy by deleting expired diagnostic and staging rows, and a row hidden from the job owner by sharing would otherwise never be purged. It has no entry point besides its scheduler. The reason is stated at the declaration.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This batch enforces the configured data-retention policy by deleting expired diagnostic and staging rows. A row hidden from the job's running user by sharing would never be purged, which would defeat the policy. The batch has no entry point other than its scheduler and takes no user input.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/shared/async/RetentionPurgeBatch.cls:16 - declaration</li> <li>base/main/default/classes/shared/async/RetentionPurgeScheduler.cls:29-33 - sole enqueuer</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_scheduler.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_scheduler.htm</a></li> </ul> |

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| <b>R-079</b><br>Sharing<br><span>False positive</span> | <b>Paths 17</b><br>At scan:<br>classes/RefundReconciliationBatch.cls:44<br>Submitted source:<br>classes/donation/async/RefundReconciliationBatch.cls:44 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservice or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: public without sharing class RefundReconciliationBatch</p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the scan's own rule exempts batch and asynchronous Apex: this is a Batchable with no other entry point, and reconciling gateway refunds against every donation would silently skip records hidden from the job owner if it ran with sharing. Its reason is stated at the declaration.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This is a batch with no public surface beyond the Batchable interface. It reconciles refunds recorded in Stripe against every donation in the org. If it ran with the sharing of the job owner, refunds on records that user cannot see would be silently skipped. No caller can pass it record Ids.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/async/RefundReconciliationBatch.cls:44 - declaration</li> <li>base/main/default/classes/donation/async/RefundReconciliationScheduler.cls:39 - sole enqueuer</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_scheduler.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_scheduler.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                                            | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>R-080</b><br>Sharing<br><span>False positive</span> | <b>Paths 18</b><br>At scan:<br>classes/ReconciliationHeartbeat.cls:11<br>Submitted source:<br>classes/donation/async/ReconciliationHeartbeat.cls:11 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public without sharing class ReconciliationHeartbeat implements Schedulable {</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim that this entry point should run with sharing is false because the scan's own rule says batch and asynchronous Apex should not; this is a Schedulable whose only job is to count every stuck staging row in the org, a count that would silently under-report under the sharing of whichever user scheduled it. Its reason is stated at the declaration.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This is a Schedulable with no other entry point. It counts every non-terminal donation staging row in the org and alerts when any is stuck. A sharing-restricted count would depend on who owns the scheduled job, and would under-report. Its only writes are a Transaction_Log__c row and the alert, and nothing is returned to a user.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/async/ReconciliationHeartbeat.cls:11 - declaration + rationale</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:495 - scheduled at install</li> <li>base/main/default/classes/setup/controller/SetupController.cls:2231 - admin re-schedule (SetupController granted to Fundraising_Admin only)</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_scheduler.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_scheduler.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                             | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>R-081</b><br>Sharing<br><span>False positive</span> | <b>Paths 19</b><br>At scan:<br>classes/PricebookService.cls:25<br>Submitted source:<br>classes/event/service/PricebookService.cls:25 | <b>SCANNER'S CLAIM</b><br><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app. Flagged statement: public without sharing class PricebookService {</i><br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the class is not an entry point and its only work is creating and reading the package's own event price book and its entries, standard objects a 2GP permission set cannot grant, during install (system user), admin ticket-tier saves and guest ticket purchases; none of those running users can hold sharing on Pricebook2. The reason is stated at the declaration.<br><br><b>SUPPORTING DETAIL</b><br>PricebookService only creates and looks up the package's event price book and its PricebookEntry rows for ticket products. It runs during install, when an admin saves ticket tiers, and during a guest ticket purchase. None of those contexts can rely on the running user's price-book sharing. It handles no donor data, and it only receives Product2 Ids that its callers have read from records they already gated. with sharing is not an option here: in the package install context a with sharing class cannot resolve the standard price book, which made the install-time seed fail when it was tried (2026-08-09).<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/event/service/PricebookService.cls:25 - declaration</li> <li>base/main/default/classes/event/controller/EventTicketAdminController.cls:156-176,197-263 - requireCampaignWritable gates; product Id derived from the saved tier</li> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:1544,1588 - install callers</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                                    | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>R-082</b><br>Sharing<br><span>False positive</span> | <b>Paths 20</b><br>At scan:<br>classes/PostInstallHandler.cls:18<br>Submitted source:<br>classes/setup/controller/PostInstallHandler.cls:18 | <b>SCANNER'S CLAIM</b><br><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: public without sharing class PostInstallHandler implements InstallHandler {<br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because an InstallHandler is not a user entry point: it runs once, as a package-specific system user with no permission set, to seed org-wide configuration that must exist regardless of who installed the package. Salesforce's own false-positive guidance accepts install context, and the reason is stated at the declaration.<br><br><b>SUPPORTING DETAIL</b><br>This is the package InstallHandler. It seeds settings, themes, record-type mappings, the event price book and content templates, and schedules the package's background jobs. That data is org-wide configuration, which must be created regardless of the install user's sharing. Its public helpers are also re-run from the admin-only Settings console to repair a missing seed or a job owned by a deactivated user. They take no record Ids and are idempotent.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/controller/PostInstallHandler.cls:1-4 - rationale</li> <li>base/main/default/classes/setup/controller/SetupController.cls:834-847 - admin Fix button calls scheduleJobs(true)/seedRecordTypeMappings</li> <li>base/main/default/classes/setup/controller/SetupController.cls:1772-1784 - saveGeneralSettings re-owns jobs via scheduleJobs(true)</li> <li>base/main/default/permissionsets/Fundraising_Admin.permissionset-meta.xml - SetupController granted only here</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

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| <b>R-083</b><br>Sharing<br><span>False positive</span> | <b>Paths 21</b><br>At scan:<br>classes/NpspSyncService.cls:14<br>Submitted source:<br>classes/donation/service/NpspSyncService.cls:14 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: public without sharing class NpspSyncService implements INpspSyncService {</p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the class is not an entry point: it runs only from the platform-event subscriber (the Automated Process user) and the reconciliation jobs, both asynchronous contexts the scan's own rule exempts, and it writes NPSP objects the package cannot grant any user permissions on. Its reason is stated at the declaration.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This service mirrors finalized donations into the optional NPSP objects (allocations and recurring donations). It runs from system contexts: the finalization platform-event subscriber and the reconciliation jobs that finalize donations. The records it writes belong to another package, and this package's permission sets cannot grant access to them. It never takes a record Id from a user: every Id comes from a donation or schedule the same transaction just finalized. NPSP rejections are logged and swallowed, so they cannot roll back a captured payment.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donation/service/NpspSyncService.cls:1-14 - header + declaration</li> <li>base/main/default/classes/donation/service/DonationFinalizedSubscriberHandler.cls:23-24 - syncAllocations caller</li> <li>base/main/default/classes/recurring/service/RecurringDonationService.cls:965-974 - its own doc comment: callers are StagingReconciliationBatch and StripePaymentSweep 'running as their batch owner, NOT the Automated Process user'</li> <li>base/main/default/classes/donation/service/DonationService.cls:593-595 - completeScheduleActivation from finalizeSucceededStaging</li> <li>No PlatformEventSubscriberConfig in base/ - the subscriber runs as the default Automated Process user</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

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| <b>R-084</b><br>Sharing<br><div>False positive</div> | <b>Paths 22</b><br>At scan:<br>classes/GuestRequestService.cls:13<br>Submitted source:<br>classes/guestrequest/service/GuestRequestService.cls:13 | <b>SCANNER'S CLAIM</b><br><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public without sharing class GuestRequestService implements IGuestRequestService {</code><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the service is reached only after the donor portal has authorized the donor by a verified access code, or from the background sweep, and the guest user holds no sharing on the request object; the staff methods it serves filter their rows through user-mode reads. The reason is stated at the declaration.<br><b>SUPPORTING DETAIL</b><br>This service queues and processes public requests (attendee naming, refund requests) submitted from the donor portal. It is only reached after the portal has authorized the donor, and by the background sweep. The staff methods it also serves read and close requests for a gift. They are gated by the Process_Refunds custom permission, and Guest_Request__c ships with a Public Read/Write internal sharing model, so running <code>without sharing</code> gives staff no visibility they would not already have.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>• base/main/default/classes/guestrequest/service/GuestRequestService.cls:8-13 - rationale</li> <li>• base/main/default/classes/donation/controller/RefundController.cls:95-105 - staff caller gated by Process_Refunds</li> <li>• base/main/default/objects/Guest_Request__c/Guest_Request__c.object-meta.xml:11 - <code>&lt;sharingModel&gt;ReadWrite&lt;/sharingModel&gt;</code></li> <li>• base/main/default/classes/donor/service/DonorPortalService.cls:1286,1269 - submit callers</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li>• <a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

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| <b>R-085</b><br>Sharing<br><span>False positive</span> | <b>Paths 23</b><br>At scan:<br>classes/EventRegistrationController.cls:9<br>Submitted source:<br>classes/fundraising/controller/EventRegistrationController.cls:9 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: public without sharing class EventRegistrationController {</p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because this is a guest-facing Experience Cloud controller: the guest user has no sharing that could expose an event campaign, so with sharing would make the public page empty. Reads are gated on an active, published campaign resolved by slug or Id, the single write is a staging row, and the reason is stated at the declaration.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This is the public event-registration controller for an Experience Cloud site, and the guest user has no sharing that would expose an event campaign. Every method is narrow. Reads are gated on an active, published campaign resolved by slug or Id. The write is gated on the event being open for sales. The two status methods require the unguessable access token issued at checkout. Errors are sanitized through GuestAction. Staff hold the same class grant, but reach nothing beyond what an anonymous visitor can.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/fundraising/controller/EventRegistrationController.cls:25-112 - the five @AuraEnabled methods</li> <li>base/main/default/classes/fundraising/controller/EventRegistrationController.cls:856-879 - requireActiveCampaign/requireOpenCampaign</li> <li>base/main/default/classes/fundraising/controller/EventRegistrationController.cls:781-807 - token-gated status/receipt methods</li> <li>base/main/default/permissionsets/Fundraising_GuestDonor.permissionset-meta.xml:36 - guest grant</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

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| <b>R-086</b><br>Sharing<br><span>False positive</span> | <b>Paths 24</b><br>At scan:<br>classes/DonorPortalService.cls:48<br>Submitted source:<br>classes/donor/service/DonorPortalService.cls:48 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: public without sharing class DonorPortalService implements IDonorPortalService {</p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the class serves an Experience Cloud guest user, who holds no record sharing on Contact, Account or the access-code object, so with sharing would return nothing; every guest method is instead authorized by possession of a verified access code, and the staff twins re-read their records in user mode. The reason is stated at the declaration.</p> <p><b>SUPPORTING DETAIL</b></p> <p>The class is without sharing so the Experience Cloud guest user, which has no sharing on Contact, Account or Portal_Access_Token__c, can resolve a donor from an access code and mint a portal session; every guest method is authorized by possession of the code or session it produces and trusts no client-supplied record Id. The class also holds two staff methods, issueAccessCodeForDonor and createAccessCodeForDonor, which take a record Id from a Lightning component. Both staff methods re-read the donor in the staff user's mode (USER_MODE selectors, resolveStaffDonorEmail) and refuse a donor the user cannot see before anything is minted or returned, and the class header states that its without sharing must not become a way to reach records outside the caller's access.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/donor/service/DonorPortalService.cls:38-47 - header rationale for without sharing and the staff-twin exception; :48 declaration</li> <li>base/main/default/classes/donor/controller/DonorPortalAdminController.cls:156 / :175 - USER_MODE reads of the Contact / Person Account before the service is called</li> <li>docs/security/appexchange-security-review-readiness.md F-10; FP-3 sharing census</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                             | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>R-087</b><br>Sharing<br><span>False positive</span> | <b>Paths 25</b><br>At scan:<br>classes/CommunityService.cls:13<br>Submitted source:<br>classes/setup/service/CommunityService.cls:13 | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i></p> <p>Flagged statement: <code>public virtual without sharing class CommunityService implements ICommunityService {</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the service is reachable only from the administrator-only settings console, and the one record-shared object it reads is User, where a Private user-sharing model can hide the site's guest user from the administrator and turn every guest-setup check into a false failure; all other reads are setup objects with no sharing model. The reason is stated at the declaration.</p> <p><b>SUPPORTING DETAIL</b></p> <p>CommunityService backs the admin Settings console's Experience Cloud panel. The one record-shared object it reads is User: it looks up the site's guest user, which a Private User sharing model can hide from the admin, turning every guest check into a false "no guest user". Its other reads (Network, BackgroundOperation, PermissionSetAssignment, ObjectPermissions and SetupEntityAccess) are setup objects with no record sharing, and each carries its own SYSTEM_MODE reason. The only caller is the admin-only settings controller. It returns status flags and names only.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/setup/service/CommunityService.cls:6-13 - rationale</li> <li>base/main/default/classes/setup/controller/SetupController.cls:44-45 - communityServiceInstance, the only reference (e.g. :2811)</li> <li>grep: CommunityService referenced only by SetupController, Application and TestDataFactory among non-test classes (others are comments)</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                                              | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>R-088</b><br>Sharing<br><span>False positive</span> | <b>Paths 26</b><br>At scan:<br>classes/AttendeeNamingHandler.cls:14<br>Submitted source:<br>classes/guestrequest/handler/AttendeeNamingHandler.cls:14 | <b>SCANNER'S CLAIM</b><br><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public without sharing class AttendeeNamingHandler implements IGuestRequestHandler {</code><br><b>WHY THE CLAIM IS FALSE</b><br>The claim is false because the handler runs asynchronously from the request queue, a context the scan's own rule exempts, where no sharing relationship to the donor's order exists; the authorization (verified portal access code plus ownership of the order) was performed synchronously before the request was queued. The reason is stated at the declaration.<br><b>SUPPORTING DETAIL</b><br>This handler fills in guest names on the seats of one ticket order, for a request the donor submitted through the portal. It runs asynchronously from the request queue, where no sharing relationship to the donor's order exists. The authorization was performed when the request was accepted (portal session or code, then the order Id recorded server-side). The payload is size-capped and parsed defensively.<br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/classes/guestrequest/handler/AttendeeNamingHandler.cls:10-14 - rationale</li> <li>base/main/default/classes/guestrequest/handler/GuestRequestRouter.cls:28 - sole constructor site</li> <li>base/main/default/classes/guestrequest/handler/AttendeeNamingHandler.cls:19 - MAX_SEATS cap on untrusted payload</li> </ul> <b>SOURCES</b> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_scheduler.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_scheduler.htm</a></li> </ul> |

| Reported vulnerability                                 | Location                                                                                                                             | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>R-089</b><br>Sharing<br><span>False positive</span> | <b>Paths 27</b><br>At scan: <code>classes/AppLogger.cls:9</code><br>Submitted source:<br><code>classes/shared/AppLogger.cls:9</code> | <p><b>SCANNER'S CLAIM</b></p> <p><i>All entry points to an app (Global or Controller classes) must use the 'with sharing' keyword. Classes without this keyword run without sharing if they are entry points to your code, or with the sharing policy of the caller. Do not omit the sharing declaration as this hides critical security information in side-effects that can change when code is refactored. Only declare classes as 'without sharing' if they are not entry points to your app and if they only modify objects whose security is managed by your code (such as wizard state, or fields in a site). It is a common misconception to believe that batch apex or async apex must run with the global keyword. This is not true, the only classes that must be global are those that expose webservices or are intended to be used by extension packages. All async apex should run as public in order to avoid creating privileged entry points to your app.</i> Flagged statement: <code>public without sharing class AppLogger {</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim is false because the class only inserts diagnostic rows, performs no queries and returns no records, so sharing could not expose any data through it; it must succeed from every context, including a guest user and a failure caused by sharing itself, or the error being logged would be lost. The reason is stated at the declaration.</p> <p><b>SUPPORTING DETAIL</b></p> <p>AppLogger only inserts diagnostic Log__c rows. It performs no queries and returns no records, so there is no data a caller could read through it. It must succeed from any context, including the guest user and failures caused by sharing, or the error being logged is lost. Log__c is read-only to staff and not granted to the guest user.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/classes/shared/AppLogger.cls:1-9 - rationale</li> <li>base/main/default/classes/shared/AppLogger.cls:125 - the only DML: insert as system Log__c</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm">https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_classes_keywords_sharing.htm</a></li> </ul> |

## Apex VF Dynamic JS Resource — 2 paths, 2 locations

**What this query claims:** *Static resources must be used load third-party javascript code*

| Reported vulnerability                                        | Location                                                                                                   | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>R-090</b><br>Apex VF Dynamic JS Resource<br>False positive | <b>Paths 1</b><br>At scan: pages/stripePayment.page:52<br>Submitted source:<br>pages/stripePayment.page:53 | <p><b>SCANNER'S CLAIM</b></p> <p><i>Static resources must be used load third-party javascript code</i> Flagged statement: <code>&lt;script src="https://js.stripe.com/v3/"&gt;&lt;/script&gt;</code></p> <p><b>WHY THE CLAIM IS FALSE</b></p> <p>The claim that the script resource is dynamic is false because the src is a literal <code>https://js.stripe.com/v3/</code> string with no merge field, parameter or stored value contributing to it; and the rule's remedy (a static resource) is prohibited for this library, because Stripe requires Stripe.js to be loaded from <code>js.stripe.com</code> for PCI compliance and does not permit self-hosting.</p> <p><b>SUPPORTING DETAIL</b></p> <p>This is a static <code>&lt;script&gt;</code> tag with a literal <code>https://js.stripe.com</code> URL; no merge field, parameter or stored value contributes to the src, so it is not a dynamically constructed resource. Stripe's documentation requires Stripe.js to be loaded directly from <code>https://js.stripe.com</code> and not bundled or self-hosted, so it cannot be moved into a static resource; <code>js.stripe.com</code> is shipped as a packaged CSP Trusted Site. The page exists precisely to keep Stripe.js out of the LWC runtime: the LWC frames this Visualforce page by relative URL and talks to it by <code>postMessage</code>, Stripe Elements tokenizes card data in Stripe's own nested <code>iframe</code>, and only a payment-method id is returned to Apex. The page has no Apex controller and renders no server value into its script: the <code>postMessage</code> peer is the page's own origin (<code>window.location.origin</code>), and inbound messages are accepted only from that origin and from <code>window.parent</code>. Moving the script into a static resource was reviewed again on 2026-10-01 and is not permitted: Stripe's integration guidance states that Stripe.js must always be loaded directly from <code>js.stripe.com</code> and must not be included in a bundle or self-hosted, as a condition of PCI compliance.</p> <p><b>EVIDENCE</b></p> <ul style="list-style-type: none"> <li>base/main/default/pages/stripePayment.page:48-53 - justification comment + static script tag</li> <li>base/main/default/pages/stripePayment.page:1-7 - <code>&lt;apex:page&gt;</code> declares no controller; the page contains no <code>{!...}</code> merge field</li> <li>base/main/default/pages/stripePayment.page:89 - <code>var PARENT_ORIGIN = window.location.origin;</code> (the page is always framed same-origin)</li> <li>base/main/default/pages/stripePayment.page:134 - message listener drops any event whose origin <code>!== PARENT_ORIGIN</code> or whose source <code>!== window.parent</code></li> <li>base/main/default/pages/stripePayment.page:173-174 - <code>returnUrl</code> built with <code>encodeURIComponent</code> on <code>publishableKey</code> and <code>stagingId</code></li> <li>base/main/default/lwc/dfForm/dfForm.js:441 - parent side checks <code>event.origin === ALLOWED_ORIGIN (window.location.origin)</code></li> <li><code>https://docs.stripe.com/js/including</code> - 'it should always be loaded directly from <code>https://js.stripe.com</code>, rather than included in a bundle or hosted yourself'</li> <li><code>https://docs.stripe.com/security/guide</code> - Stripe.js CSP directives: <code>script-src https://*.js.stripe.com, https://js.stripe.com; frame-src https://*.js.stripe.com, https://js.stripe.com, https://hooks.stripe.com; connect-src</code></li> </ul> |

| Reported vulnerability | Location | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |          | <p>https://api.stripe.com</p> <ul style="list-style-type: none"><li>• base/main/default/cspTrustedSites/Stripe_js.cspTrustedSite-meta.xml - endpointUrl https://js.stripe.com, isActive true, connect-src + frame-src, context All (CspTrustedSite metadata has no script-src flag)</li><li>• base/main/default/cspTrustedSites/ - also Stripe_api (api.stripe.com), Stripe_hooks (hooks.stripe.com), Stripe_m, Stripe_m_network; no *.js.stripe.com entry (Stripe lists it as optional, for performance)</li></ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"><li>• <a href="https://docs.stripe.com/js/including">https://docs.stripe.com/js/including</a></li><li>• <a href="https://docs.stripe.com/security/guide">https://docs.stripe.com/security/guide</a></li><li>• <a href="https://docs.stripe.com/sdks/stripejs-versioning">https://docs.stripe.com/sdks/stripejs-versioning</a></li><li>• <a href="https://stripe.com/guides/pci-compliance">https://stripe.com/guides/pci-compliance</a></li></ul> |

| Reported vulnerability                                        | Location                                                                                                                  | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>R-091</b><br>Apex VF Dynamic JS Resource<br>False positive | <b>Paths 2</b><br>At scan:<br>pages/stripePaymentReturn.page:39<br>Submitted source:<br>pages/stripePaymentReturn.page:40 | <b>SCANNER'S CLAIM</b><br>Static resources must be used load third-party javascript code Flagged statement: <code>&lt;script src="https://js.stripe.com/v3/"&gt;&lt;/script&gt;</code><br><br><b>WHY THE CLAIM IS FALSE</b><br>The claim that the script resource is dynamic is false because the src is a literal https://js.stripe.com/v3/ string with nothing dynamic contributing to it; and Stripe requires Stripe.js to be loaded from js.stripe.com rather than bundled as a static resource, so the rule's remedy is not available for this library.<br><br><b>SUPPORTING DETAIL</b><br>This is a static <code>&lt;script&gt;</code> tag with a literal https://js.stripe.com URL, the second of the two Stripe.js loads in the package; nothing dynamic contributes to the src. Stripe redirects the donor here after an off-site authentication step (3D Secure, bank redirect), and the page needs Stripe.js to call <code>retrievePaymentIntent/retrieveSetupIntent</code> ; Stripe's documentation requires Stripe.js to be loaded from js.stripe.com rather than self-hosted, which is why it is not a static resource. URL parameters read by the page (pk and the Stripe intent client secret) are passed only to Stripe.js calls, and status text is written with <code>textContent</code> . The page has no Apex controller. Its only server-rendered value is the 'Return to the site' link, whose href is the org's saved Experience site URL read through a <code>\$Setup</code> merge field, rendered only when it begins with https://, and never taken from the query string. Moving the script into a static resource was reviewed again on 2026-10-01 and is not permitted: Stripe's integration guidance states that Stripe.js must always be loaded directly from js.stripe.com and must not be included in a bundle or self-hosted, as a condition of PCI compliance.<br><br><b>EVIDENCE</b> <ul style="list-style-type: none"> <li>base/main/default/pages/stripePaymentReturn.page:35-40 - justification comment + static script tag</li> <li>base/main/default/pages/stripePaymentReturn.page:1-7 - <code>&lt;apex:page&gt;</code> declares no controller</li> <li>base/main/default/pages/stripePaymentReturn.page:82-87 - only merge fields: the return link, rendered only when <code>\$Setup.Settings__c.Experience_Site_URL__c</code> BEGINS 'https://', value from the same setting (never from a URL parameter)</li> <li>base/main/default/pages/stripePaymentReturn.page:91-96 - pk and intent client secrets read from <code>URLSearchParams</code>; pk only passed to <code>Stripe(pk)</code>; the <code>opp</code> parameter is never read</li> <li>base/main/default/pages/stripePaymentReturn.page:109-115 - <code>PARENT_ORIGIN = window.location.origin</code>; <code>postMessage</code> targets only that origin</li> <li>base/main/default/pages/stripePaymentReturn.page:117-124 - <code>showStatus</code> writes title/message/icon via <code>textContent</code></li> <li>https://docs.stripe.com/js/including - 'it should always be loaded directly from https://js.stripe.com, rather than included in a bundle or hosted yourself'</li> <li>https://docs.stripe.com/security/guide - Stripe.js CSP directives: <code>script-src https://*.js.stripe.com, https://js.stripe.com; frame-src https://*.js.stripe.com, https://js.stripe.com, https://hooks.stripe.com; connect-src https://api.stripe.com</code></li> </ul> |

| Reported vulnerability | Location | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |          | <ul style="list-style-type: none"> <li>base/main/default/cspTrustedSites/Stripe_js.cspTrustedSite-meta.xml - endpointUrl https://js.stripe.com, isActive true, connect-src + frame-src, context All (CspTrustedSite metadata has no script-src flag)</li> <li>base/main/default/cspTrustedSites/ - also Stripe_api (api.stripe.com), Stripe_hooks (hooks.stripe.com), Stripe_m, Stripe_m_network; no *.js.stripe.com entry (Stripe lists it as optional, for performance)</li> </ul> <p><b>SOURCES</b></p> <ul style="list-style-type: none"> <li><a href="https://docs.stripe.com/js/including">https://docs.stripe.com/js/including</a></li> <li><a href="https://docs.stripe.com/security/guide">https://docs.stripe.com/security/guide</a></li> <li><a href="https://docs.stripe.com/sdks/stripejs-versioning">https://docs.stripe.com/sdks/stripejs-versioning</a></li> <li><a href="https://stripe.com/guides/pci-compliance">https://stripe.com/guides/pci-compliance</a></li> </ul> |

## 5. Code-quality findings

The quality rules are reported per path like the security queries. Each path below is classified by the file its flagged statement sits in, and each group carries one disposition.

| Rule                             | Group                                               | Paths      | Status         |
|----------------------------------|-----------------------------------------------------|------------|----------------|
| Multiple Trigger On same sObject | One trigger per object — scanner misfire            | 5          | False positive |
| DML Statements Inside Loops      | Unit of Work commit — one statement per object type | 12         | By design      |
| DML Statements Inside Loops      | Batch job — per-row isolation by design             | 9          | By design      |
| DML Statements Inside Loops      | Error logging — guarded insert                      | 48         | By design      |
| DML Statements Inside Loops      | Single statement after the loop                     | 27         | By design      |
| DML Statements Inside Loops      | Test fixture                                        | 1          | By design      |
| <b>Total</b>                     |                                                     | <b>102</b> |                |

### Multiple Trigger On same sObject — 5 paths

**What this rule says:** *This rule identifies the presence of multiple Apex triggers associated with the same sObject. Simply having triggers on the same sObject is not an issue. However, it is possible to have redundant code or logic executed in the multiple triggers. Examining multiple triggers may be a way to enhance and improve the code. Please be aware that there is no guarantee the order in which multiple triggers on the same sObject execute.*

**One trigger per object — scanner misfire** — 5 paths (False positive). The package has exactly one trigger per object; each is a single line that hands off to its FFLib domain class (`fflib_SObjectDomain.triggerHandler`). The reported triggers are on five different objects (Contact, Opportunity, the recurring-donation object, the ticket-type object and the payment-account object). Reproduce with `grep -h "^trigger" base/main/default/triggers/*.trigger | awk '{print $4}' | sort | uniq -d`, which prints nothing.

Files: `ContactTrigger.trigger` (1), `EventTicketTypeTrigger.trigger` (1), `OpportunityTrigger.trigger` (1), `PaymentAccountTrigger.trigger` (1), `RecurringDonationTrigger.trigger` (1)

## DML Statements Inside Loops — 97 paths

**What this rule says:** *This rule identifies Apex where DML statements execute within a loop. It is a best practice to ensure no SOQL or SOSL statements are executed inside a loop and instead, that the query/search statements retrieve all the necessary data in a single execution.*

**Unit of Work commit — one statement per object type — 12 paths** (By design). `fplib_SObjectUnitOfWork.commitWork()` and its `SystemDML` strategy iterate over the registered object types and issue one DML statement per type, passing the full list of records of that type. The number of statements is bounded by the number of object types registered, not by the number of records, which is the bulkification pattern the Unit of Work exists to provide.

Files: `fplib_SObjectUnitOfWork.cls` (8), `SystemDML.cls` (4)

**Batch job — per-row isolation by design — 9 paths** (By design). The two jobs write per record on purpose. `StripePaymentSweep` runs with a batch scope of 1 and caps repairs at 5 per run, so each execution writes for one payment account. `StagingReconciliationBatch` runs with a scope of 5, makes all its Stripe callouts first, then writes each staging row's outcome separately so that one row's failure does not roll back the outcome already recorded for another. Both stay far inside the per-transaction DML limit.

Files: `StripePaymentSweep.cls` (5), `StagingReconciliationBatch.cls` (4)

**Error logging — guarded insert — 48 paths** (By design). `AppLogger.write()` inserts one `Log__c` row per call. The callers inside loops log from catch blocks, anomaly branches and per-row outcome notices; in one path (`DonationService.publishErrorText`) the loop only joins the event bus's error messages into one string, which a single `AppLogger.error` call then records. Before inserting, `write()` returns when the entry's level is below the org's configured log-level threshold (or logging is off), and when the transaction has no DML statements left (`Limits.getLimitDmlStatements() <= Limits.getDmlStatements()`), so logging cannot raise a limit exception that would mask the error being recorded. The insert itself is wrapped in try/catch.

Files: `AppLogger.cls` (48)

**Single statement after the loop — 27 paths** (By design). The DML statement is not inside the loop the scanner names. In each path the loop builds the list that one bulk statement then writes (`insert as system newStd, Database.insert(toCreate, false, AccessLevel.USER_MODE), Database.update(toUpdate, false, ...)`), generates a token character by character before a single insert, or joins the record-type mappings into one encoded string that a single `upsert` of the settings record then saves; in the remaining paths the loop is in a test method that calls the service more than once. The rule follows data from the loop into any method that contains a DML statement, including a call made once after the loop has ended, so a bulk call that receives the list the loop built is reported as DML inside the loop. The statement runs once per call whatever the number of records.

Files: `PricebookService.cls` (10), `CampaignPostService.cls` (7), `EventPurchaseService.cls` (4), `DonorPortalService.cls` (3), `AttendeeNamingHandler.cls` (1), `DonationService.cls` (1), `SettingsService.cls` (1)

**Test fixture — 1 paths** (By design). The loop and the DML are in test code (`TestDataFactory` and `_Test` classes) that builds a small, fixed set of records for a test. Test classes do not run in subscriber transactions.

Files: `AttendeeNamingHandler_Test.cls` (1)

## Appendix A — every reported path

| Query                         | Path | Source (scan)                             | Sink (scan)                               | Response | Status         |
|-------------------------------|------|-------------------------------------------|-------------------------------------------|----------|----------------|
| Apex CRUD Violation           | 1    | classes/SetupController.cls:642           | classes/CommunityService.cls:513          | R-001    | False positive |
| Apex CRUD Violation           | 2    | classes/fflib_QueryFactory.cls:360        | classes/PaymentSetupGuideService.cls:1918 | R-002    | False positive |
| Apex CRUD Violation           | 3    | classes/fflib_QueryFactory.cls:360        | classes/ReceiptService.cls:103            | R-003    | False positive |
| Apex CRUD Violation           | 4    | classes/fflib_QueryFactory.cls:360        | classes/ReceiptService.cls:149            | R-004    | False positive |
| Apex CRUD Violation           | 5    | classes/fflib_QueryFactory.cls:360        | classes/ReceiptService.cls:734            | R-005    | False positive |
| Apex CRUD Violation           | 6    | classes/fflib_QueryFactory.cls:360        | classes/ReceiptService.cls:603            | R-006    | False positive |
| Apex CRUD Violation           | 7    | classes/fflib_IDGenerator.cls:37          | classes/PaymentSetupGuideService.cls:1918 | R-002    | False positive |
| Apex CRUD Violation           | 8    | classes/fflib_IDGenerator.cls:37          | classes/ReceiptService.cls:603            | R-006    | False positive |
| Apex CRUD Violation           | 9    | classes/fflib_IDGenerator.cls:37          | classes/ReceiptService.cls:518            | R-007    | False positive |
| Apex CRUD Violation           | 10   | classes/fflib_IDGenerator.cls:37          | classes/ReceiptService.cls:103            | R-003    | False positive |
| Apex CRUD Violation           | 11   | classes/fflib_IDGenerator.cls:37          | classes/DonorPortalService.cls:483        | R-008    | False positive |
| Apex CRUD Violation           | 12   | classes/fflib_IDGenerator.cls:37          | classes/DonorPortalService.cls:548        | R-009    | False positive |
| Apex CRUD Violation           | 13   | classes/fflib_IDGenerator.cls:37          | classes/ReceiptService.cls:149            | R-004    | False positive |
| Apex CRUD Violation           | 14   | classes/fflib_IDGenerator.cls:37          | classes/ReceiptService.cls:329            | R-010    | False positive |
| Apex CRUD Violation           | 15   | classes/SetupController.cls:479           | classes/PaymentSetupGuideService.cls:1918 | R-002    | False positive |
| Apex CRUD Violation           | 16   | classes/RefundController.cls:22           | classes/ReceiptService.cls:408            | R-011    | False positive |
| Apex CRUD Violation           | 17   | classes/fflib_ApexMocks.cls:115           | classes/ReceiptService.cls:329            | R-010    | False positive |
| Apex CRUD Violation           | 18   | classes/fflib_ApexMocks.cls:115           | classes/ReceiptService.cls:408            | R-011    | False positive |
| Apex CRUD Violation           | 19   | classes/ReceiptAdminController.cls:38     | classes/ReceiptService.cls:173            | R-012    | False positive |
| Apex CRUD Violation           | 20   | classes/DonationMonitorController.cls:57  | classes/ReceiptService.cls:103            | R-003    | False positive |
| Apex CRUD Violation           | 21   | classes/fflib_QueryFactory.cls:218        | classes/RefundRequestHandler.cls:138      | R-013    | False positive |
| Apex CRUD Violation           | 22   | classes/fflib_QueryFactory.cls:218        | classes/ReceiptService.cls:518            | R-007    | False positive |
| Apex CRUD Violation           | 23   | classes/DonorPortalAdminController.cls:40 | classes/DonorPortalService.cls:483        | R-008    | False positive |
| Apex CRUD Violation           | 24   | classes/DonorPortalAdminController.cls:40 | classes/DonorPortalService.cls:548        | R-009    | False positive |
| Apex CRUD Violation           | 25   | classes/fflib_ApexMocks.cls:69            | classes/ReceiptService.cls:329            | R-010    | False positive |
| Apex CRUD Violation           | 26   | classes/fflib_ApexMocks.cls:69            | classes/ReceiptService.cls:408            | R-011    | False positive |
| Apex CRUD Violation           | 27   | classes/DonorPortalAdminController.cls:50 | classes/DonorPortalService.cls:483        | R-008    | False positive |
| Apex CRUD Violation           | 28   | classes/DonorPortalAdminController.cls:50 | classes/DonorPortalService.cls:548        | R-009    | False positive |
| Apex CRUD ContentDistribution | 1    | classes/CampaignPostService.cls:456       | classes/CampaignPostService.cls:555       | R-014    | False positive |
| Apex CRUD ContentDistribution | 2    | classes/CampaignPostService.cls:537       | classes/CampaignPostService.cls:555       | R-014    | False positive |

| Query                         | Path | Source (scan)                                        | Sink (scan)                                     | Response | Status         |
|-------------------------------|------|------------------------------------------------------|-------------------------------------------------|----------|----------------|
| Apex CRUD ContentDistribution | 3    | classes/CampaignPostService.cls:654                  | classes/AppLogger.cls:125                       | R-015    | False positive |
| Apex CRUD ContentDistribution | 4    | classes/CampaignPostService.cls:657                  | classes/AppLogger.cls:125                       | R-015    | False positive |
| Apex CRUD ContentDistribution | 5    | classes/SettingsService.cls:571                      | classes/SettingsService.cls:345                 | R-016    | False positive |
| Apex CRUD ContentDistribution | 6    | classes/CampaignPostService.cls:678                  | classes/AppLogger.cls:125                       | R-015    | False positive |
| Apex CRUD ContentDistribution | 7    | classes/SettingsService.cls:616                      | classes/SettingsService.cls:345                 | R-016    | False positive |
| Client DOM Stored XSS         | 1    | lwc/publicDonorDashboard/publicDonorDashboard.js:316 | lwc/pfGivingStatement/pfGivingStatement.html:34 | R-017    | False positive |
| CRUD Delete                   | 1    | classes/RetentionPurgeBatch.cls:90                   | classes/RetentionPurgeBatch.cls:90              | R-018    | False positive |
| FLS Create                    | 1    | classes/ReconciliationHeartbeat.cls:60               | classes/ReconciliationHeartbeat.cls:59          | R-019    | False positive |
| FLS Create                    | 2    | classes/AppLogger.cls:126                            | classes/AppLogger.cls:125                       | R-015    | False positive |
| FLS Create                    | 3    | classes/AppLogger.cls:127                            | classes/AppLogger.cls:125                       | R-015    | False positive |
| FLS Create                    | 4    | classes/AppLogger.cls:128                            | classes/AppLogger.cls:125                       | R-015    | False positive |
| FLS Create                    | 5    | classes/AppLogger.cls:129                            | classes/AppLogger.cls:125                       | R-015    | False positive |
| FLS Create                    | 6    | classes/AppLogger.cls:130                            | classes/AppLogger.cls:125                       | R-015    | False positive |
| FLS Create                    | 7    | classes/AppLogger.cls:131                            | classes/AppLogger.cls:125                       | R-015    | False positive |
| FLS Create                    | 8    | classes/StripePaymentSweep.cls:204                   | classes/StripePaymentSweep.cls:203              | R-020    | False positive |
| FLS Create                    | 9    | classes/StripePaymentSweep.cls:205                   | classes/StripePaymentSweep.cls:203              | R-020    | False positive |
| FLS Create                    | 10   | classes/StripePaymentSweep.cls:206                   | classes/StripePaymentSweep.cls:203              | R-020    | False positive |
| FLS Create                    | 11   | classes/StripePaymentSweep.cls:207                   | classes/StripePaymentSweep.cls:203              | R-020    | False positive |
| FLS Create                    | 12   | classes/StripePaymentSweep.cls:208                   | classes/StripePaymentSweep.cls:203              | R-020    | False positive |
| FLS Create                    | 13   | classes/StagingReconciliationBatch.cls:214           | classes/StagingReconciliationBatch.cls:281      | R-021    | False positive |
| FLS Create                    | 14   | classes/StagingReconciliationBatch.cls:250           | classes/StagingReconciliationBatch.cls:281      | R-021    | False positive |
| FLS Create                    | 15   | classes/SettingsService.cls:327                      | classes/SettingsService.cls:345                 | R-016    | False positive |
| FLS Create                    | 16   | classes/RefundReconciliationBatch.cls:340            | classes/RefundReconciliationBatch.cls:344       | R-022    | False positive |
| FLS Create                    | 17   | classes/RefundReconciliationBatch.cls:341            | classes/RefundReconciliationBatch.cls:344       | R-022    | False positive |
| FLS Create                    | 18   | classes/SettingsService.cls:1014                     | classes/SettingsService.cls:1026                | R-023    | False positive |
| FLS Create                    | 19   | classes/SettingsService.cls:1016                     | classes/SettingsService.cls:1026                | R-023    | False positive |
| FLS Create                    | 20   | classes/PostInstallHandler.cls:1512                  | classes/PostInstallHandler.cls:1515             | R-024    | False positive |
| FLS Create                    | 21   | classes/PostInstallHandler.cls:1554                  | classes/PostInstallHandler.cls:1557             | R-025    | False positive |
| FLS Create                    | 22   | classes/PostInstallHandler.cls:1598                  | classes/PostInstallHandler.cls:1601             | R-026    | False positive |
| FLS Create                    | 23   | classes/PostInstallHandler.cls:213                   | classes/PostInstallHandler.cls:260              | R-027    | False positive |
| FLS Create                    | 24   | classes/StagingReconciliationBatch.cls:213           | classes/StagingReconciliationBatch.cls:281      | R-021    | False positive |
| FLS Create                    | 25   | classes/PostInstallHandler.cls:214                   | classes/PostInstallHandler.cls:260              | R-027    | False positive |
| FLS Create                    | 26   | classes/PostInstallHandler.cls:215                   | classes/PostInstallHandler.cls:260              | R-027    | False positive |

| Query      | Path | Source (scan)                              | Sink (scan)                                | Response | Status         |
|------------|------|--------------------------------------------|--------------------------------------------|----------|----------------|
| FLS Create | 27   | classes/PostInstallHandler.cls:221         | classes/PostInstallHandler.cls:260         | R-027    | False positive |
| FLS Create | 28   | classes/SettingsService.cls:324            | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Create | 29   | classes/RefundReconciliationBatch.cls:338  | classes/RefundReconciliationBatch.cls:344  | R-022    | False positive |
| FLS Create | 30   | classes/SettingsService.cls:558            | classes/SettingsService.cls:563            | R-028    | False positive |
| FLS Create | 31   | classes/SettingsService.cls:559            | classes/SettingsService.cls:563            | R-028    | False positive |
| FLS Create | 32   | classes/SettingsService.cls:560            | classes/SettingsService.cls:563            | R-028    | False positive |
| FLS Create | 33   | classes/SettingsService.cls:572            | classes/SettingsService.cls:581            | R-029    | False positive |
| FLS Create | 34   | classes/SettingsService.cls:573            | classes/SettingsService.cls:581            | R-029    | False positive |
| FLS Create | 35   | classes/SettingsService.cls:574            | classes/SettingsService.cls:581            | R-029    | False positive |
| FLS Create | 36   | classes/SettingsService.cls:575            | classes/SettingsService.cls:581            | R-029    | False positive |
| FLS Create | 37   | classes/SettingsService.cls:576            | classes/SettingsService.cls:581            | R-029    | False positive |
| FLS Create | 38   | classes/PostInstallHandler.cls:1501        | classes/PostInstallHandler.cls:1503        | R-030    | False positive |
| FLS Create | 39   | classes/PostInstallHandler.cls:1534        | classes/PostInstallHandler.cls:1537        | R-031    | False positive |
| FLS Create | 40   | classes/PostInstallHandler.cls:1578        | classes/PostInstallHandler.cls:1581        | R-032    | False positive |
| FLS Create | 41   | classes/StagingReconciliationBatch.cls:212 | classes/StagingReconciliationBatch.cls:281 | R-021    | False positive |
| FLS Create | 42   | classes/EventPurchaseService.cls:587       | classes/EventPurchaseService.cls:590       | R-033    | False positive |
| FLS Create | 43   | classes/SettingsService.cls:1011           | classes/SettingsService.cls:1026           | R-023    | False positive |
| FLS Create | 44   | classes/PostInstallHandler.cls:1509        | classes/PostInstallHandler.cls:1515        | R-024    | False positive |
| FLS Create | 45   | classes/PostInstallHandler.cls:1551        | classes/PostInstallHandler.cls:1557        | R-025    | False positive |
| FLS Create | 46   | classes/PostInstallHandler.cls:1595        | classes/PostInstallHandler.cls:1601        | R-026    | False positive |
| FLS Create | 47   | classes/CampaignMemberSyncService.cls:68   | classes/CampaignMemberSyncService.cls:80   | R-034    | False positive |
| FLS Create | 48   | classes/StagingReconciliationBatch.cls:186 | classes/StagingReconciliationBatch.cls:281 | R-021    | False positive |
| FLS Create | 49   | classes/StagingReconciliationBatch.cls:187 | classes/StagingReconciliationBatch.cls:281 | R-021    | False positive |
| FLS Create | 50   | classes/SettingsService.cls:355            | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Create | 51   | classes/SettingsService.cls:383            | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Create | 52   | classes/SettingsService.cls:391            | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Create | 53   | classes/SettingsService.cls:421            | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Create | 54   | classes/SettingsService.cls:430            | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Create | 55   | classes/PostInstallHandler.cls:1670        | classes/PostInstallHandler.cls:1645        | R-035    | False positive |
| FLS Create | 56   | classes/SettingsService.cls:2319           | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Create | 57   | classes/SetupController.cls:2682           | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Create | 58   | classes/SetupController.cls:2723           | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Create | 59   | classes/SetupController.cls:2778           | classes/SettingsService.cls:345            | R-016    | False positive |

| Query      | Path | Source (scan)                       | Sink (scan)                         | Response | Status         |
|------------|------|-------------------------------------|-------------------------------------|----------|----------------|
| FLS Create | 60   | classes/SettingsService.cls:382     | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 61   | classes/SettingsService.cls:390     | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 62   | classes/PostInstallHandler.cls:1254 | classes/PostInstallHandler.cls:758  | R-036    | False positive |
| FLS Create | 63   | classes/SettingsService.cls:2318    | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 64   | classes/SetupController.cls:2722    | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 65   | classes/SettingsService.cls:389     | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 66   | classes/PostInstallHandler.cls:1253 | classes/PostInstallHandler.cls:758  | R-036    | False positive |
| FLS Create | 67   | classes/SettingsService.cls:2317    | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 68   | classes/SetupController.cls:2721    | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 69   | classes/PricebookService.cls:142    | classes/PricebookService.cls:125    | R-037    | False positive |
| FLS Create | 70   | classes/PricebookService.cls:142    | classes/PricebookService.cls:135    | R-038    | False positive |
| FLS Create | 71   | classes/PricebookService.cls:143    | classes/PricebookService.cls:125    | R-037    | False positive |
| FLS Create | 72   | classes/PricebookService.cls:143    | classes/PricebookService.cls:135    | R-038    | False positive |
| FLS Create | 73   | classes/PostInstallHandler.cls:1252 | classes/PostInstallHandler.cls:758  | R-036    | False positive |
| FLS Create | 74   | classes/SettingsService.cls:2316    | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 75   | classes/SetupController.cls:2718    | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 76   | classes/PostInstallHandler.cls:1251 | classes/PostInstallHandler.cls:758  | R-036    | False positive |
| FLS Create | 77   | classes/SetupController.cls:2717    | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 78   | classes/SettingsService.cls:300     | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 79   | classes/PostInstallHandler.cls:1250 | classes/PostInstallHandler.cls:758  | R-036    | False positive |
| FLS Create | 80   | classes/PostInstallHandler.cls:1660 | classes/PostInstallHandler.cls:1645 | R-035    | False positive |
| FLS Create | 81   | classes/SetupController.cls:2716    | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 82   | classes/PostInstallHandler.cls:1249 | classes/PostInstallHandler.cls:758  | R-036    | False positive |
| FLS Create | 83   | classes/PostInstallHandler.cls:1653 | classes/PostInstallHandler.cls:1645 | R-035    | False positive |
| FLS Create | 84   | classes/PostInstallHandler.cls:1654 | classes/PostInstallHandler.cls:1645 | R-035    | False positive |
| FLS Create | 85   | classes/PostInstallHandler.cls:1655 | classes/PostInstallHandler.cls:1645 | R-035    | False positive |
| FLS Create | 86   | classes/PostInstallHandler.cls:1656 | classes/PostInstallHandler.cls:1645 | R-035    | False positive |
| FLS Create | 87   | classes/PostInstallHandler.cls:1657 | classes/PostInstallHandler.cls:1645 | R-035    | False positive |
| FLS Create | 88   | classes/SetupController.cls:2715    | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 89   | classes/PostInstallHandler.cls:1248 | classes/PostInstallHandler.cls:758  | R-036    | False positive |
| FLS Create | 90   | classes/SetupController.cls:2714    | classes/SettingsService.cls:345     | R-016    | False positive |
| FLS Create | 91   | classes/PostInstallHandler.cls:1220 | classes/PostInstallHandler.cls:758  | R-036    | False positive |
| FLS Create | 92   | classes/SetupController.cls:2713    | classes/SettingsService.cls:345     | R-016    | False positive |

| Query      | Path | Source (scan)                              | Sink (scan)                                | Response | Status         |
|------------|------|--------------------------------------------|--------------------------------------------|----------|----------------|
| FLS Create | 93   | classes/PostInstallHandler.cls:1216        | classes/PostInstallHandler.cls:758         | R-036    | False positive |
| FLS Create | 94   | classes/SetupController.cls:2712           | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Create | 95   | classes/PostInstallHandler.cls:1215        | classes/PostInstallHandler.cls:758         | R-036    | False positive |
| FLS Create | 96   | classes/SetupController.cls:2711           | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Create | 97   | classes/PostInstallHandler.cls:1206        | classes/PostInstallHandler.cls:758         | R-036    | False positive |
| FLS Create | 98   | classes/SetupController.cls:2710           | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Create | 99   | classes/PostInstallHandler.cls:1205        | classes/PostInstallHandler.cls:758         | R-036    | False positive |
| FLS Create | 100  | classes/SetupController.cls:2709           | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Create | 101  | classes/PostInstallHandler.cls:1204        | classes/PostInstallHandler.cls:758         | R-036    | False positive |
| FLS Create | 102  | classes/PostInstallHandler.cls:1203        | classes/PostInstallHandler.cls:758         | R-036    | False positive |
| FLS Create | 103  | classes/PostInstallHandler.cls:1202        | classes/PostInstallHandler.cls:758         | R-036    | False positive |
| FLS Create | 104  | classes/PostInstallHandler.cls:1201        | classes/PostInstallHandler.cls:758         | R-036    | False positive |
| FLS Create | 105  | classes/PostInstallHandler.cls:1200        | classes/PostInstallHandler.cls:758         | R-036    | False positive |
| FLS Create | 106  | classes/PostInstallHandler.cls:1199        | classes/PostInstallHandler.cls:758         | R-036    | False positive |
| FLS Create | 107  | classes/PostInstallHandler.cls:1198        | classes/PostInstallHandler.cls:758         | R-036    | False positive |
| FLS Create | 108  | classes/PostInstallHandler.cls:1197        | classes/PostInstallHandler.cls:758         | R-036    | False positive |
| FLS Create | 109  | classes/PostInstallHandler.cls:1196        | classes/PostInstallHandler.cls:758         | R-036    | False positive |
| FLS Update | 1    | classes/StagingReconciliationBatch.cls:269 | classes/StagingReconciliationBatch.cls:275 | R-039    | False positive |
| FLS Update | 2    | classes/SettingsService.cls:327            | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Update | 3    | classes/RefundReconciliationBatch.cls:340  | classes/RefundReconciliationBatch.cls:344  | R-022    | False positive |
| FLS Update | 4    | classes/RefundReconciliationBatch.cls:341  | classes/RefundReconciliationBatch.cls:344  | R-022    | False positive |
| FLS Update | 5    | classes/DonorPortalService.cls:374         | classes/DonorPortalService.cls:383         | R-040    | False positive |
| FLS Update | 6    | classes/SettingsService.cls:1014           | classes/SettingsService.cls:1026           | R-023    | False positive |
| FLS Update | 7    | classes/SettingsService.cls:1016           | classes/SettingsService.cls:1026           | R-023    | False positive |
| FLS Update | 8    | classes/PostInstallHandler.cls:1495        | classes/PostInstallHandler.cls:1498        | R-041    | False positive |
| FLS Update | 9    | classes/PostInstallHandler.cls:1512        | classes/PostInstallHandler.cls:1515        | R-024    | False positive |
| FLS Update | 10   | classes/PostInstallHandler.cls:1554        | classes/PostInstallHandler.cls:1557        | R-025    | False positive |
| FLS Update | 11   | classes/PostInstallHandler.cls:1598        | classes/PostInstallHandler.cls:1601        | R-026    | False positive |
| FLS Update | 12   | classes/SettingsService.cls:324            | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Update | 13   | classes/SettingsService.cls:355            | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Update | 14   | classes/SettingsService.cls:383            | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Update | 15   | classes/SettingsService.cls:391            | classes/SettingsService.cls:345            | R-016    | False positive |
| FLS Update | 16   | classes/SettingsService.cls:421            | classes/SettingsService.cls:345            | R-016    | False positive |

| Query      | Path | Source (scan)                         | Sink (scan)                           | Response | Status         |
|------------|------|---------------------------------------|---------------------------------------|----------|----------------|
| FLS Update | 17   | classes/SettingsService.cls:430       | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 18   | classes/SettingsService.cls:2319      | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 19   | classes/SetupController.cls:2682      | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 20   | classes/SetupController.cls:2723      | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 21   | classes/SetupController.cls:2778      | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 22   | classes/AttendeeNamingHandler.cls:106 | classes/AttendeeNamingHandler.cls:129 | R-042    | False positive |
| FLS Update | 23   | classes/AttendeeNamingHandler.cls:114 | classes/AttendeeNamingHandler.cls:129 | R-042    | False positive |
| FLS Update | 24   | classes/GuestRequestService.cls:129   | classes/GuestRequestService.cls:138   | R-043    | False positive |
| FLS Update | 25   | classes/GuestRequestService.cls:229   | classes/GuestRequestService.cls:237   | R-044    | False positive |
| FLS Update | 26   | classes/SettingsService.cls:382       | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 27   | classes/SettingsService.cls:390       | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 28   | classes/PostInstallHandler.cls:402    | classes/PostInstallHandler.cls:409    | R-045    | False positive |
| FLS Update | 29   | classes/PostInstallHandler.cls:1254   | classes/PostInstallHandler.cls:758    | R-036    | False positive |
| FLS Update | 30   | classes/SettingsService.cls:2318      | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 31   | classes/SetupController.cls:2722      | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 32   | classes/AttendeeNamingHandler.cls:105 | classes/AttendeeNamingHandler.cls:129 | R-042    | False positive |
| FLS Update | 33   | classes/GuestRequestService.cls:128   | classes/GuestRequestService.cls:138   | R-043    | False positive |
| FLS Update | 34   | classes/GuestRequestService.cls:228   | classes/GuestRequestService.cls:237   | R-044    | False positive |
| FLS Update | 35   | classes/SettingsService.cls:389       | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 36   | classes/PostInstallHandler.cls:1253   | classes/PostInstallHandler.cls:758    | R-036    | False positive |
| FLS Update | 37   | classes/SettingsService.cls:2317      | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 38   | classes/SetupController.cls:2721      | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 39   | classes/GuestRequestService.cls:227   | classes/GuestRequestService.cls:237   | R-044    | False positive |
| FLS Update | 40   | classes/PostInstallHandler.cls:1252   | classes/PostInstallHandler.cls:758    | R-036    | False positive |
| FLS Update | 41   | classes/SettingsService.cls:2316      | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 42   | classes/SetupController.cls:2718      | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 43   | classes/PostInstallHandler.cls:1251   | classes/PostInstallHandler.cls:758    | R-036    | False positive |
| FLS Update | 44   | classes/SetupController.cls:2717      | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 45   | classes/PostInstallHandler.cls:1250   | classes/PostInstallHandler.cls:758    | R-036    | False positive |
| FLS Update | 46   | classes/SetupController.cls:2716      | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 47   | classes/PostInstallHandler.cls:1249   | classes/PostInstallHandler.cls:758    | R-036    | False positive |
| FLS Update | 48   | classes/SetupController.cls:2715      | classes/SettingsService.cls:345       | R-016    | False positive |
| FLS Update | 49   | classes/PostInstallHandler.cls:1248   | classes/PostInstallHandler.cls:758    | R-036    | False positive |

| Query                            | Path | Source (scan)                             | Sink (scan)                        | Response | Status         |
|----------------------------------|------|-------------------------------------------|------------------------------------|----------|----------------|
| FLS Update                       | 50   | classes/SetupController.cls:2714          | classes/SettingsService.cls:345    | R-016    | False positive |
| FLS Update                       | 51   | classes/PostInstallHandler.cls:1220       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| FLS Update                       | 52   | classes/SetupController.cls:2713          | classes/SettingsService.cls:345    | R-016    | False positive |
| FLS Update                       | 53   | classes/PostInstallHandler.cls:1216       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| FLS Update                       | 54   | classes/SetupController.cls:2712          | classes/SettingsService.cls:345    | R-016    | False positive |
| FLS Update                       | 55   | classes/PostInstallHandler.cls:1215       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| FLS Update                       | 56   | classes/SetupController.cls:2711          | classes/SettingsService.cls:345    | R-016    | False positive |
| FLS Update                       | 57   | classes/PostInstallHandler.cls:1206       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| FLS Update                       | 58   | classes/SetupController.cls:2710          | classes/SettingsService.cls:345    | R-016    | False positive |
| FLS Update                       | 59   | classes/PostInstallHandler.cls:1205       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| FLS Update                       | 60   | classes/SetupController.cls:2709          | classes/SettingsService.cls:345    | R-016    | False positive |
| FLS Update                       | 61   | classes/PostInstallHandler.cls:1204       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| FLS Update                       | 62   | classes/PostInstallHandler.cls:1203       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| FLS Update                       | 63   | classes/PostInstallHandler.cls:1202       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| FLS Update                       | 64   | classes/PostInstallHandler.cls:1201       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| FLS Update                       | 65   | classes/PostInstallHandler.cls:1200       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| FLS Update                       | 66   | classes/PostInstallHandler.cls:1199       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| FLS Update                       | 67   | classes/PostInstallHandler.cls:1198       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| FLS Update                       | 68   | classes/PostInstallHandler.cls:1197       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| FLS Update                       | 69   | classes/PostInstallHandler.cls:1196       | classes/PostInstallHandler.cls:758 | R-036    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 1    | classes/CurrencyResolver.cls:260          | classes/CurrencyResolver.cls:265   | R-046    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 2    | classes/FundraiserDetailController.cls:39 | classes/AccountsSelector.cls:311   | R-047    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 3    | classes/FundraiserDetailController.cls:39 | classes/ContactsSelector.cls:228   | R-048    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 4    | classes/DonorDetailController.cls:56      | classes/AccountsSelector.cls:311   | R-047    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 5    | classes/DonorDetailController.cls:56      | classes/ContactsSelector.cls:228   | R-048    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 6    | classes/SetupController.cls:632           | classes/CommunityService.cls:269   | R-049    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 7    | classes/SetupController.cls:632           | classes/SettingsService.cls:345    | R-016    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 8    | classes/SetupController.cls:642           | classes/CommunityService.cls:269   | R-049    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 9    | classes/SetupController.cls:642           | classes/CommunityService.cls:513   | R-001    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 10   | classes/SetupController.cls:642           | classes/CommunityService.cls:328   | R-050    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 11   | classes/SetupController.cls:642           | classes/CommunityService.cls:569   | R-051    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 12   | classes/SetupController.cls:642           | classes/CommunityService.cls:608   | R-052    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 13   | classes/SetupController.cls:642           | classes/CommunityService.cls:407   | R-053    | False positive |

| Query                            | Path | Source (scan)                               | Sink (scan)                                      | Response | Status         |
|----------------------------------|------|---------------------------------------------|--------------------------------------------------|----------|----------------|
| Apex SOQL SOSL USER MODE MISSING | 14   | classes/SetupController.cls:397             | classes/SettingsService.cls:1026                 | R-023    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 15   | classes/SettingsService.cls:354             | classes/SettingsService.cls:345                  | R-016    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 16   | classes/SetupController.cls:2674            | classes/SettingsService.cls:345                  | R-016    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 17   | classes/TestDataFactory.cls:778             | classes/EventPurchaseService_Test.cls:906        | R-054    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 18   | classes/TestDataFactory.cls:778             | classes/EventPurchaseService_Test.cls:261        | R-055    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 19   | classes/TestDataFactory.cls:778             | classes/EventPurchaseService_Test.cls:578        | R-056    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 20   | classes/TestDataFactory.cls:778             | classes/EventPurchaseService_Test.cls:610        | R-057    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 21   | classes/TestDataFactory.cls:778             | classes/EventPurchaseService_Test.cls:641        | R-058    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 22   | classes/TestDataFactory.cls:778             | classes/EventPurchaseService_Test.cls:672        | R-059    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 23   | classes/TestDataFactory.cls:778             | classes/EventRegistrationController_Test.cls:171 | R-060    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 24   | classes/DonationService.cls:499             | classes/ReceiptSequenceSelector.cls:63           | R-061    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 25   | classes/EventPurchaseService.cls:1123       | classes/ReceiptSequenceSelector.cls:63           | R-061    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 26   | classes/RecurringDonationService.cls:1188   | classes/ReceiptSequenceSelector.cls:63           | R-061    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 27   | classes/SetupController.cls:334             | classes/SettingsService.cls:345                  | R-016    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 28   | classes/OpportunitiesSelector.cls:189       | classes/CurrencyResolver.cls:264                 | R-062    | False positive |
| Apex SOQL SOSL USER MODE MISSING | 29   | classes/RecordTypeService.cls:133           | classes/ReceiptSequenceSelector.cls:63           | R-061    | False positive |
| Sharing                          | 1    | classes/RecurringDonationsSelector.cls:11   | classes/RecurringDonationsSelector.cls:11        | R-063    | False positive |
| Sharing                          | 2    | classes/RecurringDesignationsSelector.cls:4 | classes/RecurringDesignationsSelector.cls:4      | R-064    | False positive |
| Sharing                          | 3    | classes/OpportunitiesSelector.cls:19        | classes/OpportunitiesSelector.cls:19             | R-065    | False positive |
| Sharing                          | 4    | classes/fflib_SObjectUnitOfWork.cls:191     | classes/fflib_SObjectUnitOfWork.cls:191          | R-066    | False positive |
| Sharing                          | 5    | classes/EventTicketTypesSelector.cls:6      | classes/EventTicketTypesSelector.cls:6           | R-067    | False positive |
| Sharing                          | 6    | classes/EventRegistrationController.cls:196 | classes/EventRegistrationController.cls:196      | R-068    | False positive |
| Sharing                          | 7    | classes/EmailSenderService.cls:10           | classes/EmailSenderService.cls:10                | R-069    | False positive |
| Sharing                          | 8    | classes/DonationService.cls:7               | classes/DonationService.cls:7                    | R-070    | False positive |
| Sharing                          | 9    | classes/DonationDesignationsSelector.cls:8  | classes/DonationDesignationsSelector.cls:8       | R-071    | False positive |
| Sharing                          | 10   | classes/CurrencyResolver.cls:8              | classes/CurrencyResolver.cls:8                   | R-072    | False positive |
| Sharing                          | 11   | classes/CampaignMemberSyncService.cls:9     | classes/CampaignMemberSyncService.cls:9          | R-073    | False positive |
| Sharing                          | 12   | classes/SystemDML.cls:30                    | classes/SystemDML.cls:30                         | R-074    | False positive |
| Sharing                          | 13   | classes/StripePaymentSweep.cls:26           | classes/StripePaymentSweep.cls:26                | R-075    | False positive |
| Sharing                          | 14   | classes/StagingReconciliationBatch.cls:31   | classes/StagingReconciliationBatch.cls:31        | R-076    | False positive |
| Sharing                          | 15   | classes/SettingsService.cls:543             | classes/SettingsService.cls:543                  | R-077    | False positive |
| Sharing                          | 16   | classes/RetentionPurgeBatch.cls:16          | classes/RetentionPurgeBatch.cls:16               | R-078    | False positive |
| Sharing                          | 17   | classes/RefundReconciliationBatch.cls:44    | classes/RefundReconciliationBatch.cls:44         | R-079    | False positive |

| Query                       | Path | Source (scan)                             | Sink (scan)                               | Response | Status         |
|-----------------------------|------|-------------------------------------------|-------------------------------------------|----------|----------------|
| Sharing                     | 18   | classes/ReconciliationHeartbeat.cls:11    | classes/ReconciliationHeartbeat.cls:11    | R-080    | False positive |
| Sharing                     | 19   | classes/PricebookService.cls:25           | classes/PricebookService.cls:25           | R-081    | False positive |
| Sharing                     | 20   | classes/PostInstallHandler.cls:18         | classes/PostInstallHandler.cls:18         | R-082    | False positive |
| Sharing                     | 21   | classes/NpsSyncService.cls:14             | classes/NpsSyncService.cls:14             | R-083    | False positive |
| Sharing                     | 22   | classes/GuestRequestService.cls:13        | classes/GuestRequestService.cls:13        | R-084    | False positive |
| Sharing                     | 23   | classes/EventRegistrationController.cls:9 | classes/EventRegistrationController.cls:9 | R-085    | False positive |
| Sharing                     | 24   | classes/DonorPortalService.cls:48         | classes/DonorPortalService.cls:48         | R-086    | False positive |
| Sharing                     | 25   | classes/CommunityService.cls:13           | classes/CommunityService.cls:13           | R-087    | False positive |
| Sharing                     | 26   | classes/AttendeeNamingHandler.cls:14      | classes/AttendeeNamingHandler.cls:14      | R-088    | False positive |
| Sharing                     | 27   | classes/AppLogger.cls:9                   | classes/AppLogger.cls:9                   | R-089    | False positive |
| Apex VF Dynamic JS Resource | 1    | pages/stripePayment.page:52               | pages/stripePayment.page:52               | R-090    | False positive |
| Apex VF Dynamic JS Resource | 2    | pages/stripePaymentReturn.page:39         | pages/stripePaymentReturn.page:39         | R-091    | False positive |

## Appendix B — every code-quality path

| Rule                             | Path | First node (scan)                           | Flagged statement (scan)                    | Group                                               | Status         |
|----------------------------------|------|---------------------------------------------|---------------------------------------------|-----------------------------------------------------|----------------|
| Multiple Trigger On same sObject | 1    | triggers/ContactTrigger.trigger:2           | triggers/ContactTrigger.trigger:2           | One trigger per object — scanner misfire            | False positive |
| Multiple Trigger On same sObject | 2    | triggers/OpportunityTrigger.trigger:2       | triggers/OpportunityTrigger.trigger:2       | One trigger per object — scanner misfire            | False positive |
| Multiple Trigger On same sObject | 3    | triggers/RecurringDonationTrigger.trigger:2 | triggers/RecurringDonationTrigger.trigger:2 | One trigger per object — scanner misfire            | False positive |
| Multiple Trigger On same sObject | 4    | triggers/EventTicketTypeTrigger.trigger:1   | triggers/EventTicketTypeTrigger.trigger:1   | One trigger per object — scanner misfire            | False positive |
| Multiple Trigger On same sObject | 5    | triggers/PaymentAccountTrigger.trigger:1    | triggers/PaymentAccountTrigger.trigger:1    | One trigger per object — scanner misfire            | False positive |
| DML Statements Inside Loops      | 1    | classes/fflib_SObjectUnitOfWork.cls:1037    | classes/SystemDML.cls:35                    | Unit of Work commit — one statement per object type | By design      |
| DML Statements Inside Loops      | 2    | classes/fflib_SObjectUnitOfWork.cls:1037    | classes/fflib_SObjectUnitOfWork.cls:208     | Unit of Work commit — one statement per object type | By design      |
| DML Statements Inside Loops      | 3    | classes/fflib_SObjectUnitOfWork.cls:1037    | classes/fflib_SObjectUnitOfWork.cls:286     | Unit of Work commit — one statement per object type | By design      |
| DML Statements Inside Loops      | 4    | classes/fflib_SObjectUnitOfWork.cls:1046    | classes/SystemDML.cls:40                    | Unit of Work commit — one statement per object type | By design      |
| DML Statements Inside Loops      | 5    | classes/fflib_SObjectUnitOfWork.cls:1046    | classes/fflib_SObjectUnitOfWork.cls:218     | Unit of Work commit — one statement per object type | By design      |
| DML Statements Inside Loops      | 6    | classes/fflib_SObjectUnitOfWork.cls:1046    | classes/fflib_SObjectUnitOfWork.cls:290     | Unit of Work commit — one statement per object type | By design      |
| DML Statements Inside Loops      | 7    | classes/fflib_SObjectUnitOfWork.cls:1056    | classes/SystemDML.cls:45                    | Unit of Work commit — one statement per object type | By design      |
| DML Statements Inside Loops      | 8    | classes/fflib_SObjectUnitOfWork.cls:1056    | classes/fflib_SObjectUnitOfWork.cls:229     | Unit of Work commit — one statement per object type | By design      |
| DML Statements Inside Loops      | 9    | classes/fflib_SObjectUnitOfWork.cls:1056    | classes/fflib_SObjectUnitOfWork.cls:299     | Unit of Work commit — one statement per object type | By design      |
| DML Statements Inside Loops      | 10   | classes/fflib_SObjectUnitOfWork.cls:1070    | classes/SystemDML.cls:50                    | Unit of Work commit — one statement per object type | By design      |
| DML Statements Inside Loops      | 11   | classes/fflib_SObjectUnitOfWork.cls:1070    | classes/fflib_SObjectUnitOfWork.cls:240     | Unit of Work commit — one statement per object type | By design      |
| DML Statements Inside Loops      | 12   | classes/fflib_SObjectUnitOfWork.cls:1070    | classes/fflib_SObjectUnitOfWork.cls:294     | Unit of Work commit — one statement per object type | By design      |
| DML Statements Inside Loops      | 13   | classes/StripePaymentSweep.cls:149          | classes/StripePaymentSweep.cls:203          | Batch job — per-row isolation by design             | By design      |

| Rule                        | Path | First node (scan)                          | Flagged statement (scan)                   | Group                                   | Status    |
|-----------------------------|------|--------------------------------------------|--------------------------------------------|-----------------------------------------|-----------|
| DML Statements Inside Loops | 14   | classes/StripePaymentSweep.cls:149         | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 15   | classes/StripePaymentSweep.cls:149         | classes/EventPurchaseService.cls:590       | Single statement after the loop         | By design |
| DML Statements Inside Loops | 16   | classes/StagingReconciliationBatch.cls:92  | classes/StagingReconciliationBatch.cls:234 | Batch job — per-row isolation by design | By design |
| DML Statements Inside Loops | 17   | classes/StagingReconciliationBatch.cls:92  | classes/StagingReconciliationBatch.cls:240 | Batch job — per-row isolation by design | By design |
| DML Statements Inside Loops | 18   | classes/StagingReconciliationBatch.cls:92  | classes/StagingReconciliationBatch.cls:275 | Batch job — per-row isolation by design | By design |
| DML Statements Inside Loops | 19   | classes/StagingReconciliationBatch.cls:92  | classes/StagingReconciliationBatch.cls:281 | Batch job — per-row isolation by design | By design |
| DML Statements Inside Loops | 20   | classes/StagingReconciliationBatch.cls:92  | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 21   | classes/StagingReconciliationBatch.cls:92  | classes/EventPurchaseService.cls:590       | Single statement after the loop         | By design |
| DML Statements Inside Loops | 22   | classes/GuestRequestService.cls:140        | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 23   | classes/RecurringRenewalBatch.cls:153      | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 24   | classes/StripePaymentSweep.cls:86          | classes/StripePaymentSweep.cls:191         | Batch job — per-row isolation by design | By design |
| DML Statements Inside Loops | 25   | classes/StripePaymentSweep.cls:86          | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 26   | classes/NpsSyncService.cls:82              | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 27   | classes/NpsSyncService.cls:148             | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 28   | classes/CampaignMemberSyncService.cls:150  | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 29   | classes/CampaignMemberSyncService.cls:152  | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 30   | classes/CampaignDesignationService.cls:171 | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 31   | classes/AttendeeNamingHandler.cls:193      | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 32   | classes/RefundReconciliationBatch.cls:217  | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 33   | classes/ReceiptService.cls:313             | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 34   | classes/ReceiptService.cls:392             | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 35   | classes/FundraiserAdminController.cls:519  | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 36   | classes/CommunityService.cls:698           | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 37   | classes/EventPurchaseService.cls:911       | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 38   | classes/EventPurchaseService.cls:917       | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 39   | classes/RecurringRenewalBatch.cls:80       | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 40   | classes/StripePaymentSweep_Test.cls:207    | classes/StripePaymentSweep.cls:191         | Batch job — per-row isolation by design | By design |
| DML Statements Inside Loops | 41   | classes/StripePaymentSweep_Test.cls:207    | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 42   | classes/RecurringRenewalBatch.cls:66       | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 43   | classes/EventTicketAdminController.cls:258 | classes/AppLogger.cls:125                  | Error logging — guarded insert          | By design |

| Rule                        | Path | First node (scan)                                  | Flagged statement (scan)                  | Group                                   | Status    |
|-----------------------------|------|----------------------------------------------------|-------------------------------------------|-----------------------------------------|-----------|
| DML Statements Inside Loops | 44   | classes/EventTicketAdminController.cls:258         | classes/PricebookService.cls:125          | Single statement after the loop         | By design |
| DML Statements Inside Loops | 45   | classes/EventTicketAdminController.cls:258         | classes/PricebookService.cls:135          | Single statement after the loop         | By design |
| DML Statements Inside Loops | 46   | classes/RefundReconciliationBatch.cls:283          | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 47   | classes/EventRegistrationController.cls:591        | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 48   | classes/EventPurchaseService.cls:1195              | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 49   | classes/NpsSyncService.cls:114                     | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 50   | classes/GuestRequestService.cls:124                | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 51   | classes/AttendeeNamingHandler_Test.cls:237         | classes/AttendeeNamingHandler_Test.cls:40 | Test fixture                            | By design |
| DML Statements Inside Loops | 52   | classes/AttendeeNamingHandler_Test.cls:237         | classes/AttendeeNamingHandler.cls:129     | Single statement after the loop         | By design |
| DML Statements Inside Loops | 53   | classes/SecureTokens.cls:30                        | classes/DonationService.cls:887           | Single statement after the loop         | By design |
| DML Statements Inside Loops | 54   | classes/CancelGoalReachedRecurringQueueable.cls:89 | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 55   | classes/RefundReconciliationBatch.cls:177          | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 56   | classes/ReceiptService.cls:514                     | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 57   | classes/CampaignPostService.cls:679                | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 58   | classes/PostInstallHandler.cls:804                 | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 59   | classes/AttendeeNamingHandler.cls:133              | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 60   | classes/CampaignPostService.cls:332                | classes/CampaignPostService.cls:678       | Single statement after the loop         | By design |
| DML Statements Inside Loops | 61   | classes/CommunityService.cls:728                   | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 62   | classes/CampaignMemberSyncService.cls:65           | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 63   | classes/StripePaymentSweep.cls:113                 | classes/StripePaymentSweep.cls:203        | Batch job — per-row isolation by design | By design |
| DML Statements Inside Loops | 64   | classes/StripePaymentSweep.cls:113                 | classes/EventPurchaseService.cls:590      | Single statement after the loop         | By design |
| DML Statements Inside Loops | 65   | classes/AttendeeNamingHandler.cls:185              | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 66   | classes/CompanyMatchService.cls:47                 | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 67   | classes/EventPurchaseService.cls:419               | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 68   | classes/DonorPortalService.cls:452                 | classes/DonorPortalService.cls:443        | Single statement after the loop         | By design |
| DML Statements Inside Loops | 69   | classes/CommunityService.cls:793                   | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 70   | classes/PostInstallHandler.cls:1754                | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 71   | classes/CampaignPostService.cls:613                | classes/CampaignPostService.cls:678       | Single statement after the loop         | By design |
| DML Statements Inside Loops | 72   | classes/DonationService.cls:652                    | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 73   | classes/EventPurchaseService.cls:1049              | classes/AppLogger.cls:125                 | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 74   | classes/CampaignPostService.cls:286                | classes/CampaignPostService.cls:678       | Single statement after the loop         | By design |
| DML Statements Inside Loops | 75   | classes/CampaignPostService.cls:302                | classes/CampaignPostService.cls:678       | Single statement after the loop         | By design |

| Rule                        | Path | First node (scan)                              | Flagged statement (scan)             | Group                                   | Status    |
|-----------------------------|------|------------------------------------------------|--------------------------------------|-----------------------------------------|-----------|
| DML Statements Inside Loops | 76   | classes/EventPurchaseService.cls:832           | classes/PricebookService.cls:125     | Single statement after the loop         | By design |
| DML Statements Inside Loops | 77   | classes/EventPurchaseService.cls:832           | classes/PricebookService.cls:135     | Single statement after the loop         | By design |
| DML Statements Inside Loops | 78   | classes/RecurringRenewalBatch.cls:426          | classes/AppLogger.cls:125            | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 79   | classes/NpsSyncService.cls:56                  | classes/AppLogger.cls:125            | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 80   | classes/RecordTypeMappingService.cls:112       | classes/SettingsService.cls:345      | Single statement after the loop         | By design |
| DML Statements Inside Loops | 81   | classes/StripeGateway.cls:533                  | classes/StripePaymentSweep.cls:203   | Batch job — per-row isolation by design | By design |
| DML Statements Inside Loops | 82   | classes/StripeGateway.cls:533                  | classes/EventPurchaseService.cls:590 | Single statement after the loop         | By design |
| DML Statements Inside Loops | 83   | classes/RecurringRenewalBatch_Test.cls:722     | classes/AppLogger.cls:125            | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 84   | classes/PricebookService_Test.cls:182          | classes/PricebookService.cls:125     | Single statement after the loop         | By design |
| DML Statements Inside Loops | 85   | classes/PricebookService_Test.cls:182          | classes/PricebookService.cls:135     | Single statement after the loop         | By design |
| DML Statements Inside Loops | 86   | classes/CampaignPostService.cls:589            | classes/CampaignPostService.cls:678  | Single statement after the loop         | By design |
| DML Statements Inside Loops | 87   | classes/RecurringDonationService_Test.cls:1399 | classes/AppLogger.cls:125            | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 88   | classes/RefundReconciliationBatch.cls:628      | classes/AppLogger.cls:125            | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 89   | classes/CampaignPostService.cls:605            | classes/CampaignPostService.cls:678  | Single statement after the loop         | By design |
| DML Statements Inside Loops | 90   | classes/AttendeeNamingHandler.cls:92           | classes/AppLogger.cls:125            | Error logging — guarded insert          | By design |
| DML Statements Inside Loops | 91   | classes/CampaignPostService.cls:256            | classes/CampaignPostService.cls:678  | Single statement after the loop         | By design |
| DML Statements Inside Loops | 92   | classes/EventTicketAdminController.cls:210     | classes/PricebookService.cls:125     | Single statement after the loop         | By design |
| DML Statements Inside Loops | 93   | classes/EventTicketAdminController.cls:210     | classes/PricebookService.cls:135     | Single statement after the loop         | By design |
| DML Statements Inside Loops | 94   | classes/DonorPortalService_Test.cls:456        | classes/DonorPortalService.cls:392   | Single statement after the loop         | By design |
| DML Statements Inside Loops | 95   | classes/DonorPortalService_Test.cls:456        | classes/DonorPortalService.cls:383   | Single statement after the loop         | By design |
| DML Statements Inside Loops | 96   | classes/EventPurchaseService.cls:772           | classes/PricebookService.cls:125     | Single statement after the loop         | By design |
| DML Statements Inside Loops | 97   | classes/EventPurchaseService.cls:772           | classes/PricebookService.cls:135     | Single statement after the loop         | By design |